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LILA

ON THE NATURE OF REALITY

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Dedication

This book was being created as a kind of internal inquiry, covering the topics interesting for me personally and serving as a way of sorting out things and systemizing them. Thus, I was not and am not quite sure anybody will ever read it and, therefore, not sure such kind of book allows any dedication. However, I still want to add it.

With all my heart, I dedicate this book:

- To my former wife Anne, whom I once chose to love, and think I still love and respect in spite of the fact that our paths diverged.
- To my daughters Vanda and Karolina, my mother Marina, and my brother Anton.
- To my dear friend Michael, who talked to me even when everybody else stopped.
- To all those whom I chose to love and admire in the past several years of my life. Whose presence I appreciate even if there's a little or no chance at all to be with them.

Clear Thinking and Language

I want to keep this book's thinking and language clear, free from the burden of previously created concepts whose relation to what actually happens in the world, what is observed, and what is experienced is unclear. People are quite capable of creating fantasies about the world, and I believe many of these fantasies are much more conditioned by the states of their creators' minds than by data and evidence.

Introduction - What This Book Is About

This book searches for a model of Reality having more explanatory power and fewer gaps than Materialism. It is composed of the following major chapters:

- **Problems to Solve**: exploration of 12 major anomalies in the Materialistic paradigm, coming from the diverse areas of physics (quantum mechanics and relativity), studies of ASC (altered states of consciousness), and biology/information.
- **Nature of Reality**: defining the criteria for a good model and the cornerstone problem with further formulating the **MALm (Mind at Large model)** and showing how MALm solves the anomalies described in the previous chapter and some extra problems.
- **Dialog with Other Models**: the most subjective and at the same time the most nuanced and emotional part in which MALm has its dialog with other models created by humankind throughout its history (Buddhism, Advaita, Materialism, and others).
- **What is in All That for Me**: small concluding chapter building a bridge from model to practice and sharing the author's personal experience and criteria for this practice.
- **Wrong Model:** small chapter-reminder about models being maps, not the area, and the potential of every model to be improved or replaced with new data and nuances' arrival.

While reading this book, a reader will probably find some unevenness of its parts, so that while there are genuine connections between these parts, some of them may look more thoroughly worked through and populated with more data and details, and some may continue the ideas in a shortened form. This is probably related to some features of the author's mind and does not really matter a lot. However, by critics or opponents, every "week" part may be used to dismantle the entire premise—that should never be the case. In spite of any local issues, the non-local observation presented by the book provides an unbreakable logical system, to which I intellectually do not see any alternatives.

Problems to Solve

In this chapter, we'll agree with an American historian and philosopher of science, Thomas Samuel Kuhn, on an episodic model of the history of science where periods of conceptual continuity and cumulative progress, referred to as periods of "normal science," were interrupted by periods of revolutionary science. The discovery of "anomalies" accumulating and precipitating revolutions in science leads to new paradigms.

After that, we'll show a set of 12 major anomalies currently presented for the major Materialistic paradigm. Those anomalies will come from physics (quantum mechanics and relativity), studies of ASC (altered states of consciousness), and biology/information. The presence of those anomalies, their nature and amount, is the foundation for the conclusion that a better paradigm should be searched for and formulated.

Experience, Experiencer, and Mistakes

Whatever we know about the world, how do we access this information? We may create concepts, but in good cases, we base them on data. But how do we access this data? Or, to formulate it differently, what is this data by its core nature? Of course, if we start thinking about any of our knowledge in terms of what it is we know, we will find a lot of different types of it: we may know how to make a cup of tea, know how to swim in the warm sea feels like, know how the fear of the sudden loud sound happens, know the neighbor's secret he tries to hide, know how to solve this fancy equation with "x," know yesterday's visit to the cinema with friends, know boundaries of our bodies, know what this book was about and how fun it was to sit in the father's armchair reading it, know that the smell of its paper connects with the smell of spicy sausages that were sent together with it by the grandma. We also know our dreams and fantasies, we know our hopes, and we know tricky or boring concepts and ideas made by philosophers or other weirdos. We know geometry, we know how our eyes follow someone we like, and we know how our thoughts persistently pursue the subject that we would like to forget.

We can easily unify all those types of data: they are all our experiences. That is what happens before any ideas or concepts. It is a given that cannot be denied because to deny it, you first need to have it. And the experience requires the experiencer, whatever it is or may be. I will try to show with this book that experience is the answer to the question of "sense of life," and the experiencer is the answer to the question of the "nature of reality."

To further that goal, let us pay attention once again to the content of the experience, particularly to its intuitions—non-deliberative insights or understandings that arise without the need for conscious reasoning or logical analysis and are felt strongly compelling. What we know for sure from the history of humankind is that intuitions can be wrong.

Our ancestors believed the Earth was flat because the horizon looked flat, and there was obviously no way the ocean's water could stick to a ball. But since about 330 BC, when Aristotle provided strong empirical evidence for a spherical Earth, many have changed that obvious intuition into something else. Spherical? Ok, but with the sun revolving around the Earth and not the other way around—everyone could see that with their own eyes, didn't they?

It is not only about the ancestors, though. We tend to think of matter as solid, but modern physics reveals that atoms, the building blocks of all matter, are composed of 99.9% empty space. To put this into perspective, the nucleus of an atom is roughly 100,000 times smaller in diameter than the atom itself, which includes the electron cloud. Electrons, in turn, are far smaller than the nucleus and occupy an almost negligible fraction of the atom's total space.

Despite this, electrons move extremely quickly, making the first step to the impression of a solid structure. Plus, the atom's structure is maintained by powerful forces that come into play when objects interact. Specifically, the electrons in one atom repel those in another due to the electromagnetic force. This repulsion prevents atoms from passing through one another, giving another input to the perception of solidity. Valence electrons—those in the outermost energy level of an atom—can be shared between atoms to form chemical bonds. However, the Pauli exclusion principle states that no two electrons in an atom can occupy the same energy level or orbital simultaneously. When atoms come into contact, this principle forces valence electrons to move to higher energy levels, which requires energy. This energy barrier explains the resistance to atomic overlap.

Further, elementary particles, which are what atoms are composed of, come into existence from energy, and their characteristics, such as mass, are actually the form of energy themselves. They also have a dual nature of being not only particles but also probability waves that behave differently depending on whether they are observed or not.

This is how the intuition of solid matter meets data on something, which is a complex fluid combination of interactions, energies, and forces. It meets and continues to remain in force, successfully driving our everyday lives.

Another even more compelling and untouchable intuition is the one of absolute time and its being separate from space. However, the experimentally proved constancy of the speed of light led to an understanding of the interdependence between motion through space and the passage of time, counterintuitive concepts like time dilation and length contraction, which have also been experimentally verified. Moreover, complex calculations within the framework of related theory (special relativity) led even to more unexpected E=mc^2 mass-energy equivalence results, showing that mass and energy are not separate entities but different forms of the same thing.

If we shift the focus of attention from the external to the internal, we will find a similar picture of wrong intuitions here. One of the most obvious will be the mind knowing itself. As a huge amount of evidence and experimental data showed, this thing does actually have an unconscious part (thanks, Freud) with a very fuzzy border with non-personal content and experience (thanks, Jung).

Another mind-related intuition driving our lives and wrong by evidence is the feeling of stable personal identity shared by one from birth to death, obviously contradicting quite different states of a 2-year-old baby and (the same personality?) his or her 38-year-old continuation as an adult, and this is if not to mention dramatic internal state changes (including worldview, beliefs, sexuality) throughout life. We need to add here a bright phenomenon of dissociative identity disorder and other dissociative disorders, such as dissociative amnesia or depersonalization-derealization disorder.

The above relates to everyday life intuitions that are dramatically important for all of us. To enhance the picture, we should highlight the role of scientists. This unique group of individuals transcends direct observation and intuitive understanding, consistently posing unconventional questions and delving into the less obvious. So, what do we have throughout time in their zone of responsibility? The list of superseded scientific theories contains hundreds of items.

The mentioned understanding of the presence of the unconscious part in the mind, in combination with the related experiments and evidence, allows us to erase the seeming boundary between "irrational" intuitions and "rational" scientific research. If we take into consideration how people's intuitions are related to those people's experiences, current states, and cultural context, then we will need to deduce that intuitions are most probably unconscious conclusions informed by a person’s accumulated experiences, knowledge, and patterns recognized over time. On the other hand, if we look at the life descriptions of many famous scientists, specifically at the histories of their great discoveries, we will find out how obviously important their personal intuitions actually were in the process.

So, why did we speak sequentially about experience as the only given and mistakes in intuitions and theories? We did that to see that experience is, in its essence, a complex mixture coming from different sources: we do not experience the world or ourselves; we experience the mixture of that and our concepts of that. In other words, we need to distinguish reality and (many) pictures of reality. This would be quite a useless and even harmful thing to keep in mind in everyday life, but really useful in the context of this book.

One more thing to mention is that I do not postulate the separation of reality as something "outer" and us "observing" it. We are obviously part of that reality, too. This means we are the reality trying to picture itself in multiple ways. Does it not sound like purposeful observation and experimentation, aiming to have as many perspectives as possible and enjoying the comparison of differences, unveiling more and more aspects of what is being observed?

This gives direction to this book and also prevents us from thinking that it is some type of search for truth. We will obviously not create anything new and will be searching among all pictures for the best approximation with the fewest contradictions and maximum explanatory power. Paradoxically, it requires us to search for anomalies or unsolved problems—these are the things that highlight the pictures of reality they belong to. Before going into this, we need to understand the meaning of these anomaly presences.

Paradigms and Anomalies

In 1962, an American historian and philosopher of science, Thomas Samuel Kuhn, published his famous work The Structure of Scientific Revolutions, first as a monograph in the International Encyclopedia of Unified Science and then as a book by the University of Chicago Press [1].

Its publication was a landmark event in the history, philosophy, and sociology of science. Kuhn challenged the then prevailing view of progress in science in which scientific progress was viewed as "development-by-accumulation" of accepted facts and theories. Kuhn argued for an episodic model in which periods of conceptual continuity and cumulative progress, referred to as periods of "normal science," were interrupted by periods of revolutionary science. The discovery of "anomalies" accumulating and precipitating revolutions in science leads to new paradigms. New paradigms then ask new questions of old data, move beyond the mere "puzzle-solving" of the previous paradigm, alter the rules of the game, and change the "map" directing new research:

> "The operations and measurements that a scientist undertakes in the laboratory are not "the given" of experience but rather "the collected with difficulty." They are not what the scientist sees—at least not before his research is well advanced and his attention focused. Rather, they are concrete indices to the content of more elementary perceptions, and as such, they are selected for the close scrutiny of normal research only because they promise the opportunity for the fruitful elaboration of an accepted paradigm. Far more clearly than the immediate experience from which they, in part, derive, operations and measurements are paradigm-determined. Science does not deal in all possible laboratory manipulations. Instead, it selects those relevant to the juxtaposition of a paradigm with the immediate experience that that paradigm has partially determined. As a result, scientists with different paradigms engage in different concrete laboratory manipulations." [1, p. 216]

Kuhn illustrates his concept of paradigm shifts through historical examples, such as the evolution of our understanding of solutions. Eighteenth-century chemists believed that homogeneous mixtures, like water and alcohol, were chemical compounds. This belief was based on observations: these substances didn't spontaneously separate and could be mixed in any proportion, much like compounds.

However, this paradigm shifted with the advent of Dalton's atomic theory. Dalton's theory postulated that atoms combine in fixed, whole-number ratios. This contradicted the prevailing view of solutions as compounds, as their constituents could combine in varying proportions.

Kuhn uses the Copernican Revolution as another prominent example of a paradigm shift. The Ptolemaic system, with its Earth-centered universe and complex system of cycles and epicycles, struggled to accurately predict planetary movements. Copernicus proposed a heliocentric model but, ironically, initially required more cycles and epicycles than the Ptolemaic system, offering no immediate improvement in accuracy. This lack of predictive power understandably led to the rejection of his model.

Galileo's insights into motion played a crucial role in enabling this paradigm shift. Challenging the Aristotelian notion that motion requires continuous force, Galileo hypothesized that objects naturally maintain their speed in the absence of friction. This seemingly counterintuitive idea, while lacking empirical proof at the time, laid the groundwork for a new understanding of motion.

Kepler, abandoning the traditional reliance on circular orbits, proposed that planets move in ellipses. This revolutionary concept, initially met with skepticism, gradually gained traction. The combined influence of Galileo's insights into motion and Kepler's elliptical orbits, though initially lacking definitive proof, gradually shifted the prevailing scientific worldview.

Finally, Newton's groundbreaking work, unifying motion and planetary motion through his laws of motion and gravitation, solidified the paradigm shift initiated by Galileo and Kepler.

As the accumulation of these and other facts, Kuhn's model of scientific development outlines five distinct phases:

1. Pre-paradigm Phase: In this initial stage, no single theory enjoys widespread acceptance. Competing schools of thought and a lack of consensus on fundamental concepts characterize this period. Scientific inquiry often takes the form of lengthy treatises due to the absence of shared assumptions and methodologies. However, as a particular framework gains traction and a consensus emerges regarding research methods, terminology, and the nature of significant questions, a dominant paradigm begins to emerge. This consolidation often leads to the marginalization of dissenting viewpoints.
2. Normal Science Phase: Once a dominant paradigm is established, "normal science" commences. This phase is characterized by meticulous puzzle-solving within the established framework. Scientific progress occurs through incremental advancements and the refinement of existing knowledge.
3. Crisis Phase: As normal science progresses, anomalies—observations that contradict the established paradigm—may emerge. While many anomalies can eventually be explained within the existing framework, persistent and accumulating anomalies can create a sense of crisis within the scientific community.
4. Scientific Revolution Phase: If persistent crises erode confidence in the existing paradigm, a period of revolutionary upheaval may ensue. During this phase, the field's fundamental assumptions are re-examined, and new paradigms emerge.
5. Post-Revolution Phase: Following a scientific revolution, a new paradigm gains dominance, and normal science resumes. Scientists now work within the framework of this new paradigm, addressing new puzzles and expanding knowledge within its established boundaries.

Science may go through these cycles repeatedly, though Kuhn notes that it is good for science that such shifts do not occur often or easily. Once a paradigm shift has taken place, textbooks are rewritten. Often, the history of science, too, is rewritten, being presented as an inevitable process leading up to the current, established framework of thought.

Since Kuhn considered problem solving (or "puzzle solving") to be a central element of science, he told the following about the candidate paradigm:

> "First, the new candidate must seem to resolve some outstanding and generally recognized problem that can be met in no other way. Second, the new paradigm must promise to preserve a relatively large part of the concrete problem-solving ability that has accrued to science through its predecessors. Novelty, for its own sake, is not a desideratum in the sciences as it is in so many other creative fields. As a result, though new paradigms seldom or never possess all the capabilities of their predecessors, they usually preserve a great deal of the most concrete parts of past achievement, and they always permit additional concrete problem-solutions besides." [1, p. 169]

In the second edition of the book, Kuhn added a postscript to further elaborate on his views on scientific progress. He presented a thought experiment: imagine an observer encountering a series of scientific theories, each representing a different stage in the historical development of a particular field. If these theories were presented without any chronological order, Kuhn argued that it would still be possible to reconstruct their historical sequence. This is because more recent theories, by their very nature, tend to be more effective and comprehensive in addressing the puzzles and challenges that scientists typically encounter within their field.

Being well-grounded in data, Kuhn's ideas have been credited with producing the kind of "paradigm shift" Kuhn discussed. Since the book's publication, over one million copies have been sold, including translations into sixteen different languages. In 1987, it was reported to be the twentieth-century book most frequently cited in the period 1976–1983 in the arts and the humanities.

Considering this, I will not hesitate to use Kuhn's concept as a tool for exploring modern anomalies and understanding their ability to reveal modern paradigms (pictures of reality) and detect a phase in which we are now in different spheres—whether it is a normal, crisis, or paradigm shift period.

Quantum mechanics

Quantum mechanics, a theory describing the strange behavior of matter and energy at the atomic and subatomic level, has gifted us with some of the most baffling yet profoundly impactful discoveries in the history of science. For decades, phenomena like the wave-particle duality of matter, the interconnectedness of entangled particles, and the seemingly paradoxical influence of the future on the past have captivated physicists and philosophers alike. These aren't just fleeting curiosities; they are robust experimental results repeatedly confirmed and harnessed in emerging technologies. And yet, despite their enduring presence on the scientific stage and despite their proven utility, these quantum phenomena continue to challenge our fundamental understanding of reality. They expose the limitations of our classical intuitions, forcing us to confront the deepest questions about the nature of space, time, and the very fabric of existence. In the following sections, we will delve into three of these perplexing phenomena – the double-slit experiment, quantum entanglement, and the delayed-choice quantum eraser – exploring their experimental details and, more importantly, grappling with the profound conceptual challenges they present, challenges that, even after generations of scrutiny, still leave us searching for a truly satisfying interpretation.

I would be putting myself at a great disadvantage if I were to state directly my understanding of the meaning of these experiments and what they tell us. Instead, I simply take the details of the experiments and present them to anyone with any lengthy description of reality, with the goal of asking one question: how does your vision fit in with the data from these experiments?

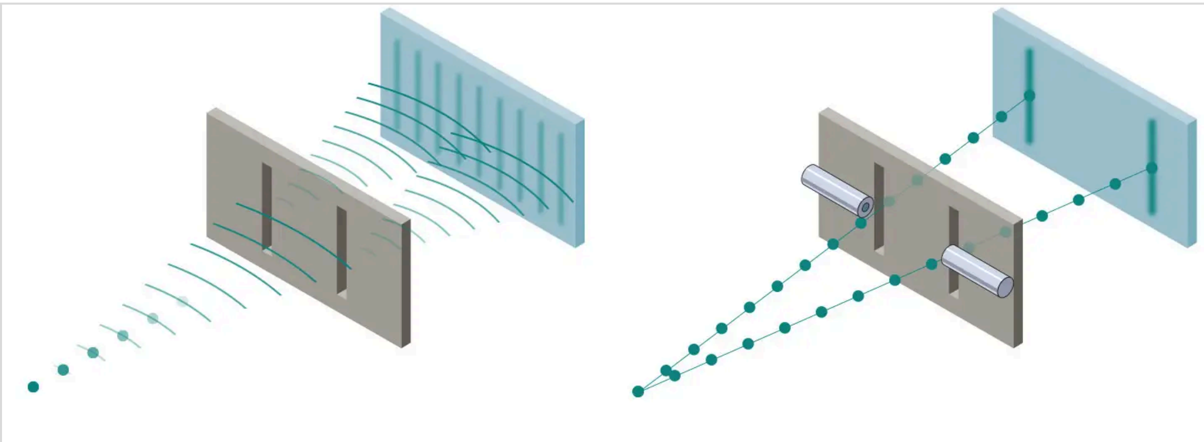
Double slit experiment

One of the most famous experiments in physics is the double slit experiment. This type of experiment was first performed by Thomas Young in 1801 as a demonstration of the wave behavior of visible light. In 1927, Davisson and Germer and, independently, George Paget

Thomson and his research student Alexander Reid demonstrated that electrons show the same behavior, which was later extended to atoms and molecules.

In the basic version of this experiment, a coherent light source, such as a laser beam, illuminates a plate pierced by two parallel slits, and the light passing through the slits is observed on a screen behind the plate. The wave nature of light causes the light waves passing through the two slits to interfere, producing bright and dark bands (interference pattern) on the screen—a result that would not be expected if light consisted of classical particles. However, the light is always found to be absorbed on the screen at discrete points, as individual particles (not waves); the interference pattern appears via the varying density of these particle hits on the screen.

Furthermore, versions of the experiment that include detectors at the slits (before and after) find that each detected photon passes through one slit (as would a classical particle) and not through both slits (as would a wave). However, such experiments demonstrate that particles do not form the interference pattern if one detects which slit they pass through.



An important version of this experiment involves single particle detection. Illuminating the double slit with low intensity detects single particles as white dots on the screen. Remarkably, however, an interference pattern emerges when these particles are allowed to build up one by one, which means each individual particle interferes with itself.

Despite being on stage for more than 200 years, the experiment does not have any particular meaning proven in any way—instead, it is often used to highlight the differences and similarities between the various interpretations of quantum mechanics.

The standard interpretation of the double-slit experiment is that the pattern is a wave phenomenon, representing interference between two probability amplitudes (meaning a wave in the case of a particle is not the physically existing wave but the wave of probabilities of particle presence in a specific part of space), one for each slit. Low-intensity experiments demonstrate that the pattern is filled in one particle detection at a time. Any change to the apparatus designed to detect a particle at a particular slit alters the probability amplitudes, and the interference disappears. This interpretation is independent of any conscious observer.

Niels Bohr interpreted quantum experiments like the double-slit experiment using the concept of complementarity. In Bohr's view, quantum systems are not classical, but measurements can only give classical results. Certain pairs of classical properties will never be observed in a quantum system simultaneously: the interference pattern of waves in the double-slit experiment will disappear if particles are detected at the slits. Modern quantitative versions of the concept allow for a continuous tradeoff between the visibility of the interference fringes and the probability of particle detection at a slit.

In the Copenhagen interpretation, which is a collection of views about the meaning of quantum mechanics stemming from the work of Niels Bohr, Werner Heisenberg, Max Born, and others, included the idea that quantum mechanics was intrinsically indeterministic, with calculated probabilities and the act of "observing" or "measuring" an object irreversible, and no truth could be attributed to an object, except according to the results of its measurement. In this interpretation, complementarity means a particular experiment can demonstrate particle behavior (passing through a definite slit) or wave behavior (interference) but not both at the same time. The question of which slit a particle travels through has no meaning when there is no detector.

According to the relational interpretation of quantum mechanics, first proposed by Carlo Rovelli, observations such as those in the double-slit experiment result specifically from the interaction between the observer (measuring device) and the object being observed (physically interacted with), not any absolute property possessed by the object. In the case of an electron, if it is initially "observed" at a particular slit, then the observer–particle (photon–electron) interaction includes information about the electron's position. This partially constrains the particle's eventual location on the screen. If it is "observed" (measured with a photon) not at a particular slit but rather at the screen, then there is no "which path" information as part of the interaction, so the electron's "observed" position on the screen is determined strictly by its probability function. This makes the resulting pattern on the screen the same as if each individual electron had passed through both slits.

There are multiple variants of the many-worlds interpretation. The unifying theme is that physical reality is identified with a wavefunction, and this wavefunction always evolves unitarily, i.e., following the Schrödinger equation with no collapses. Consequently, there are many parallel universes that only interact with each other through interference. David Deutsch argues that the way to understand the double-slit experiment is that in each universe, the particle travels through a specific slit, but its motion is affected by interference with particles in other universes, and this interference creates the observable fringes.[98] David Wallace, another advocate of the many-worlds interpretation, writes that in the familiar setup of the double-slit experiment, the two paths are not sufficiently separated for a description in terms of parallel universes to make sense.

Applying Thomas Kuhn's concept of paradigms, the 200-year history of the experiment, and yet an absent understanding of what it tells us about the world highlights the probable crisis in some worldview (picture of reality, paradigm). Philosophically, the problem reaches its peak in the "shut up and calculate" approach, suggesting one not to worry about the interpretation or the meaning of quantum mechanics but rather focus on the mathematical formalism and the empirical predictions that it makes.

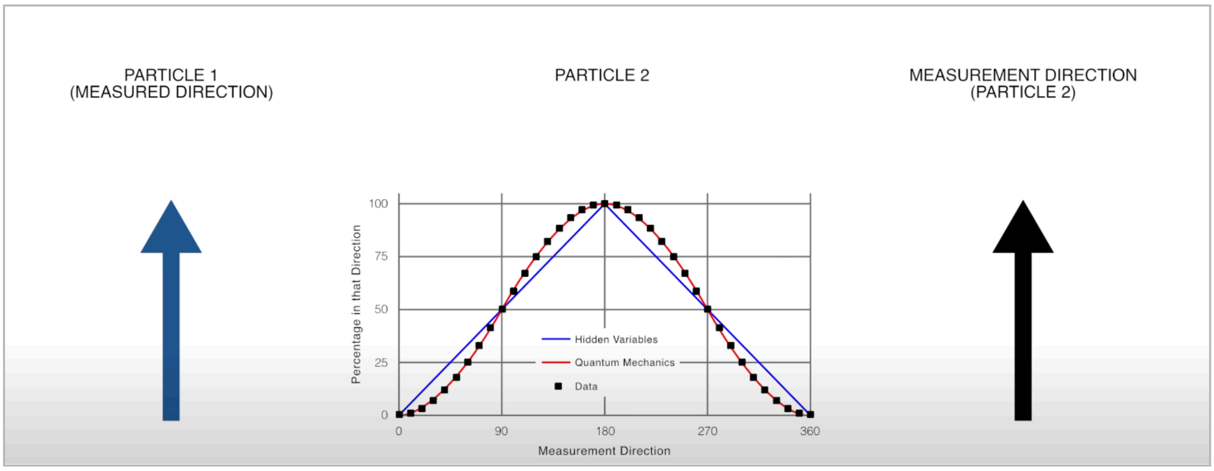
Quantum entanglement

Quantum entanglement is another phenomenon in the world of particles that has challenged some aspects of different worldviews for years. The term describes two or more particles being linked together so that their specific properties are correlated, no matter how far apart they are—if you measure the properties of one particle, you instantly know the properties of the other—whereas the particles don't have definite properties until measured.

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen published a paper, followed by several papers from Erwin Schrödinger, exploring this strange behavior. Einstein and others found this "spooky action at a distance," as Einstein called it, impossible because it seemed to violate the local realism view of causality, stating that an object is influenced directly only by its immediate surroundings and that for a cause at one point to have an effect at another point, something in the space between those points must mediate the action. To exert an influence, something, such as a wave or particle, must travel through the space between the two points, carrying the influence.

It is interesting to mention that Einstein, who, 30 years earlier, shook the conventional worldview with his famous idea of space and time being relative (special theory of relativity) and the following conclusion of mass being a form of energy, in this case, resisted the next worldview revolution, willing to protect the picture of reality that already was weird enough thanks to him. To protect local realism, he and others suggested the idea of local hidden variables, i.e., properties contained within the individual particles themselves and not arising due to measurement—as a pair of gloves put into two boxes then driven a thousand miles away leading to when we open one box and making sure it contains the right glove, we can be 100% sure the opening the second box will reveal the left glove.

Later, however, by John Bell starting in 1964 and others, the mathematical model was created showing that hidden variable presence and their absence produce different predictions. Between 1980 and 1982, the first experiments, followed by many subsequent ones, were conducted to provide the actual data matching one of these predictions—the prediction that there were no hidden variables.

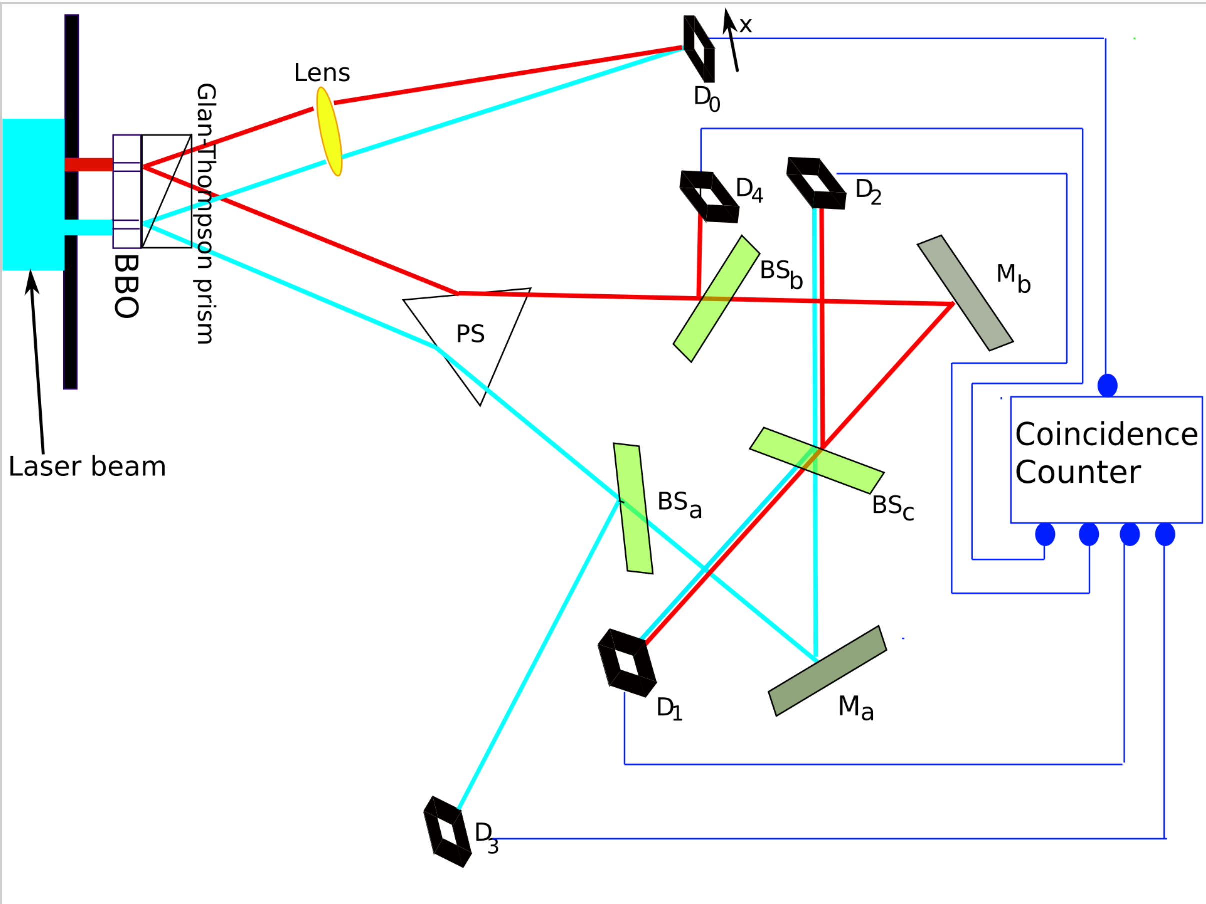


For Einstein's gloves, that meant that neither of them was right or left before the box was opened and that both of them were parts of the system ignoring space. As well as in the case of a double-slit experiment, the meaning of that data remains a subject of debate lying within approximately the same set of interpretations.

Applying Thomas Kuhn's concept of paradigms, the experiment's almost 50-year history and yet an absent understanding of what it tells us about the world highlight the probable crisis in some worldview (picture of reality, paradigm).

Delayed choice quantum eraser experiment

The delayed choice quantum eraser experiment is a modification of the double slit experiment utilizing additionally the phenomenon of quantum entanglement. The experiment has many variations, the most famous of which is from 1999 by Dr. Kim and others [2].



The experiment [3] starts with an argon laser randomly firing individual photons either through slit A or slit B of a double-slit apparatus. In the illustration, the photon paths are color-coded as red or light blue lines to indicate which slit the photon came through (red indicates slit A, light blue indicates slit B). As there is no measuring device at either slit, we cannot know which slit the laser fired through until they pass through and are detected on the other side.

On passing the slits, a photon (from either slit) is converted by a nonlinear optical crystal BBO (beta barium borate) into two identical, orthogonally polarized entangled photons with 1/2 the frequency of the original photon. The paths followed by these orthogonally polarized photons are caused to diverge by the Glan–Thompson prism: the top one goes to detector D_0. So, if the photons only hit D_0, we do not know the which-path information since when the photon arrives, it could have either come from path A or path B. Because we do not have the path information, it should produce an interference pattern, as the particle would act in a way as if it goes through both slits, being the wave of possibilities instead of one of these possibilities.

However, the other entangled photon goes the other way, and because it is entangled, it will affect the result of its twin that went to D_0. The other photon from either A or B goes through the PS prism that sends it along divergent paths depending on whether it came from slit A or slit B. It hits either beam splitter BS_A or BS_B. At both of these, there is a 50/50 chance it will either pass through or bounce off and go to either D_3 or D_4.

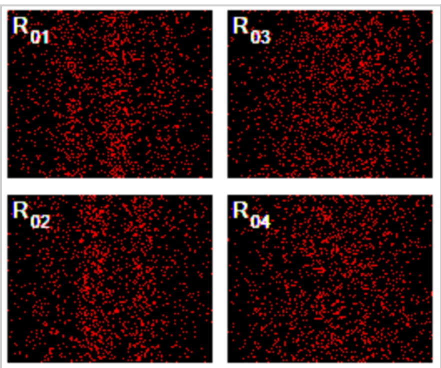
If the photon hits either one of these detectors, we obtain which-path information because the only way the photon could hit D_3 is if it came out of B, and the only way it could hit D_4 is if

it came out of A. As we have the path information, it should produce a clump pattern.

If the photon passes through BS_A or BS_B, it will bounce off mirror M_A or $M_{B,}$ and with a 50/50 chance, either pass through or bounce off BS_C and go to either D_1 or D_2. If the photon hits either one of these detectors, we lose which-path information since when the photon arrives, it could have either come from path A or path B. Because we do not have the path information, it should produce an interference pattern.

Here comes the first important note on the experiment: some argue that physical interaction from the detector is what is causing the collapse. But all the detectors of the experiment physically interact with photons; however, D_3 and D_4 "cause" the collapse, and D_1 and D_2 "do not." All the detectors are equal—the only difference between them is not physical—what we, conscious observers, know about the experiment setup. Thus, who- or whatever this observer is, the one's knowledge is an integral part of the experimental setup.

Further, the optical path length measured from slit to D_1, D_2, D_3, and D_4 is 2.5 m longer than the optical path length from slit to D_0. This means that any information that one can learn from the photons on any other detector is approximately 8 ns later than what one can learn from its entangled pair on D_0. Thus, D_0 presents the information that only arises later, and both entangled photons are parts of the system ignoring time. It is important to note that information is not just revealed but created at detectors other than D_0 (considering hidden variable absence, see quantum entanglement description in the previous chapter).



By using a coincidence counter, the experimenters were able to isolate the entangled signal from photo-noise, recording only events where both entangled photons were detected (after compensating for the 8 ns delay) on two detectors simultaneously: jointly between D_0 and D_1, D_2, D_3, D_4 (R_{01}, R_{02}, R_{03}, R_{04}). It is important to note that isolating these events is not a methodological error like many critics of the experiment claim [4], [5], but a necessary part of it. $R_{01\text{-}04}$ subsets of data isolated with a coincidence counter have nothing in common with "randomness"; they are exactly the data the experiment aims to get.

Hard Problem of Consciousness

In 1995, in his paper "Facing up to the problem of consciousness." [6], Australian philosopher and cognitive scientist David Chalmers named "hard" one of the problems related to the study of consciousness. Precisely, he argued that one question remained unanswered in all theories, claiming that experience arises from physical processes: "How exactly does it arise?"

This question contrasted in Chalmers' view with the "easy problems" of explaining why and how physical systems give a human being the ability to discriminate, integrate information, and perform behavioral functions such as watching, listening, speaking (including generating an utterance that appears to refer to personal behavior or belief), and so forth. The easy problems are amenable to functional explanation—that is, explanations that are mechanistic or behavioral—since each physical system can be explained (at least in principle) purely by reference to the "structure and dynamics" that underpin the phenomenon. But:

> "... even when we have explained the performance of all the cognitive and behavioral functions in the vicinity of experience—perceptual discrimination, categorization, internal access, verbal report—there may still remain a further unanswered question: Why is the performance of these functions accompanied by experience?" [6]

The hard problem of consciousness has scholarly antecedents considerably earlier than Chalmers. Chalmers himself noted that "a number of thinkers in the recent and distant past" have "recognized the particular difficulties of explaining consciousness." [7] He stated that all his original 1995 paper contributed to the discussion was "a catchy name, a minor reformulation of philosophically familiar points."

However, we can go far back in time and deep into ancient traditions of thought such as Hinduism or Buddhism to surprisingly find out that these traditions did not care too much about what Chalmers formulated as a "problem of consciousness"—not because these traditions did not care about consciousness but because they did not care much about the matter—not a shock if to consider the fact that they did not have good tools to explore matter while having an excellent tool to explore consciousness—attentive and not distracted introspection. But time passed, and the West found an extremely fun toy to play with—matter. As the tools to play with this comfortably measurable subject developed, the fun ran higher—the game was so enthralling that theories built on its basis almost forgot about consciousness. And it was a nearly miraculous event—science has managed to exclude from the picture of the world the element through which ALL the scientific data (and all the data at all) came.

With time, however, the feature of self-reflection returned people with their minds on the stage. Works on the history and philosophy of science unexpectedly showed that science was made by people, and these people had minds [1]. These unpleasant findings, combined with even more unpleasant discoveries provided by the behavior of nature at and below the scale of atoms [2], [3], led both to consciousness' return in focus and to an extraordinarily high level of resistance against this return.

This resistance is politically correctly named "a wide philosophical and scientific response to David Chalmers' formulation of the hard problem of consciousness." Below, we will go through different areas of this resistance. We can with confidence predict that—considering the fact that the "new" unpleasant element in the picture is consciousness—the resistance will take multiple forms of intellectual exercises aimed to exclude that element from the view again. But this prediction itself does not automatically make arguments invalid. This validity should be logically checked to understand if the "hard problem of consciousness" may or may not be excluded from the list of problems of the current dominant scientific paradigm.

So, one view is that the hard problem is just another easy problem because every fact about the mind is a fact about the performance of various functions or behaviors. But how exactly does the performance of functions or their combinations produce subjective experience? They say: "Once all the relevant functions and behaviors have been accounted for, there will not be any facts left over in need of explanation." But how exactly are any functions relevant to experience? These remain our open questions to Paul and Patricia Churchland, Daniel Dennett, Keith Frankish, and Thomas Metzinger.

Another view is that the capacity to recognize a type of experience when it occurs in one's own mental life explains phenomenal consciousness. But, before its type is recognized, why and how exactly does experience occur in the way it does? This remains our open question to Peter Carruthers. "Since consciousness is a representation, and representation is fully functionally analyzable, there is no hard problem of consciousness." But how exactly does this particular representation arise from the physical parameters of the perceived world?

Another view is that the main arguments for the existence of a hard problem—philosophical zombies, Mary's room, and Nagel's bats—are only persuasive if one already assumes that "consciousness must be independent of the structure and function of mental states, i.e., that there is a hard problem." But does the hard problem say anything about consciousness' independence of the structure and function of mental states? And does the hard problem not exist without listed arguments? One does not need any of these arguments to ask the question: if we are claiming that experience arises from physical processes, then "How exactly does it arise?" These remain our open questions to Glenn Carruthers and Elizabeth Schier.

Another view is that "Of course, an explanation isn't the same as an experience, but that's because the two are completely independent categories, like colors and triangles. It is obvious that I cannot experience what it is like to be you, but I can potentially have a complete explanation of how and why it is possible to be you." But modern data, all of a sudden, shows [14] that in specific circumstances, it is possible to experience what it is like to be you—and at the same time, the hard problem is exactly the statement that we have no progress for a long time in any attempts to explain how this mental "being you" (or "me") is related to physical correlates in the brain or physical properties of perceived objects. And why this explanation is not only incomplete so far but also does not show any signs of progress—this remains our open question to Massimo Pigliucci.

Another view is that the hard problem is misguided as it asks how consciousness can emerge from matter, whereas, in fact, sentience emerges from the evolution of living organisms. But how exactly does it emerge? And this remains our open question to Peter Hacker.

Another view is that it is mistaken not only to believe there is a hard problem of consciousness but to believe phenomenal consciousness (conscious experience) exists at all. But how exactly can anyone have such a point of view as a part of their experience if this experience does not exist? I could never imagine anyone going so far in an attempt to win a debate that one using an argument "I actually do not experience." And the question of how exactly they experience this debate in the absence of experience remains our open question to Daniel Dennett, Georges Rey, and Keith Frankish.

Another view is that phenomenal consciousness (conscious experience) is an illusion. But isn't any illusion something that we are experiencing? And if so, does it not mean that any content of experience can be distorted (i.e., can be an illusion) but not the fact of experience itself? And this remains our open question to Daniel Dennett and his beard.

Here, before going to the observation of the next views, I would like to quote Chalmers' own answer to that:

> "The reality of consciousness is more certain than any theoretical commitments (to, for example, physicalism) that may be motivating the illusionist to deny the existence of consciousness. The reason for this is because we have direct "acquaintance" with consciousness, but we do not have direct acquaintance with anything else (including anything that could inform our beliefs in consciousness being an illusion). In other words, consciousness can be known directly, so the reality of consciousness is more certain than any philosophical or scientific theory that says otherwise." [8]

We, however, proceed to the next view. It is that the hard problem stems from human psychology and is therefore not indicative of a real gap between consciousness and the physical world; it reflects only how the human mind tends to conceptualize the relationship between mind and matter, but not into what the true nature of this relationship actually is. But cannot it be the same applied to ANY knowledge obtained via "human psychology," i.e., to any scientific theory and any worldview at all? And if so, is the hard problem returning in a normal row with all the other known problems—solved or unsolved to some degree at the moment? What should we do with the idea that a hard problem is a concept, and some other theories are not, and somehow deal with the true nature directly? And these remain our open questions to David Papineau, Joseph Levine, and Janet Levine.

Another view is that an explanatory gap does exist, and no answer to date is satisfactory, but the gap will inevitably be closed in the future; just study and wait. The future is uncertain, no question about that, but "hard" in the problem name deals with the already recorded long history of the problem that remains unsolved for a suspiciously long time with no visible progress in any attempt to address it. It indicates that there is a good chance that we deal with the case when the performed activities have nothing to do with the goal. Like if we want to fill the bottle with water and pour it, but the bottle suspiciously remains empty, there is a chance that we pour water past the bottle. And if so, what is the chance that if we continue doing so, the prediction that the bottle will inevitably be filled in the future is wrong? Is not the hard problem itself the thought of the possibility of movement in a completely wrong direction, saying that there are signs of it? And these remain our open questions to Daniel Stoljar and others.

Another view is that consciousness is either a non-physical substance separate from the brain or a non-physical property of the physical brain. But if separate, then how exactly do they interact while being obviously correlated? It is a form of question marked as the hard problem, and it remains untouched. And if it is a non-physical property of the physical brain, then how exactly can physical have non-physical properties? It is a form of the question marked as the hard problem too, and it remains untouched. And these remain our open questions to René Descartes, Martine Nida-Rümelin, and Frank Jackson.

Another view is that the fundamental nature of reality is neither mental nor physical; in other words, it is "neutral." But how exactly does this neutrality give rise to the experience of physical objects and psychological states? This remains our open question to William James and Bertrand Russell.

To summarize this brief review of philosophical attempts to deny the hard problem of consciousness, I would say that the opposing questions to each and every one of these attempts that we presented above "debunk all the debunkings," turning them into nothing. And in the presence of this "nothing," i.e., in the absence of anything dismissing, the hard problem of consciousness remains an integral part of any theory claiming that experience arises from physical processes.

Besides the validity of the philosophical responses to the hard problem, its presence in the current scientific studies of consciousness and its insolubility within models built by these studies are also worth checking. Thus, we will try to check these things in the following brief overview.

In the late 1980s, cognitive psychologists Bernard Baars and Stan Franklin proposed global workspace theory (GWT) [9] as a cognitive architecture that was supposed to work as a theory of consciousness as well. The authors explain the theory with the metaphor of a theater, with conscious processes represented by an illuminated stage. This theater integrates inputs from a variety of unconscious and otherwise autonomous networks in the brain and then broadcasts them to unconscious networks (processing modules within the brain represented in the metaphor by a broad, unlit "audience"). Attention acts as a spotlight, bringing some of the unconscious activity into conscious awareness. The global workspace is a functional hub of broadcast and integration that allows information to be disseminated across modules. With time and research, global workspace theory has garnered significant support from a variety of experimental data in cognitive psychology and neuroscience and proved itself to be a good model for describing the mechanisms by which information becomes globally available and how this broadcast allows for flexible and adaptive behavior. It explains the function of consciousness in making information accessible to various modules. However, the hard problem asks why this global availability should be accompanied by subjective experience (qualia) at all. Why isn't all this information processing happening in the dark, without any "feeling" or "what-it's-like-ness"? The theory doesn't bridge the "explanatory gap" between the neural correlates of consciousness (the brain activity that consistently accompanies conscious experience) and the qualitative nature of experience. Knowing that certain neural networks are active and broadcasting information doesn't inherently tell us why that activity feels like anything in particular (e.g., the redness of red, the pain of a burn). Therefore, while being one of the leading modern theories of consciousness, GWT is generally considered not to directly solve and even address the hard problem of consciousness, with many proponents of it acknowledging this limitation.

Another widely known and discussed theory of consciousness is integrated information theory (IIT), pioneered by neuroscientist Giulio Tononi in 2004 and further developed by neuroscientist Christof Koch [10]. It proposes that consciousness is fundamentally identical to a specific kind of information: integrated information. IIT also attempts to explain the specific quality of an experience (what it feels like, or qualia) by the specific structure of the integrated information. And as it looks like renaming consciousness to "integrated information" and experience to "structure of integrated information," the hard problem turns into the question of how integrated information and its structure relate to physical processes (in the brain or wherever else). IIT answers that integrated information and its structure are not merely correlates of physical processes but are the phenomenal aspect of those processes. And that looks like renaming observable correlation to "aspect." What does "phenomenal aspect" mean? IIT answers that this means "subjective, first-person experience." Thus, using the connection of terms suggested by IIT and reverting renames, we get "consciousness and subjective experience are correlated with physical processes." As—after playing with words is reverted—we just go back to the obvious initial observation with no extra explanation of the nature of this correlation, IIT starts to look like a kind of linguistic joke, and not at all a harmless one: while global workspace theory (GWT) directly acknowledged its limitation (being not able to address the hard problem), IIT pretends to solve it while careful analysis of the suggested concepts shows endless walking in terminological circles with 0 explanatory power.

The next theory worth mentioning is "orchestrated objective reduction" (Orch-OR), or the quantum theory of mind, proposed by English Nobel Laureate in Physics and philosopher of science, Roger Penrose, and American anesthesiologist Stuart Hameroff [11]. It states that consciousness originates at the quantum level inside neurons. The mechanism is held to be a quantum process called objective reduction that is orchestrated by cellular structures called microtubules, which form the cytoskeleton around which the brain is built. However, while the theory describes a physical mechanism—quantum processes in microtubules leading to objective reduction—Orch-OR doesn't inherently explain why or how these specific physical events should be accompanied by subjective experience, or, in other words, contains no solution to the "hard problem." The theory's authors attempt to fill this gap by proposing that the objective reduction (OR) events are not just random collapses of quantum states. Instead, they suggest that each OR event is associated with a fundamental "proto-conscious" experience or a basic unit of qualia. Yet, this proposal does not include any explanation of how exactly they are associated, and the explanatory gap remains. Penrose and Hameroff do not stop here and try to fill this next-level gap with further explanation: they utilize the proto-panpsychist ideas, suggesting that rudimentary forms of consciousness or experiential qualities might be intrinsic to the fundamental structure of reality, possibly linked to the geometry of spacetime at the Planck scale. However, this does not seem to eliminate the gap, because in that picture, both physical properties and experiential qualities are the intrinsic elements of reality, which is a very reasonable conclusion, as we observe both of these. However, the hard problem is exactly about the explanation of how these two things are related—no statement of correlation solves this.

In 2011, Graziano and Kastner proposed the attention schema theory (AST) of consciousness (or subjective awareness) [12]. The theory proposes that the brain is an information-processing device, and it has the capacity to focus its processing resources more on some signals than on others. That focus may be on select, incoming sensory signals, or it may be on internal information such as specific, recalled memories. That ability to process select information in a focused manner is sometimes called attention. The brain not only uses the process of attention, but it also builds a set of information, or a representation, descriptive of attention. That representation, or internal model, is the attention schema, which provides the requisite information that allows the machine to make claims about consciousness. Thus, the theory proposes that our feeling of subjective experience is a construct of the brain—a useful, albeit simplified, model of attention. The hard problem, from this perspective, is a consequence of the way our brains represent themselves and their attentional processes. It suggests that we are convinced we have qualia because our internal model of attention leads us to that conclusion. However, even if the brain creates a model of attention, why does the perception of that model feel like anything at all? Why isn't it just a set of internal representations being processed without any accompanying "inner movie" or subjective awareness? In other words, AST does not provide a definitive answer to the fundamental question of why certain physical processes should be accompanied by subjective experience.

We could continue this row of theories and models, but we can also stop at any moment. The hard problem of consciousness itself, as an idea, is a prediction that any model, theory, or experiment based on the concept of a genuine distinction between mind and matter will never be able to build any bridge between the two. The reason for that is precisely this distinction: we can build a bridge between land and other land, but we can never connect with the bridge the land and its absence.

Applying Thomas Kuhn's concept of paradigms, the hard problem's almost 30-year history (with indefinitely long predecessors) and yet an absence of ANY progress in solving it highlights the probable crisis in some worldview (picture of reality, paradigm).

Altered States of Consciousness

Altered states of consciousness (ASCs) refer to any condition that significantly deviates from one's normal waking state of consciousness. These states are almost always temporary and involve a change in the overall pattern of subjective experience, where an individual perceives their mental functioning as distinctly different from their usual norms.

While often associated with drugs, many ASCs occur naturally or can be induced through various non-pharmacological methods. The study of ASCs is a growing field in a contemporary context, while the history of ASCs and reports on their formal structure and factual content date back several thousand years.

In recent decades, several factors have driven a resurgence of scientific, clinical, and general interest in ASCs. After years of prohibition, there's renewed interest in the therapeutic potential of psychedelics (e.g., psilocybin, MDMA, LSD) for conditions like depression, PTSD, anxiety, and addiction. This has led to numerous clinical trials and a push for medical legalization in various contexts. Advances in communication and translation have led to the widespread adoption of mindfulness-based practices in mainstream society, healthcare, and education, and have spurred research into the neurobiological and psychological effects of meditation, a well-known method for inducing ASCs. The study of ASCs now involves collaboration across fields like psychology, neuroscience, anthropology, pharmacology, philosophy, and religious studies, leading to a more holistic understanding.

Beyond psychedelics and meditation, there's growing interest in understanding and utilizing other factors causing ASCs, such as breathwork, sensory deprivation, dreaming (especially lucid dreaming), and extreme physical states. In addition, all this nowadays is backgrounded by neuroscience and neuroimaging: techniques like fMRI, EEG, and MEG allow researchers to observe brain activity in real time during ASCs, providing objective data on what's happening neurologically. This helps to move the study of ASCs beyond purely subjective reports.

While modern science provides new tools, human engagement with and understanding of ASCs is ancient and deeply embedded in human culture. Dating back tens of thousands of years, shamanic traditions across diverse cultures worldwide have intentionally induced ASCs (through drumming, chanting, fasting, hallucinogenic plants, etc.) to communicate with other realms, heal, and gain knowledge.

Practices like yoga and meditation, which aim to achieve profound altered states of consciousness (e.g., Samadhi in Yoga, various Jhanas in Buddhism), date back millennia. Texts like the Yoga Sutras of Patanjali (around 400 CE) meticulously describe the stages, techniques, and experiences of meditation, representing early reports of formal structures and factual content of these states.

Along with the questions raised by quantum mechanics data and the persistence of the hard problem of consciousness, huge data accumulated in the sphere of ASCs provide an additional set of anomalies for the framework of the modern prevailing materialistic paradigm. It is important to note that all three problems listed in this chapter form a "continuum" context, meaning they represent not "separate cases" but a logically consistent and unified realm.

Dissociation-Association Coincidence Problem

In this chapter, we will discuss two opposite processes related to the mind and subjective experience. We will start with the one that linguistically seems "negative" (but is actually NOT from certain perspectives)—dissociation—and then continue with the opposite one—association—that linguistically seems "positive" (but is absolutely NOT from certain perspectives). In a common use in language, the words "association" ("A" is connected to "B") and "dissociation" ("A" was connected to "B" but is not anymore) are very generic and can be applied to multiple spheres. Without losing these meanings, in this book, to stick the terms to the sphere of the mind, we will have to define them more specifically. In the definition of "dissociation," we will follow word-for-word a Brazilian-Dutch philosopher, Bernardo Kastrup:

> "Dissociation is when certain cognitive bridges are not present. They may have been present, and then they are no longer present, or they may have never been there. A cognitive bridge is what allows you to go from one mental content to another because of a certain association. For instance, if you think of inflatable colorful balloons, you think of a birthday party—that is a cognitive bridge—an association between two completely different experiential states: one perceiving or imagining colorful air balloons, and the other is everything related to a birthday party. That birthday party, in turn, can lead to an association with the parents, your mother, your father, and childhood. So these are cognitive associations. They are the glue of mind; they are the glue that keeps a mental space sort of

> integrated and gives us a sense of a unified self: we are not just a disjoint little experiential states here and there, like a memory here, a self-image there—no, we are sort of a coherent integrated web of mental states that are mutually associated." [13]

Using this definition, we will go through the data to check how common **dissociation** is for the mind. The first thing we will touch on is forgetting things. No study ever found an individual who never faced forgetting. There are around 8 billion people on Earth, and we can, with no doubt, extrapolate data to say this means we have 8 billion of 8 billion cases of the phenomenon's representation. This number alone is evidence of an extreme prevalence of dissociation in the sphere of the mind, but we will continue the investigation a little further. Considering our schema with "A" and "B" and the connection between them, it is important to note that forgetting is not eliminating "B" (something that we forgot), but a temporary loss of connection to it (as recalling information accompanies the process). So, this will be good to demonstrate how much information is remembered on perceiving, and with what level of detail.

In this regard, interesting data is provided by the research on NDEs (near-death experiences), specifically by the aspect of such experiences called "life review"—a widely reported phenomenon in NDEs, where individuals vividly and rapidly re-experience their life history. It's often described as "having their life flash before their eyes," but it's typically far more profound and detailed than a simple memory recall. The review can encompass an entire lifetime, often feeling like it happens in an instant, yet with incredible detail. It's not necessarily a linear playback but can be panoramic or involve multiple events simultaneously. The memories are often rich with sensory details (sights, sounds, smells) and are deeply emotional. Experiencers also report re-feeling the emotions of the events. However, the obvious feature of NDE life review is that it is not just a replay of past events with a high level of detail, but is a transformation of that having memories as the basis: for example, many experiencers describe observing their life from an objective, third-person viewpoint, as if watching a movie or play of their own existence, or they describe not just their own recalled emotions, but also the emotions of those they interacted with; the experience of time during a life review is often described as altered, with events unfolding simultaneously or with a different temporal logic than in waking life.

While the scientific community continues to explore the neurological and psychological underpinnings of NDEs, metaphysical debates on their nature are not important to our current goal, which is to show how much information is actually recorded, dissociated, and may be re-associated in specific circumstances or states of consciousness. The extreme level of detail in a life review during an NDE refers to this goal.

Our everyday recall of past events is highly selective and often fragmented. We remember snippets, general impressions, and significant milestones. We certainly don't have instant, panoramic access to every single detail of our lives, including every conversation, thought, and subtle interaction. The life review in NDEs, however, is frequently described as encompassing an astonishing breadth and depth of detail that one might not consciously recall in normal waking life. Many NDErs describe the life review as being "more real than real," more vivid and intense than any waking memory. This suggests a quality of experience that transcends ordinary perception and memory processing. The information isn't just "remembered"; it's re-experienced with an immersive intensity; however, memory (lost and recalled) obviously stays the foundation of this re-experience. If every detail of one's life is indeed accessible in such a review, it implies an incredibly comprehensive and permanent storage of all experiences, thoughts, and emotions, as well as an incredibly high level and activity of a dissociation process.

It also greatly highlights the question about why so much information seems dissociated and inaccessible in our normal waking state, which problem is addressed by various psychological and neurological theories. The key theories and concepts on this include, first of all, the necessity of forgetting (dissociation): our brains are constantly bombarded with sensory input. If we consciously remembered every single detail of every moment, our minds would be overwhelmed and inefficient. Forgetting is not a flaw; it's a crucial adaptive mechanism that allows us to filter out irrelevant information, prioritize what's important, and focus on current tasks and future planning.

A significant portion of our memories is stored unconsciously (implicit memory). This includes procedural memories (how to ride a bike, tie shoes), conditioned responses, and even emotional associations. We "know" these things, but we can't consciously verbalize every step or the exact moment we learned them. Also, some psychological schools convincingly demonstrate that dissociation is a common protective mechanism that acts in response to overwhelming stress or trauma by pushing it out of conscious awareness.

This way, from exploring the phenomenon of forgetting, we come to an understanding that not only is forgetting an extremely common observable event in the mind, but the division of mind itself into conscious and unconscious parts is dissociation by its nature: that includes not only forgetting information that once was consciously available, but also processing information that never even comes into awareness, although obviously presented and processed. Examples of that include motor skills and habits, like walking, typing, or driving a familiar route: once learned, these complex actions can be performed without conscious thought about each individual movement. You don't consciously think "lift left foot, swing right arm" when walking; your mind executes the entire sequence autonomously. Another thing is that your mind is constantly filtering out irrelevant sensory input (e.g., the feeling of your clothes on your skin, background noise) so you can focus on what's important. The filtering itself happens without conscious effort and is an autonomous process.

While understanding spoken language, we effortlessly comprehend sentences and extract meaning, even though the underlying processes of parsing grammar, identifying words, and interpreting nuances are incredibly complex and occur largely unconsciously. Producing speech, meaning formulating sentences and articulating words, generally happens without conscious deliberation over each grammatical rule or muscle movement.

In the face of danger, your body's physiological response (increased heart rate, adrenaline release) is largely automatic and unconscious. Certain stimuli can trigger emotional responses (e.g., fear, joy) before you have consciously processed why you feel that way.

Another thing to mention is bodily functions that are controlled by the autonomic nervous system, which is a part of the nervous system that operates below the level of conscious awareness. Breathing is perhaps the most obvious example. While you can consciously control your breath (hold it, breathe faster or slower), the vast majority of the time, your body breathes for you without any conscious thought. The respiratory control center in your brainstem monitors oxygen and carbon dioxide levels in your blood and adjusts your breathing rate and depth accordingly. Your heart continuously pumps blood throughout your body, day and night, without any conscious instruction from you. The sinoatrial (SA) node, often called the "natural pacemaker," generates electrical impulses that regulate your heart rate, influenced by the autonomic nervous system. From the moment you swallow food, a complex series of processes – peristalsis (muscle contractions that move food through the digestive tract), enzyme secretion, nutrient absorption – all occur automatically and unconsciously. Your body constantly adjusts blood vessel constriction and heart rate to maintain a stable blood pressure, crucial for delivering oxygen and nutrients to all tissues. This is an unconscious, finely tuned process. Also, we can mention body temperature regulation: when you're too hot, you sweat; when you're too cold, you shiver. These are unconscious physiological responses aimed at maintaining your core body temperature within a narrow range.

While these are often categorized as "physiological" rather than "cognitive," the brain (and thus the broader concept of the "mind") is the ultimate control center. The brainstem, hypothalamus, and other subcortical structures are responsible for monitoring internal conditions and sending signals via the autonomic nervous system to maintain homeostasis (internal balance). The amount of analyzed and processed information is enormous.

Here we step from forgetting and division of mind into conscious and unconscious to another extremely widely spread example of dissociation that all of us experience every day of our lives—dreams. In dreams, we experience the creation of entire subjective realities, complete with objects, environments, and even seemingly autonomous characters, all originating from our own minds during sleep. Also, we normally experience a self (first-person character of our dream) perceiving that reality and its events. Both this character and all the objects of the dream are dissociated from their source—a dreamer.

Dreams are inherently personal and unique to the individual. While there might be common dream themes or archetypes across humanity, the specific details, emotions, and narratives are deeply subjective. A sleeping mind being able to construct such complex and immersive worlds highlights its incredible capacity for imagination, memory recall, and synthesis, even in a state of reduced external input. Dreams can range from photorealistic to highly abstract or symbolic. The mind draws upon stored memories, sensory experiences, and even unconscious associations to build these settings. Sometimes these environments are familiar, other times utterly novel. The "first-person character" is often referred to as the dream ego. It's the "you" in the dream, experiencing and reacting to the dream world. While it feels like our waking self, it can sometimes behave differently, possess different abilities, or be subject to dream logic.

While the dream content originates from the dreamer's mind, there's often a disconnect. We don't typically experience ourselves creating the dream in real-time; rather, we experience it as if it's unfolding externally, much like waking reality. This dissociation allows for surprise, fear, joy, and other emotions, as if the events are happening *to* us. The dissociation "from the source" is what allows for the sense of autonomy of dream characters and the perceived independence of dream events. If we were fully conscious of our authorship, the dream experience would be fundamentally different, perhaps more akin to conscious visualization or daydreaming.

In the context of dream dissociation, it is particularly interesting to mention lucid dreaming, where the dreamer becomes aware that they are dreaming and can sometimes exert control over the dream. In these instances, the dissociation is temporarily bridged or overcome.

Why does our mind create these elaborate realities? Theories range from memory consolidation and emotional regulation to problem-solving, creative expression, and even purely epiphenomenal mind/brain activity. Many psychological perspectives, particularly Jungian and Freudian, emphasize the symbolic nature of dreams, suggesting that dream elements represent deeper unconscious thoughts, desires, or conflicts.

While we will touch on dreams in other chapters of this book, for now, it is important to show that dreams are based on the dissociation process.

I want to remind you here that our intention in this chapter was to talk about dissociation and association. We started with dissociation and up to the moment we were talking about cases of forgetting, conscious-unconscious processing, and dreaming—all extremely common, all culturally estimated as "normal." At that point, we are going to step into the area, culturally highlighted as "abnormal" or "disrupting order". This controversial area highlights both an applicability of dissociation almost to anything that happens in an inner life, and the extremely important fact, that inner life in its core is built on association—whatever our everyday mind may be dissociated from, its inner unity is built on association, some aspects of which can be altered by different factors and circumstances, resulting in some degree of inability to effectively act in an everyday life, or in some level of suffering, which turns these alterations into something reasonably estimated as disorder, diagnosis and something that requires treatment.

What we are talking about is called dissociative disorders and drug-induced dissociative experiences. We will stay away from a long and ongoing debate on causes, mechanisms (both psychological and physiological), and treatment, as well as away from diagnostic difficulties and phenomena classification. We will put this phenomenon on stage, having in mind that they are much less widespread than the ones we touched on before: what is interesting in them is their brightness, giving us clear evidence of a power lying at the foundation of the observed processes, making it more difficult to ignore what is happening. And secondly, and I would even say, firstly, it is the light that these phenomena shed on the importance of association as the foundation of the unity of the inner life of a personal self.

Dissociative disorders are defined as mental health conditions that involve experiencing a detachment or disconnection from thoughts, feelings, memories, identity, or sense of self. They often develop as a coping mechanism in response to trauma. The main types of dissociative disorders recognized by the American Psychiatric Association (APA) and common in clinical practice are dissociative identity disorder (DID), dissociative amnesia, and depersonalization/derealization disorder. We also need to mention that, additionally, there are such categories as other specified dissociative disorder (OSDD) and unspecified dissociative disorder (UDD). The first is used when an individual experiences dissociative symptoms that cause significant distress or impairment but do not fully meet the criteria for any of the specific dissociative disorders. The second is used when dissociative symptoms cause significant distress or impairment, but there isn't enough information to make a specific diagnosis.

We will not touch extended categories, just mentioning that they clearly demonstrate the fact of how wide the spectrum of events and their variations is. We will also not touch dissociative amnesia, as we already talked about forgetting in the "normal realm" and can consider an amnesia to be a severe implementation of the same mechanism (turning at that intensity something that is extremely useful and necessary for everyday functioning into something preventing and destroying this functioning). Thus, we are going to explore the remaining: dissociative identity and depersonalization/derealization disorders.

Dissociative identity disorder (DID), also known as multiple personality disorder (MPD), is characterized by the presence of at least two personality states or "alters." Each alter is characterized by a separate autobiographical memory, independent initiative, and a sense of ownership over individual behavior. Alters can vary widely in different other aspects: they may have different names, ages (e.g., a child alter, a teenage alter, an adult alter), genders, mannerisms, voices, likes/dislikes, memories, and even different skills or physical traits (e.g., one alter might be right-handed, another left-handed). The condition involves—for the periods of another alter's activities—total memory gaps, meaning they include gaps in consciousness, basic bodily functions, perception, and all behaviors. These gaps are not just "forgetting where I put my keys." They are forgetting entire conversations, significant periods of time, major life events, or even how they got from one place to another. They might find notes in their own handwriting that they don't recall writing, or discover items they've purchased but have no memory of buying. This amnesia between identities may be asymmetrical; identities may or may not be aware to a different degree of what is known by another. Also, the communication and even conflict between alters may occur: individuals may experience "voices" in their head that are distinct from their own thoughts, representing the different alters talking to each other or to the "host." There can be intense internal debates, arguments, or even commands coming from different parts of the self.

With all these characteristics, dissociative identity disorder challenges and deepens our understanding of what "personality" actually means. It provides a factual base to confront some philosophical and psychological aspects of a view of personality. Typically, we tend to think of personality as a singular, consistent, integrated (associated) whole that defines us across situations and over time. DID suggests that personality, at least under extreme conditions, can be fundamentally non-unitary. It demonstrates that the various elements we associate with personality—memories, emotions, beliefs, preferences, skills, mannerisms, and self-perception—can be compartmentalized and operate somewhat independently.

Normally, our sense of self is tied to a continuous stream of consciousness and memory, a stable "I" that persists through time. From DID's perspective, if one alter has memories and skills that another does not, and experiences itself as a distinct "I," what defines that "I"? Is it the unique set of memories? The particular emotional responses? The specific behavioral patterns? DID shows that these elements can be bundled differently within one person, leading to multiple experiences of "I" or "we."

The common view regarding the role of memories is that they form the foundation of our identity. "I am who I am because of what I've experienced and remembered." The profound amnesia in DID demonstrates a disconnect between memory and conscious awareness within the same individual. An alter might have full memory of an event that the "host" or another alter has no conscious access to. This means that parts of one's "personality" or "self" can exist and function without the conscious awareness of other parts. This raises questions about what constitutes "knowing oneself" when large chunks of one's experiences are not consciously accessible.

Next question: is personality fixed or fluid? In the typical view, while personality can evolve over time, it's generally seen as relatively stable from adolescence onward. However, the rapid and involuntary shifts between alters in DID, with their distinct traits and behaviors, highlight the fluidity of personality. It shows how profoundly personality can change based on internal states and triggers. Even for non-DID individuals, this might prompt reflection on how much our "stable" personality is just a consistent pattern of what our various internal "parts" or "states" choose to express or are allowed to express.

In the context of our research on dissociation, DID is important to show how effective dissociation is in building mind phenomena, and how much it is internally interconnected with association that builds any inner integrity.

The next thing that brights shine depicting the dissociation phenomena is a depersonalization-derealization disorder (DPDR, DDD) in which the person has persistent or recurrent feelings of depersonalization and/or derealization. Depersonalization is described as a feeling of being disconnected or detached from one's own thoughts, feelings, or body, or the experience of "unreality in one's self." Depersonalization experiences may also include feeling like an automaton or robot, feeling emotionally numb, a sense of being an outside observer of one's own thoughts, feelings, body, or actions, loss of a sense of self-identity, or feeling as if one is dreaming or in a movie. Feeling a distorted sense of one's own body may include the whole body or its parts, like limbs that feel alien or hands that look too big or small, and so on.

Derealization is described as detachment from one's surroundings. Individuals experiencing derealization perceive the surrounding world as foggy, dreamlike, surreal, or visually distorted. The experiences may also include objects seeming flat, distant, or artificial, sounds seeming muffled or unusually loud, people seeming unfamiliar or unreal, even close family or friends, a sense that time is speeding up or slowing down, or feeling as if you are in a bubble or behind a glass wall.

A crucial feature of these states is that the person feels these experiences as a distortion, even while feeling them, which differentiates them from psychotic disorders.

With all the listed characteristics, depersonalization-derealization disorder directly challenges our everyday assumption of a cohesive, stable, and embodied self. Depersonalization, particularly, demonstrates how deeply our sense of self is tied to our physical body. When individuals feel detached from their body or perceive it as alien, it shatters the intuitive "mineness" of their physical existence. It suggests that the feeling of being "in" one's body, and having ownership and agency over it, is not a given but a complex construct, potentially vulnerable to disruption. The experience of depersonalization highlights that the mind and body are not separate entities, but deeply integrated. A disruption in the felt connection to the body immediately impacts mental and emotional states, leading to feelings of unreality, numbness, and emotional detachment. A common symptom of depersonalization is emotional numbness or a feeling of being disconnected from one's own emotions. This suggests that a full, vibrant sense of self-integrity relies not just on cognitive awareness but also on the affective experience of emotions. When emotions are "blunted" or felt as alien, the self feels less real, less engaged, and less personally significant. This points to emotions being crucial for establishing a "felt reality" of the self.

Many with depersonalization report feeling like a robot or being on "autopilot," lacking control over their speech or movements. This directly questions our intuitive sense of free will and personal agency. It demonstrates that the feeling of being the author of our own actions is a fundamental component of self-integrity, and its disruption can be deeply disturbing. Some individuals with depersonalization report that their memories lack emotional valence or feel as if they didn't personally experience them. This impacts the continuity of one's personal narrative, which is a cornerstone of identity. If memories don't feel "mine," or lack the emotional "color" that links them to the present self, it weakens the fabric of self-integrity over time.

Derealization, in turn, prompts profound philosophical questions about what constitutes "reality" and how we perceive it. The fact that the external world can feel unreal, dreamlike, or distorted, even when reality testing is intact (meaning the person knows it's not actually unreal), suggests that our experience of reality is not merely a passive reception of external stimuli. Instead, our perception of reality is an active construction by the mind, influenced by emotional states, cognitive processes, and the integration of sensory information. Derealization shows how this construction can break down.

Philosophers have long debated whether there's a distinct "feeling of reality" or if reality is simply what we perceive. DDD offers strong evidence for the former. People with derealization see objects and hear sounds, but they lack that fundamental "sense of realness" or "felt reality" that accompanies typical perception. This suggests that alongside sensory input, there's an underlying affective or qualitative layer that imbues our perceptions with the conviction of being "real." When this layer is missing, reality seems flat or artificial. For many with derealization, the world loses its vibrancy, meaning, or emotional significance. This implies that our experience of reality is deeply intertwined with its value and relevance to us. If objects or people lose their emotional salience, they may also lose their "realness," suggesting that objective existence and subjective reality are not entirely separate.

Concluding the topic of intense dissociative states, we need to mention dissociative agents, a class of psychoactive drugs that induce a sense of detachment from one's body, thoughts, or surroundings. One of them is ketamine, that in low to moderate doses can induce feelings of detachment from self and environment, altered sensory perceptions (sight, sound, time, body image), numbness, disorientation, confusion, dizziness, nausea, and loss of coordination; in high doses can lead to profound sedation, immobility, amnesia, and a complete sensory detachment, often described as a "near-death" or "out-of-body" experience. This can be terrifying or, for some, blissful. Another one is PCP (phencyclidine), known for such effects as euphoria, giddiness, feelings of detachment, distorted perception of reality, numbness, impaired motor function, anxiety, and memory loss; in high doses, it can lead to severe psychological distress, including extreme panic, paranoia, aggression, and violence. And another one to mention is DMT, classified as a classical psychedelic, while still often able to produce dissociative-like effects. DMT experiences are known for their rapid onset (often within seconds when smoked) and short duration (5-30 minutes), characterized by intense, vivid visual and auditory hallucinations, profound changes in the sense of self, ego dissolution, and often a feeling of entering an alternate reality or interacting with "entities." Dissociative agents serve in a way as a possibility to artificially increase the number of observed intensive dissociative states and study them more carefully in a controllable environment.

So here we are finishing our survey of dissociation with the general conclusion about its extreme prevalence in the sphere of the mind. In the area of dissociative disorders, we also already had a chance to see how important the antagonist—association—is in building the integrity of inner life. So our next step will be to touch on the association process. We will purposely do that in a very specific context—the context of altered states of consciousness. We will also do that in a very specific way—by providing multiple examples of patterns in experiences, asking ourselves constantly whether each next pattern example is a case of usually unpresented association or not.

To reveal patterns, we will turn to the factual materials (generalized reports on the content of the ASCs), starting from the contemporary ones. Here, one of the significant sets of collected materials is the data collected by Stanislav Grof and his colleagues. Stanislav Grof is a prominent psychiatrist, author, and researcher known for his pioneering work in transpersonal psychology and the exploration of non-ordinary states of consciousness. From 1961 to 1967, Grof was a principal investigator in a psychedelic research program at the Psychiatric Research Institute in Prague, Czechoslovakia. In this program, he systematically explored the therapeutic and heuristic potential of LSD and other psychedelic substances. He personally conducted over 1,600 LSD sessions and had access to thousands more reports from colleagues. He continued this research after moving to the United States, serving as chief of psychiatric research at the Maryland Psychiatric Research Center and assistant professor of psychiatry at Johns Hopkins University in Baltimore, MD (1967-1973).

In the late 1960s, Stanislav Grof, together with Abraham Maslow and Anthony Sutich, co-founded the field of transpersonal psychology (whose *name* matters a lot in our specific investigation of the association process). Grof's work and books profoundly shaped transpersonal psychology by providing empirical observations from psychedelic states that supported the existence of transpersonal phenomena. After the legal restrictions on psychedelic research in the early 1970s, Grof, with his late wife Christina Grof, developed holotropic breathwork as a non-pharmacological method to access non-ordinary states of consciousness. They extensively taught and conducted professional training programs in holotropic breathwork worldwide, collecting more ASC evidence.

The ASCs' experience content reported in Grof's books [14], [15] fall into three broad, interconnected levels, moving from the more personal to the more universal. First (and usually first accessed in the ASCs' processes) is the biographical psychodynamic level. This level encompasses experiences related to an individual's post-natal life history. It includes vivid reliving of emotionally significant memories, unresolved conflicts, traumatic experiences from childhood, and interactions with significant figures (parents, siblings, etc.). It's the realm where traditional Freudian and other psychodynamic therapies operate. Is this a case of a usually unpresented association? It is, because the direct association with the thoughts, emotions, experiences, and memories, not accessible (dissociated) in a normal state of consciousness, takes place.

Second is the perinatal level. This profound and often intense level involves experiences related to the process of biological birth and the intrauterine period. The level brightly demonstrates the association with factual memories about biological birth from the child's perspective and the intrauterine period from the embryo's perspective, however strange that would sound. Also, it shows an extreme amount of additional information associated with a specific stage of the intrauterine period and birth by similarity of thematic content. For example, the stage of "cosmic engulfment" and "no exit": feelings of crushing pressure, suffocation, claustrophobia, overwhelming anxiety, and despair, related to the onset of labor contractions when the cervix is still closed. The feelings themselves are associated with multiple dynamic representations of the themes of existential crisis, a sense of meaninglessness, or profound paranoia. The perinatal level, thus, demonstrates not only association with the factual, usually non-accessible memories, but also associations of its factual content with the broader non-personal themes, assigned to the level and its stages by similarity.

Third is the transpersonal level, the most expansive one, where consciousness transcends the boundaries of the individual self, time, and space. Experiences on this level include encounters with universal patterns, myths, and deities from various cultures (e.g., Jungian archetypes like the Great Mother, the Wise Old Man, tricksters, heroes), vivid and often emotionally charged experiences that feel like memories from previous incarnations, identification with or direct experience of the consciousness of ancestors, access to the shared experiences and knowledge of humanity, identification (association) with other people, groups of people, animals, plants, or elements of nature, and even inorganic matter or the entire cosmos, feelings of profound unity with the universe, the divine, or all existence, often accompanied by ecstatic joy, boundless love, and a sense of ultimate meaning.

It makes sense to expand this level of description with more detailed examples, although each is not a separate case but rather a pattern with many underlying cases.

So, this is identification with an atom, a molecule, or specific cells, either as individual units or as representing a plurality, various organs, and tissues of the body. This is identification with physical objects and any unities of them (like all the grains of sand on the beach, or all buildings in a city, all tables and spoons and caps of coffee, and chairs and flowers on the veranda of an evening cafe). This is identification with physical or chemical processes, forces, and interactions. This is identification with other individuals, and also identification with a whole group of people, such as all the mothers, soldiers, or Christians in the world, and observation of groups as objects. This is identification with a single wolf or an entire pack, as well as an entire species. This is identification with the consciousness of the ecosystem, with the entirety of Life as a cosmic phenomenon, or with the entire planet. This is identification with stars, star systems, Galaxies, interstellar space, or the entire cosmos.

This is identification with memories (recalling them and feeling as one's own) of other individuals, groups, close and distant ancestors, nations and races, and evolution itself.

And, at the edge, this is a meeting with the Absolute Consciousness and identification with it; and a meeting with the Cosmic Void, Emptiness, Nothingness, Non-Existence which does not contain anything in a concrete, manifest form, but seems to contain within itself all being in its potential form; and a meeting with Absolute Consciousness and Emptiness being identical and equivalent; and a meeting with Absolute Consciousness as identical and equivalent to everything having a concrete, manifest form, the entire Universe, all being.

Is all this a case of a usually not presented association? It is, moreover, the association that takes multiple forms and levels, being the main observed characteristic of this level of experience.

From these contemporary observations, let us take a step towards the past. In a brief overview, we will try to highlight how humans have consistently sought to transcend ordinary consciousness through various means—sometimes internal, sometimes external, and often a blend of both. As previously, we will accompany it with our core question about the results produced by all the practices: "Is each next provided example a case of usually unpresented association or not?"

In pre-1500 BCE, in indigenous, shamanic cultures, a form of meditation that we meet is the ritualistic trances induced by focused attention during ceremonies (e.g., watching a fire, listening to drumming), as well as rudimentary forms of silent contemplation. These "purely" attention-altering practices were extensively accompanied by the use of psychoactive substances, which was widespread and central to many practices. Among such substances, we can mention psilocybin mushrooms, ayahuasca (DMT), peyote/San Pedro (mescaline), cannabis, opium poppy, and various entheogenic plants. To meditation and substances, also "natural" chemical and physiological procedures were added. This was breathwork in the forms of rhythmic chanting, hyperventilation (e.g., Kalahari Kung bushmen's rapid breathing to induce a trance state or ecstasy), and prolonged exhalation (e.g., didgeridoo playing). This was sensory deprivation or overload in the form of spending time in darkness or silence, or conversely, intense drumming, dancing, and rhythmic movement (e.g., ecstatic dance). These also sometimes were physical extremes, such as fasting, sleep deprivation, endurance rituals, or pain induction. All these practices and their different combination resulted in different experiences, including direct communication with spirits, ancestors, or deities; profound empathy with nature and animals; out-of-body experiences, lucid dreaming-like states, travel to other realms for healing, divination, or guidance; ego dissolution and a powerful feeling of interconnectedness with the cosmos, dissolving individual boundaries into a larger whole. Can we call such experiences cases of usually unpresented association? Yes, we can, and we should.

In ancient civilizations and religious foundations, from 1500 BCE - 500 CE (Axial Age religions), meditation practices became highly formalized. In Hinduism, they were dhyana (meditation), pranayama (breath control in Yoga), the Yoga Sutras' eight limbs, all with the focus on internalizing consciousness. In Buddhism, they were samatha (calm abiding), vipassana (insight meditation), and zazen (seated meditation), with the emphasis on mindfulness and direct observation. In Taoism, these were zuowang (sitting forgetting), neidan (internal alchemy), and various Qigong practices with a focus on internal energetic cultivation. In Judaism, Christianity, and Sufism, they were contemplative prayer (e.g., Lectio Divina, Hesychasm, Dhikr), devotional practices involving focused repetition or silent presence. Less universally central than in shamanic contexts, but present in specific traditions, the varying integration of psychoactive substances is presented in this period as well: in Hinduism, this was bhang (cannabis) used by some yogis and in rituals; in some tantric and vajrayana branches of Buddhism, this was limited, guided use of certain substances in advanced or secretive rituals, although usage of such and any substances was generally discouraged in mainstream Buddhism. In Taoism, some alchemical elixirs potentially contained psychoactive elements for longevity or spiritual insights. In Greek mystery religions, this was probable use of kykeon (possibly ergot-derived) in rites like the

Eleusinian mysteries. To meditation and substances, also "natural" chemical and physiological procedures were added, as in the previous era. This was breathwork: pranayama in Yoga (e.g., alternate nostril breathing, rapid breathing for energy), Qigong breathing in Taoism. This was fasting: common across many traditions (Judaism, Christianity, Islam, Buddhism) to purify the body and sharpen the mind. This was sensory modulation in the form of prolonged chanting, rhythmic prayer, specific postures, and sacred dance (e.g., Sufi whirling). These practices and their different combination resulted in the different experiences, including direct, ineffable experience of oneness with God (Christianity, Sufism), Brahman (Hinduism), or the Tao; cessation of suffering, realization of emptiness, and non-self, profound insight into the nature of reality, interconnectedness of all phenomena; states of deep meditative absorption, bliss, heightened awareness, and intuitive knowledge; receiving divine messages or insights; experiences of Prana or Qi flow, vitality, and internal balance. Can we call such experiences cases of usually unpresented association? Yes, for most of them, we can, and we should.

From the medieval to the modern eras, from 500 CE to the present, meditation practices continued to evolve and diversify within established religious traditions. Also, from the 20th century, their secularization and widespread popularization in the West took place. Many secular forms of meditation arose, such as transcendental meditation (TM), mindfulness-based stress reduction (MBSR), secular mindfulness, and various guided meditations. Within these eras, psychoactive substances were initially largely marginalized due to religious and state prohibition. However, in the mid-20th century, within the frame of the "psychedelic movement" (LSD, psilocybin, MDMA), they re-emerged in therapeutic and spiritual contexts (e.g., in psychedelic-assisted therapy, ceremonial ayahuasca and psilocybin use), which is currently (2025) an ongoing process. Natural and body-chemical-based physiological procedures continued their existence and development, resulting in such forms of techniques as holotropic breathwork (developed by Grof as a non-drug alternative to LSD therapy), rebirthing breathwork, Wim Hof method, various modern breathwork modalities, sensory deprivation in isolation tanks, ecstatic dance, 5Rhythms, facilitated movement practices, dietary practices, continued to be used in various spiritual paths and modern wellness. The results of such practices are extremely patterned and have already been discussed by us above.

The important impact of the information age that we need to mention here is that nowadays we have immediate and easy access to reports on all these experiences from different traditions and eras, and the possibility to analyze this data by comparing and finding patterns. Thus, asking our common question on this data: "Can we call such experiences to be cases of usually unpresented association?", we get the general answer that usually unpresented association is the main pattern of ASCs' experiences, by whatever means they were induced.

Here, we arrive at the main point of this current chapter of the book and finally can formulate what we call the "dissociation-association coincidence problem." We demonstrated that association is the basis of personal and any integrity in the sphere of the mind, and that for sanity and feeling of personal self with its boundaries, the dissociation (from everything that is "not me" or everything that is "too much to be at the same time") is the crucial and extremely common process, revealing itself in a bunch of forms and experiences.

We also demonstrated that the opposite process—association with anything and everything that is commonly dissociated (being "separate from me", "external object or information")—is the most common pattern of ASCs' experiences, and a never-ending thirst to achieve these experiences, revealed in many practices throughout humankind's entire recorded history.

Thus, we formulate the problem as a question: "Is it a coincidence that, while dissociation is an extremely common feature of the everyday mind, the most common general pattern of ASCs is association?"

Reverse Correlation Problem of Consciousness

In this chapter, we will review several contemporary studies of the correlation between states of the physical brain and corresponding altered states of consciousness induced by different factors. Our goal is to clarify and clearly see patterns in this type of data collected by different scientists using different methodologies.

We will string one fact upon another, comparing each newly added element with those previously stated.

In the 2015 research [16], Cristofori et al. investigated the causal relationship between specific brain regions and mystical experiences. The authors challenge previous theories and aim to provide empirical evidence using a unique dataset: the Vietnam head injury study (VHIS). They specifically focus on the dorsolateral prefrontal cortex (dlPFC) and the temporal cortex (TC). In the research, mystical experiences are defined as subjective encounters with a supernatural world or being, often characterized by a sense of timelessness, spacelessness, and union with an external entity. They are thought to underpin ordinary religious beliefs. There were two theories preceding the study and to be a subject of proof or disproof. One of them was a Temporal Involvement Hypothesis suggesting that mystical experiences are induced by activity in temporal lobe regions associated with emotion, abstract semantics, and imagery. Evidence includes reports from patients with right temporal lobe epilepsy. Another one was the Executive Inhibition Hypothesis proposing that mystical experiences are linked to a down-regulation of executive functions, particularly those supported by frontal brain regions like the dlPFC. This hypothesis suggests that executive functions normally inhibit or regulate mystical experiences. The study's hypotheses were that damage to the dlPFC would lead to increased mystical experiences due to its role in inhibitory control and its observed deactivation during religious and mystical contexts in previous studies, and that lesions to the temporal cortex (superior and middle temporal gyrus) would be associated with fewer mystical experiences, given the temporal cortex's theorized role in generating these experiences. Key findings include that the pTBI group reported significantly more mystical experiences than the healthy control group; VLSM analysis showed that lesions to frontal and temporal brain regions were linked with greater mystical experiences. Specifically, these regions included the dorsolateral prefrontal cortex (dlPFC) and the middle/superior temporal cortex (TC). Confirmatory analysis showed that the dlPFC group (those with lesions to the dlPFC) presented markedly increased mysticism.

The findings strongly support the Executive Inhibition Hypothesis, suggesting that the dlPFC causally contributes to the down-regulation of mystical experiences. In other words, when the dlPFC is damaged, its inhibitory control over mystical experiences is reduced, leading to an increase in reported mysticism.

Another study was published earlier, in 2010, by Cosimo Urgesi and colleagues [17]. The core premise of the study was to investigate the causative link between specific brain regions and self-transcendence (ST), a stable personality trait that measures a person's predisposition toward spiritual feelings, thoughts, and behaviors.

Previous neuroimaging studies on individuals engaged in meditation (e.g., Catholic nuns, Buddhist monks) have shown that spiritual experiences are associated with neural changes in a large cortical network, including the prefrontal cortex, cingulate cortex, temporal and parietal areas, and subcortical regions. This suggests a broad brain involvement, possibly due to different aspects of spirituality being mapped to different regions, or variations in meditation forms and expertise.

The authors noted that while meditation could induce long-lasting changes in cognitive functioning and neural activity, suggesting that practices can dynamically shape personality traits, information on the neural underpinnings of stable individual differences in spirituality was scarce.

This led to the focus on self-transcendence (ST), a dimension within Cloninger's psychobiological model of personality, which was believed to reflect an enduring tendency to identify the self as an integral part of the universe. ST has been shown to have a heritable component (estimates of 0.37-0.41) and could be related to the functioning of the serotonergic system and genetic variations. Previous correlational studies found associations between ST and metabolic activity or gray matter density in frontal, temporal, and parietal lobes, but these couldn't establish a causative role.

The study itself aimed to address this gap by combining pre- and post-neurosurgery personality assessments with advanced brain-lesion mapping techniques. This approach allows them to explore how specific brain lesions causally affect ST. The hypothesis was that selective damage to frontal and temporoparietal areas would decrease and increase ST, respectively.

The research provided compelling evidence that the inferior parietal cortex (specifically the left inferior parietal lobe and right angular gyrus) plays a crucial, active role in regulating self-transcendence, a personality trait linked to spirituality. Damage to these areas leads to a specific and rapid increase in spiritual feelings and attitudes, suggesting that these regions are involved in anchoring our sense of self within the physical boundaries and, when compromised, can lead to a more expansive, transcendent self-awareness.

If we compare this study (Urgesi et al., 2010, [17]) with the previous one (Cristofori et al., 2015, [16]), we can find that both of them explore the neurobiological underpinnings of spiritual or mystical experiences using a similar, powerful methodology: examining the effects of brain lesions on these phenomena. However, they focus on different brain regions and slightly different aspects of "spiritual" experience, leading to complementary insights.

The findings from Urgesi et al. strongly suggest that the inferior parietal lobe (including the angular gyrus) plays a critical role in how we perceive the boundaries between ourselves and the world. When these regions are damaged, this sense of self-other differentiation may weaken, leading to a more profound feeling of self-transcendence and connection with the universe. This aligns with theories suggesting mystical states involve a breakdown of usual spatial and self-boundaries.

Cristofori et al.'s work complements this by highlighting the dlPFC's role as a "brake" or "regulator." If the parietal cortex contributes to generating an expansive sense of self (or if its damage removes a constraint), the dlPFC might be responsible for inhibiting or filtering these experiences, preventing them from becoming overwhelming or delusional in healthy individuals. The finding regarding the temporal cortex is interesting, as it supports its involvement, but suggests a more complex relationship than simple activation leading to mystical states.

Urgesi et al. point to areas whose damage increases a foundational spiritual trait, while Cristofori et al. suggest that damage to frontal areas unleashes mystical experiences that might otherwise be regulated. Both indicate that these brain regions are not simply "spiritual centers" but rather are involved in fundamental cognitive processes (like self-other discrimination, executive control, and inhibition) that, when altered, can profoundly impact an individual's spiritual and mystical experiences.

Now let us add one more study to the basket. This will be the 2012 Carhart-Harris et al.'s research titled "Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin" [18]. In it, the authors investigate the neural correlates of the psychedelic state induced by psilocybin (a classic psychedelic found in "magic mushrooms") in healthy human volunteers. The researchers aimed to understand how psilocybin affects the brain, especially given its historical use in healing ceremonies and renewed therapeutic interest.

The study yielded some surprising and counterintuitive findings regarding brain activity during the psilocybin-induced psychedelic state. Among them is again the decreased brain activity: despite the profound and often intense changes in consciousness experienced during a psychedelic state, the researchers observed only decreases in cerebral blood flow and BOLD signal. This contradicts the intuitive expectation that a heightened state of consciousness might be associated with increased brain activity. The research's significant finding was the decreased activity in the anterior cingulate cortex (ACC)/medial prefrontal cortex (mPFC). Furthermore, the magnitude of this decrease strongly predicted the intensity of subjective psychedelic effects. This suggests that the mPFC's reduced activity is directly linked to the profound changes in perception, thought, and self-experience.

Psilocybin also significantly reduced the positive coupling (connectivity) between the mPFC and posterior cingulate cortex (PCC), two major hubs of the default mode network (DMN). The DMN is often associated with internally directed thought, self-reflection, and a sense of ego.

The researchers interpreted these results to suggest that the subjective effects of psychedelic drugs, like psilocybin, are linked to decreased activity and connectivity in the brain's central connector hubs. This deactivation and decoupling of critical brain networks, particularly within the DMN (mPFC and PCC), may lead to a state of "unconstrained cognition." This "unconstrained" state could explain the characteristic features of the psychedelic experience, such as altered self-perception, novel insights, and a dissolution of ego boundaries, by reducing the normal filtering and organizational functions of these key brain regions.

Despite different methodologies (lesion studies vs. fMRI) and specific substances/conditions investigated, this and the previous two studies, all three delving into the neural underpinnings of non-ordinary or altered states of consciousness, all three find a paradoxical mechanism: decreased or disrupted brain activity in specific regions leads to heightened experience. Rather than these "expanded" states arising from increased brain activity, all three studies suggest they result from a reduction or disruption of normal inhibitory or filtering brain functions.

The consistent pattern suggests that these non-ordinary experiences may not be about "activating" a "spiritual center," but rather about releasing the brain from its usual constraints, filters, and self-referential loops.

Let us continue the list with the study of 2015 of effects of ayahuasca (psychoactive brew used by indigenous cultures in the Amazon and Orinoco basins) on the human brain [19]. The research revealed significant modulations of the default mode network (DMN), which under ayahuasca displayed decreased activity in most parts of it. Beyond just activity, the study also found a decrease in functional connectivity within the PCC/Precuneus region itself. This suggests that the coordination and communication within these key DMN areas are altered.

The findings of this study support the growing body of evidence that altered states of consciousness are linked to the modulation of DMN activity and connectivity. The reduction in DMN activity and connectivity induced by ayahuasca is consistent with observations from other altered states, such as those induced by psilocybin (as seen in the previously discussed Carhart-Harris et al. article), meditation, and sleep.

This suggests a common neural mechanism whereby the disruption of the highly organized and internally focused activity of the DMN contributes to the profound shifts in perception, self-awareness, and cognitive processing characteristic of psychedelic experiences. By "quieting" or decoupling the DMN, ayahuasca may facilitate a less constrained mode of cognition and experience.

If we compare all four articles we already mentioned, we'll find that all of them provide a compelling and counterintuitive commonality: instead of "more spiritual experience = more brain activity," the prevailing theme is that the reduction, disinhibition, or disruption of activity in specific, high-level brain networks facilitates these states. This collective evidence strongly suggests that these experiences emerge from a "release phenomenon" – a loosening of the brain's usual organizational, inhibitory, or filtering functions, rather than the activation of specific "spiritual centers." Directly or indirectly, all articles touch upon how these brain mechanisms affect the sense of self and its boundaries with the external world: mystical experiences involving encounters with a "supernatural world"; self-transcendence (Urgesi) involving identifying the self as part of the "universe as a whole" and transcending bodily contingencies; psychedelic states often characterized by "ego dissolution," "unconstrained cognition," and altered self-perception, all linked to DMN modulation which is heavily involved in self-referential processing.

Having different brain change triggers as a basis, all the research produces similar brain states and conclusions. Summarizing, the collective body of work suggests that our ordinary sense of self and reality is maintained by a set of highly integrated and often inhibitory brain networks. When these networks, particularly in the frontal and parietal regions (including the DMN), are disrupted, or their activity is reduced, the brain enters a state of "disinhibition" or "unconstrained cognition," leading to the profound and often spiritual/mystical experiences reported across cultures and contexts.

Will not stop here and provide more evidence to the context of the "reverse correlation" problem. The next research we will focus on is of 2012 and is dedicated to investigating spiritual experiences that involve dissociative states, specifically focusing on psychography

[20]. In the context of the research, by "dissociation" the authors mean a mental process that causes a lack of connection in a person's thoughts, memory, feelings, actions, or sense of identity; the psychography is referred to as automatic writing, a specific type of dissociative mediumship where the medium claims that a spirit directly guides their hand to produce written text, drawing, or symbols, often without the conscious awareness or control of the medium themselves. While neither "mediumship" nor "spirits" are necessary interpretations, the focus of the research is on unusual states of consciousness, their effects, and underlying brain activities.

The primary conclusion is that experienced psychographers exhibited lower levels of activity (rCBF) in several specific brain regions during psychography compared to their normal, non-trance writing. Also, complexity scores for the psychographed content were higher than for their control (normal) writing. This suggests that the production of more complex material is occurring simultaneously with reduced brain activation.

The research shares significant commonalities with the previously analyzed articles, reinforcing several key trends, such as focus on the altered states of consciousness and self-perception and decreased/disrupted activity leading to heightened or efficient experience. The article identifies changes in regions like the left anterior cingulate and hippocampus, as well as parts of the cerebellum and temporal/precentral gyri. The anterior cingulate cortex (ACC) is a crucial part of broader executive control networks and connects with the DMN, placing it squarely within the high-level networks consistently implicated across all five studies.

In the next research, one of 2017 [21], its authors tried to reconcile previous conflicting neuroimaging findings regarding the impact of psilocybin (a 5-HT-agonist) on cerebral blood flow (brain perfusion). Earlier studies had suggested increased frontal glucose metabolism (hyperfrontality), while more recent ones had indicated decreased perfusion in various brain regions. This study sought to clarify these discrepancies by examining both relative and global cerebral blood flow changes in a dose-dependent manner.

The study yielded nuanced findings regarding psilocybin's effects on brain perfusion. When adjusting for overall (global) brain perfusion, psilocybin showed increased relative perfusion in specific right hemispheric frontal and temporal regions, and bilaterally in the anterior insula, and decreased relative perfusion in left hemispheric parietal and temporal cortices and left subcortical regions. Crucially, when examining absolute perfusion (overall blood flow), psilocybin significantly reduced absolute perfusion across various brain lobes and subcortical structures.

The study successfully addresses the previous discrepancies in the literature by demonstrating that psilocybin's effects on cerebral blood flow depend significantly on the analysis method used (relative vs. absolute perfusion). The study concludes that psilocybin, a 5-HT agonist, reduces overall absolute cerebral blood flow, but this reduction is not uniform. The brain redistributes blood flow, leading to relative increases in certain regions (like parts of the frontal lobe and insula) while other areas experience relative decreases. This highlights the importance of interpreting neuroimaging data carefully, differentiating between local (relative) and global (absolute) perfusion changes to fully understand the drug's mechanism of action.

We need to explain why this last result is important and what it adds to all the previously mentioned findings. Before we mentioned it, we had the "less is more" hypothesis largely based on findings like those from Carhart-Harris et al. (2012) and Palhano-Fontes et al. (2015), which showed decreased activity and connectivity within the Default Mode Network (DMN) under psilocybin and ayahuasca, respectively. This was interpreted as a "loosening" or "disinhibition" of the brain's usual self-referential and inhibitory functions, leading to the profound, unconstrained experiences. However, there was a historical conflict in the literature: earlier studies often suggested increases in glucose metabolism (a proxy for brain activity) in certain frontal regions (sometimes termed "hyperfrontality") during psychedelic states. This created a puzzle: how could both increased and decreased activity be true?

The research we are talking about used a technique that directly measured cerebral blood flow (perfusion). Their key insight came from analyzing the data in two ways: absolute perfusion and relative perfusion. The first way measures the total amount of blood flowing to a brain region. Lewis et al. found that psilocybin caused a significant reduction in absolute perfusion across various brain lobes and subcortical structures. This means that, overall, the brain is indeed less metabolically active in the psychedelic state. This strongly supports the "less is more" idea on a global scale – the brain isn't necessarily working harder, but rather differently, with less overall energy expenditure. The second way measures the blood flow in one region relative to the overall (global) blood flow, and this approach showed a relative increase in the blood flow in some regions.

The key point here is that the "increased activity" in Lewis et al. is an increase in relative perfusion (blood flow), which occurs within the context of an overall, global decrease in absolute perfusion. Imagine a company (your brain) with a total annual budget (absolute cerebral blood flow/energy) of $1,000,000. In your normal sober state, this $1,000,000 budget is distributed among various departments. Let's say the "Self-Regulation Department" (like the DMN) gets a large chunk, say $300,000, and the "Creative Idea Generation Department" gets $100,000. Under psilocybin, the company's total annual budget (absolute cerebral blood flow) shrinks significantly, say to $600,000 (a 40% reduction globally). Now, the "Self-Regulation Department" (DMN), which was getting $300,000, might now only get $100,000. Its absolute budget decreased dramatically. This is why the DMN appears deactivated (Carhart-Harris, Palhano-Fontes) because it's receiving much less. The "Creative Idea Generation Department," which was getting $100,000, might now get $80,000. Its absolute budget also went down. However, if you look at the relative share of the new, smaller budget, the "Creative Idea Generation Department" is now getting $80,000 out of $600,000 (about 13.3%), whereas before it was $100,000 out of $1,000,000 (10%). Relatively, its share increased, even though its absolute funding went down.

This provides a deeper, more physiologically precise explanation of how the brain achieves the "less is more" state. It shows that a global metabolic downregulation is accompanied by a redistribution of remaining resources, creating a unique functional landscape responsible for the psychedelic experience.

Let us add some more elements to our list of evidence. The next research we will explore is the neuroimaging study conducted by Carhart-Harris et al. [22] aimed to comprehensively investigate the effects of Lysergic acid diethylamide (LSD) on the human brain and correlate these neural changes with the drug's characteristic psychological effects. The research sought to provide new insights into the hallucinatory and consciousness-altering properties of psychedelics. This study provides a rich and comprehensive neurobiological explanation for the LSD experience, linking specific brain changes to its complex psychological effects. The findings on visual processing suggest that LSD leads to disinhibition and hyper-connectivity within the visual system, explaining the vivid and pervasive visual hallucinations. The findings on reduced parahippocampal-RSC connectivity further support the idea that psychedelics achieve effects like ego-dissolution by disrupting key networks involved in self-processing and the construction of meaning. This aligns with the broader "Default Mode Network deactivation" hypothesis seen with other psychedelics.

We now have seven articles to compare, and the latest one on LSD's neural correlates significantly expands our understanding of the psychedelic experience and fits well within the emerging patterns. The most consistent and counterintuitive trend of all the listed research is that reduced overall brain activity and disrupted inhibition lead to heightened experience; the "less is more" hypothesis continues to be supported and refined.

However, it is worth mentioning that in the light of the data the "less is more" concept is no longer a simple deactivation, but a sophisticated model where a global reduction in brain energy expenditure is accompanied by a reorganization of information flow, leading to disinhibition of certain systems (e.g., visual cortex) and suppression of others (e.g., DMN), which collectively allows for the emergence of novel, unconstrained, and subjectively profound experiences.

Another study we will mention aimed to investigate the neurobiological mechanisms that underlie the profound changes in consciousness induced by psychedelic drugs, specifically focusing on psilocybin [23]. It sought to characterize how psilocybin affects spontaneous cortical oscillatory power in the human brain. The study proposes that the core neurobiological mechanism underlying the subjective effects of psychedelics is a broadband cortical desynchronization. This desynchronization, likely mediated by the activation of 5-HT2A receptors on deep pyramidal cells, leads to a state where the brain's normal rhythmic organization is disrupted.

This disruption (or "desynchronization") is hypothesized to be the neural basis for the profound changes in consciousness experienced during the psychedelic state, including altered perception, thought patterns, and the sense of self. It further supports the idea that psychedelics "release" the brain from its normal constraints by reducing its synchronous, organized activity.

When added to what we analyzed previously, this research solidifies the idea that the psychedelic brain is characterized by a "disordered" yet profoundly experiential state, where the usual highly synchronous and constrained brain rhythms give way to a more desynchronized and unconstrained mode of processing.

If we imagine the brain's activity like a large orchestra, in a normal conscious state (sober brain), this orchestra is highly organized. Different sections (brain regions/networks) play their parts in synchrony (brain oscillations). The conductor (perhaps a metaphor for the DMN's organizing role) keeps everyone in time. This synchronized, organized playing creates a rich, coherent, and predictable sound. This organized activity requires energy and leads to predictable patterns.

In a psychedelic state, "decreased activity" (as seen in DMN fMRI and global CBF reduction) can be seen as volume going down significantly. They are playing more quietly, or fewer instruments are playing at full force. This is like the global absolute cerebral blood flow reduction in Lewis et al., or the general DMN deactivation in Carhart-Harris (2012) and Palhano-Fontes. The brain is spending less energy on its usual, organized performance.

But further, it is also "desynchronized": now, within that quieter orchestra, the musicians start playing out of sync with each other. The violins are off from the cellos, and the brass section is out of time with the woodwinds.

When instruments play out of sync, their individual sounds might still exist, but their collective, organized signal (the "oscillatory power" or "rhythm") becomes weaker and less coherent. The "synchronous peaks" of their waves no longer align, effectively reducing the measurable strength of those brain waves. This reduction in synchrony is a form of reduced ordered activity. It means the precise, coordinated firing that underlies efficient neural communication and integration is diminished.

Thus, the research provides an electrophysiological explanation of how the brain shifts to a "less ordered," "less inhibited," and ultimately "less energetically consuming" state, which in turn gives rise to the profound and expanded conscious experiences associated with psychedelics.

The next piece of research we'll mention is a quantitative investigation of unconsciousness caused by increased head-to-foot (+Gz) acceleration stress in healthy human subjects [24]. It aimed to measure the kinetics of unconsciousness and develop a theoretical framework to describe alterations in the central nervous system (CNS) due to reduced cerebral blood flow.

The study provides a quantitative kinetic description of acceleration-induced unconsciousness in healthy humans, thoroughly documented over an 11-year period. It highlights that this type of unconsciousness is directly due to reduced cerebral blood flow (ischemia), leading to temporary CNS alterations. At the same time, it states the occurrence of "memorable dreams" during a period of very low cerebral blood flow.

So, we have now analyzed nine relatively modern studies directly or indirectly investigating non-ordinary states of consciousness, ranging from mystical/spiritual experiences, psychedelic states, and dissociative trances to the very edge of consciousness in acceleration-induced cerebral ischemia. We can clearly see that the "Less is More" principle, suggesting that profound conscious experiences (or alterations in consciousness) are associated with reduced, disrupted, or reorganized brain activity, is the most consistent and powerful finding across all these studies.

The last step we need to make before going to conclusions is a step to the "edge" of "less" on the scale of "less and more"–the cases of full cessation of blood flow to the brain and full cessation of brain electrical activity, which together form the physiological reality during **cardiac arrest**.

During cardiac arrest, the heart stops pumping blood. This means that within seconds, blood flow to the brain essentially ceases. As the brain is highly dependent on a constant supply of oxygen and glucose from the blood, within 10-20 seconds of global cerebral ischemia (a lack of blood flow to the entire brain), consciousness is lost. Crucially, within 20-30 seconds, the brain's electrical activity, as measured by a standard electroencephalogram (EEG), typically becomes isoelectric or "flatline." This means there are no discernible brain waves, which conventionally indicates a severe suppression of cortical neural activity, very close to what is medically defined as brain death if irreversible.

Despite this physiological shutdown, a significant percentage of individuals who are successfully resuscitated from cardiac arrest (estimates vary widely, from 10-20% in some large studies) report profound, vivid, and highly organized conscious experiences known as NDEs (near-death experiences). These experiences often include out-of-body experiences (OBEs)–a sensation of floating above one's body, sometimes observing the resuscitation efforts. Experiencers report feelings of peace and well-being, often described as overwhelming joy, love, and absence of pain or fear; encounters with deceased loved ones or spiritual beings; seeing a bright light or entering a tunnel; and a sense of knowing or ultimate truth. Another common and profound element is a life review–a rapid, panoramic review of one's entire life, often from an objective, detached perspective. Also, what is often mentioned is an altered perception of time–time often feels meaningless or expanded. In rare but compelling cases, individuals report seeing or hearing things that were later verified to have objectively occurred during their unconscious period (e.g., details of medical procedures or conversations in another room).

The key here is that these experiences are often described as dramatically heightened, more real than ordinary reality, clear, lucid, and highly organized, despite the severe physiological compromise of the brain. The occurrence of NDEs during periods of apparent (or actual, at the cortical level) brain inactivity presents a major challenge to strictly materialistic views of consciousness. How can complex, lucid consciousness arise when the brain, as we understand it, is not functioning?

Various scientific hypotheses have been proposed to explain NDEs, attempting to reconcile them with brain function. They include hypoxia/anoxia, in which the lack of oxygen to the brain triggers a chaotic electrical discharge or neurochemical cascade that produces the experience. However, anoxia usually leads to confusion, delirium, or unconsciousness, not typically organized, lucid, and often positive experiences with narrative structure and veridical elements. Counter-arguments argue that perhaps the onset or offset of anoxia, rather than sustained anoxia, could trigger these specific effects. Another explanatory element is the potential release of endorphins, which suggests that the brain releases endorphins in response to extreme stress, leading to feelings of euphoria and pain relief. The challenge for this view is that while endorphins can produce pleasure, they don't explain the complex narrative, OBEs, or veridical perceptions. Also, the neurochemical surge is proposed as an explanation: a surge of specific neurochemicals (e.g., serotonin, DMT in some theories) occurs as the brain dies, leading to hallucinations, but, similar to anoxia, a chaotic chemical release might not account for the organized, consistent, and often positive nature of NDEs. The timing of such a surge relative to the EEG flatline is also a matter of debate. Another view is that while scalp EEG shows a flatline (cortical inactivity), deeper brain structures (subcortical areas) might retain some minimal activity that supports a rudimentary form of consciousness, which is then interpreted by the cortex upon recovery. However, it's difficult to experimentally verify this in humans during cardiac arrest, and the complexity of NDEs seems to require more than just rudimentary activity.

Recent researchers like Dr. Sam Parnia and colleagues (e.g., AWARE study) are conducting studies specifically monitoring brain activity (using more sensitive EEG and brain oxygenation monitors) during actual cardiac arrest and resuscitation attempts. While their findings are still evolving, they are pushing the boundaries of what can be measured at the threshold of death. They have shown some evidence of brain activity patterns (gamma waves, delta waves) emerging during resuscitation, and in some rare cases, even brief, low-level activity after the heart has stopped but before complete EEG flattening, or during the very early stages of recovery. However, definitive evidence of complex, lucid consciousness during a verified flatline EEG remains elusive and is the subject of ongoing intense debate and research.

In summary, the precise mechanism by which NDEs occur during periods of apparent or confirmed brain inactivity (like EEG flatline in cardiac arrest) is one of the most profound unanswered questions in neuroscience and the study of consciousness. It directly challenges our assumptions about the relationship between the brain and consciousness and continues to be an active area of interdisciplinary investigation.

So, if partial reduction or reorganization of brain activity correlates with "richer" experiences, could a near-total cessation of conventional brain activity, as seen in cardiac arrest, be the ultimate expression of this "less is more" principle, correlating with peak experiences like NDEs? Our analysis of previous research consistently points to a core mechanism: mystical, spiritual, and psychedelic experiences are correlated with a reduction in overall brain activity, a deactivation or decoupling of key brain networks (especially the Default Mode Network), or a loosening of inhibitory controls from areas such as the prefrontal cortex.

When a person undergoes cardiac arrest, blood flow to the brain ceases almost immediately, followed by the cessation of measurable electrical activity (EEG flatline) within seconds. This is the closest we get to a "global shutdown" of conventional brain function that is still reversible (i.e., the person can be resuscitated). From the "less is more" perspective, if partial reduction of brain filtering leads to profoundly altered states, it is a compelling, albeit speculative, logical correlation to propose that a near-total cessation of filtering and brain activity (in a state reversible by resuscitation) could lead to the peak or ultimate "unfiltered" experience – precisely what is reported in NDEs. The NDEs are often described as far more vivid, real, and profound than any psychedelic or mystical experience induced by other means.

The topic of NDEs is also something I would describe as a bloody battlefield of different worldviews. On that battlefield, unimaginable mind tricks are applied to debunk the significance of NDEs, much like in the situation with the hard problem of consciousness. And as we did before with the hard problem of consciousness, to check their validity, we will consider common "scientific" arguments (we already briefly mentioned them above) used to escape the problem of NDE presence. However, in this case, we will combine these arguments into specific sets to address them more clearly. As a preface, we will provide a detailed description of what is happening in the brain during cardiac arrest.

Studies in humans and animals have shown [29] that during induced cardiac arrest, the loss of function of both the cerebral cortex and brainstem leads to unconsciousness within seconds. All brainstem reflexes also disappear: there is no corneal reflex (blinking when the eye is touched), no gag reflex, and dilated pupils do not react to light. The respiratory center near the brainstem also ceases to function, as evidenced by respiratory arrest (apnea).

When the lack of blood flow to the brain prevents glucose and oxygen from entering, the first symptoms in neurons are an inability to maintain membrane potential, leading to the loss of neuronal function. The critical loss of electrical and synaptic activity in neurons can be viewed as a built-in protective mechanism and energy-saving response (emergency mode) within the cell. When these functions cease, the remaining energy reserves can be used only for a very short time to ensure the cell's survival. In the case of short-term oxygen starvation, dysfunction may be temporary, and recovery is possible, as neurons remain viable for several minutes.

Some studies involving humans also used electroencephalograms (EEGs) to record the electrical activity of the cerebral cortex, and in animals, the electrical activity of deep brain structures was measured. The results showed that after a very short time, the electrical activity of the cerebral cortex and deep structures completely disappears. The first symptoms of oxygen starvation were recorded, on average, 6.5 seconds after the onset of cardiac arrest. If the heart rhythm is not restored immediately, a complete loss of all electrical activity in the cerebral cortex always results in a flat EEG line after 10–20 (15 on average) seconds. When testing animals, the potential from the auditory stimulus or vital signs of the brainstem could no longer be elicited, indicating the absence of the response elicited by sound stimulation in a normally functioning brain.

Many argue that the loss of blood flow and a flat EEG line do not rule out activity somewhere within the brain, since the EEG primarily records electrical activity in the cerebral cortex. However, the issue is not whether there is some unmeasured activity somewhere in the brain, but whether there are any signs of the specific patterns of brain activity that, according to modern neuroscience, are considered necessary for consciousness. And all signs of any specific patterns of brain activity are absent from the EEGs of patients with cardiac arrest. A flat EEG line is also considered one of the primary tools in diagnosing brain death, and in such cases, the objection that activity may still be present somewhere in the brain is never raised.

This complete shutdown discredits the first set of arguments–**physiological reasons**: oxygen starvation, carbon dioxide overload, chemical reactions, abnormal electrical activity in the brain, or drug-induced delirium. None of that matters, as the brain just does not have activities correlated with conscious experience at the time to which the profound experiences are reportedly linked.

Therefore, the only suggested solution to this contradiction is that experience is generated by the mind later, when the brain resumes functioning, and memories are retrospectively attached to the time of non-functioning due to one or several **psychological reasons**: fear of death and nonexistence, personal beliefs with related expectations, and the thirst for confirmation.

This explanation is logically undefeatable, as we do know the mind is capable of doing such things, and psychological motivation sounds powerful and completely reasonable. However, it breaks down if we do not take NDE as a separate phenomenon, but instead take it as an integral part of the continuum: this entire chapter shows this continuum of more or less disrupted brain activities–when we add to this a phenomenological bridge (the content of ASC with disrupted or lowered brain activities has extreme commonality with NDE content), we understand that NDE-like experiences arise against the background of reduced brain activity without **psychological reasons** mentioned before.

Thus, we summarize our observations as a standing and alive problem for the materialistic paradigm and a question: "How is it possible that reduced brain activity leads to increased and richer, highly organized and structured experience, and completely ceased brain activity leads to the most profound degree of that experience?"

ASC Content Problem

The study of consciousness—the subjective experience of being—represents one of the final frontiers of human inquiry. While it remains a concept notoriously difficult to define, the investigation into Altered States of Consciousness (ASC) offers a critical, empirical window into its fundamental nature. For millennia, humans have pursued these non-ordinary states, whether through ritualistic practice, meditative discipline, pharmacological induction, or the simple act of dreaming.

This chapter presents a rigorous analysis of data drawn from a vast body of contemporary research. It focuses on the intractability of certain collected information when viewed through a purely reductionist lens, as presented phenomenological results appear stubbornly resistant to explanation by current physical models. Ultimately, we aim to demonstrate that certain aspects of the content of different ASCs (each individually and especially in any combination) pose a specific and unresolved problem for the materialistic worldview.

As we mentioned in the chapter about dissociation and association, one of the major patterns of ASC is an experience of going beyond the boundaries of the individual self and a direct experience of association with the elements of the usually "external" world. This includes association with both "big" and "small" to the extreme scale in both directions. Also, the united entities the reporters often talk about are grouped elements (material objects in different combinations, people and other species, etc.).

Now we will revisit these examples, devoting much more attention to the phenomenology of these experiences and their detailed content. I would like to immediately highlight the reason why going into details is important: the common materialistic explanation of the ASC content is that it represents re-formatted and subconsciously processed information obtained by an individual during their lifetime (again, consciously and subconsciously, as "the brain records everything, even if you are not aware of it") As the additional cover for the concept, its proponents use the fact that experience obviously include personal, individual content, mixed up with the rest. This cover itself can be removed immediately, as the presence of individual content (which is undeniable) does not in any way cancel out the content that, in normal circumstances (when not taken from the ASC reports, but from external world phenomenology and information carriers), is interpreted as objective. Without the cover, the idea of previously subconsciously obtained and processed information by itself looks not very convincing, and when going into the **details** of the ASC content, it falls apart completely, leaving the materialistic paradigm with another unsolved problem.

So that is why we will go into the details. As we mentioned, ASCs' experience content reported by transpersonal psychology, specifically in Stanislav Grof's books [14], [15] fall into three levels: the biographical psychodynamic level, perinatal level (related to the process of biological birth and the intrauterine period), and the transpersonal level, the most expansive one, where consciousness transcends the boundaries of the individual self, time, and space. Once again, we must emphasize that these levels do not exist in actual experience as distinct stages or separate spaces; on the contrary, the content from all levels is interpenetrating, dynamically interacting, and holographically unified.

As we have already mentioned, the presence of individual content, memories, fantasies, fears, traumas, and experiences that belong to the biographical psychodynamic level is undeniable and evident. We won't delve into that too much. Two other levels will get our full attention.

The perinatal level encompasses experiences related to the biological birth process and the intrauterine period. Strictly speaking, we could include it in the set of experiences that are individual and obtained subconsciously; however, the problem is that the brain a "recording device," is present "to varying degrees" at different stages of this experience, to the point that at the earlier stages it is "more or less present" or not present at all, while reported experiences remain detailed and vivid.

What detailed data does this level provide? This is information on organism development from the moment of conception, including detailed descriptions of intra- and intercellular processes, including biochemistry, DNA processing, formation, and development of tissues and organs. All that is verifiable information that can be compared with the current scientific knowledge of the sphere. Another set of data is the relationship with the mother's organism, deep knowledge of the interaction process, and any issues and states (both positive and negative) of the mother's biochemistry, psychology, and emotions.

Once again, this "level" of inner life is not an isolated "shelf" with factual content: like all other "levels", it represents interpenetrating, dynamically interacting, and holographically unified information from all around the spectrum of experiences and inner content, in which, however, the factual and precise information is still presented.

The most interesting level for formulating this chapter's "ASC content problem" is the transpersonal level, the most expansive one, where consciousness transcends the boundaries of the individual self and the usual object-subject distinction.

Experiences at this level include encounters with myths and deities from various cultures, often those that are very distant from the reporter's own culture, both geographically and in terms of time. One interesting and demonstrative example of that was provided by Stanislav Grof in his book "When the Impossible Happens" [25], called "the story of Otto." Otto was one of Grof's Prague clients, suffering from depression and a pathological fear of death (thanatophobia). During one of his psychedelic sessions, Otto experienced a vivid psychospiritual death and subsequent rebirth. At the climax of the experience, he saw an ominous gate to the underworld, guarded by a terrifying goddess in the form of a pig. At that moment, Otto felt an urgent need to draw a specific geometric pattern. He would persistently and rapidly draw an entire series of complex abstract designs. But as soon as each drawing was completed, Otto would tear and crumple these complex patterns, clearly dissatisfied with the result and increasingly frustrated. Clearly disappointed with his drawings, he became increasingly frustrated with his inability to "do everything right." At that time, Grof was unable to put any basis under this experience, neither from Otto's personal biography, nor from any other sphere, and just put the explanation of the episode aside.

Much later, Grof described this episode to his friend and world-famous specialist in comparative mythology, Joseph Campbell, whom many considered the greatest expert of the 20th century on mythology. In most cases, Joseph had no difficulty identifying various esoteric mythological themes and symbols that Grof could neither understand nor explain. "How interesting," Joe immediately responded. "That's clearly the Celestial Night Goddess of Death, the Devouring Mother Goddess of the Malekulan of New Guinea."

He then explained that this goddess appears as a terrifying female figure with distinctly pig-like features. According to Malekulan beliefs, this goddess sits at the entrance to the underworld and guards a complex, intricate pattern. The natives believe they will encounter this goddess during the "Death Journey." The Malekulan have an elaborate system of rituals surrounding the breeding and sacrifice of pigs. This complex ritual activity aims to overcome dependence on their human mothers and, ultimately, on the Devouring Mother Goddess. The Malekulan spend a significant portion of their lives practicing labyrinth drawing, as this skill is vital for a successful journey to the world of the dead. The goddess will not grant permission to travel beyond the edge to a person who is not able to reproduce the required labyrinth accurately.

Personally, it took me effort not only to learn any facts about the Malekulan people but even to locate them on the world map after I read some information about them. The same applies to Grof's patient Otto, which makes his story so demonstrative, specifically considering Otto's behavior with drawing geometric patterns.

Let us proceed further and consider other types of experiences in ASC, which include extremely detailed and precise information, typically experienced and described in the first person, about past events. This includes information about household items and details and data about ongoing events presented in a very unusual (for textbooks) manner, with a high level of detail and a lack of general perspective. This happens when we encounter events that typically become part of textbooks ("big", "historical" ones). Even more problematic are the personal, individual events and circumstances in cases where existing materials or remaining witnesses allow for verification of these details.

Here, I would like to provide another example from Stanislav Grof's book [25]. It is the story of Charles, who participated in the monthly workshops and holotropic breathwork sessions held at Esalen by Grof and his wife, Christina. In the process, Charles experienced various aspects of his biological birth, accompanied by fragments of dramatic scenes that seemed to belong to another time and another country. They came along with the powerful emotions and physical sensations and had a deep and intimate connection with his life; yet none of these episodes seemed to make the slightest sense in relation to the events of his current life. He saw tunnels, underground warehouses, barracks, thick walls, and ramparts that seemed to be part of a fortress built on a cliff overlooking the ocean. These visions were interspersed with images of soldiers in various situations. Charles was completely confused, as the soldiers were Spanish, and the setting was more reminiscent of Scotland or Ireland.

As the process progressed, the scenes became increasingly dramatic, and Charles became increasingly involved. Many scenes were set against the backdrop of brutal battle and bloody carnage. Surrounded by soldiers, Charles felt like a priest; at one point, a very moving vision included a Bible and a cross. At that moment, he saw a signet ring on his finger and could clearly discern the initials engraved on it.

A talented artist, he decided to document this process, although at the time he didn't understand what was happening. He produced a whole series of graphic works, and very powerful and emotionally charged finger drawings. Some depicted parts of the fortress, others scenes of the massacre, and several depicted Charles's own experiences, including the scene where he is pierced by an English soldier's sword, then thrown from the ramparts to die on the shore.

As he pieced together the pieces of this story, Charles found more and more meaningful connections between various aspects of it and his current life. He began to suspect that the drama of the Spanish priest, which had occurred in the distant past, could be the source of many of his emotional and psychosomatic symptoms, as well as problems in communicating with others. A turning point came when, on an impulse, he decided to vacation in Ireland. Upon returning, while reviewing slides he had taken on the west coast of Ireland, he discovered twelve consecutive photographs of the same landscape. Charles was astonished, as he had no memory of taking these pictures, and the view captured in the slides was nothing special.

Being a pragmatic person, he took a very rational and analytical approach to this strange situation. He looked at the map and reconstructed the origin and direction of his photographs. Karl discovered that the place that had caught his attention was the ruins of an old fortress called "Dún an Óir" or Forte de Oro ("Golden Fortress"). From where he was photographing, the remains of the fortress were barely visible to the naked eye, and Charles had to peer closely for a long time to find them on the slide. Suspecting a connection between this strange behavior and his experiences during primal psychotherapy and holotropic breathwork sessions, Karl decided to study the history of Dún an Óir, trying to find some explanation.

To his surprise, he discovered that in 1580, a small Spanish expeditionary force landed in Ireland, near Smerwick Bay, to assist the Irish rebellion led by the Earl of Desmond. After the Irish joined them, the detachment's strength grew to 600 men. They formed the garrison of the fort of Dún an Óir, which was soon besieged by a large English force under Lord Grey. Walter Raleigh, who accompanied Lord Grey, acted as a mediator and negotiated with the Spaniards. He promised them free passage from the fort if they opened the gates and surrendered to the English. The Spaniards agreed to these terms and surrendered, but the English did not keep their promise. Upon entering the fort, they mercilessly slaughtered all the Spaniards and threw their bodies over the walls—into the sea and onto the shore.

Despite this stunning confirmation of the story he had painstakingly reconstructed from fragmentary memories, Charles was not satisfied. He continued researching libraries until he found a document about the Battle of Dún an Óir. This document reported that the Spanish detachment included a priest, whom the English killed along with the others. The priest's initials matched those that Charles had seen on the signet ring and had captured in one of his drawings.

I am providing this one example, saying that it is one of many, for which it is very difficult to assume that they were the result of subconsciously obtained and recalled data. However, we can still imagine (although not without difficulty) how the Battle of Dún an Óir becomes part of a film or book and is subconsciously processed by Charles. As we mentioned earlier, explanations based on "subconscious processing" are more problematic when it comes to personal-level events and circumstances that are not included in any noticeable historical events. There is no doubt that a very powerful type of report of this kind is children's spontaneous memories of other people's lives, which they perceive as "their own."

The most prominent academic research on such memories comes from the Division of Perceptual Studies (DOPS) at the University of Virginia School of Medicine. This research, initiated by the late psychiatrist Dr. Ian Stevenson [26] and continued by child psychiatrist Dr. Jim Tucker [27], has collected and studied over 2,500 cases worldwide.

The children typically begin talking about the alleged previous life around age 2 to 4 and usually stop around age 6 or 7, with a significant percentage (often over 70%) of the previous personalities reportedly dying by unnatural, sudden, or violent means. The memories are generally spontaneous and not elicited through hypnosis. Children in these cases often display unusual behaviors like phobias related to the deceased person's mode of death (e.g., a child who remembers drowning having a severe fear of water) or play that mimics the previous personality's occupation or death scene. A substantial number of cases include birthmarks or birth defects that appear to be congruent with the fatal wounds or marks on the deceased person, sometimes with supporting post-mortem reports.

While many cases have been found in cultures that believe in reincarnation (like India and Sri Lanka), cases have also been documented in Western countries and families that do not hold such beliefs.

I will provide some examples here, which I again claim to be only a few of the many, while even one is enough to request an explanation.

First is the famous case of James Leininger, born in Louisiana in 1998. The spontaneous memories began manifesting around the age of two, primarily through intense, recurrent nightmares of a plane crash. James would wake up screaming, shouting phrases like, "Airplane crash! The plane is on fire! Little man can't get out!" He would often kick his legs as if trying to push his way out of a confined space. He provided a series of specific details about the plane and the battle, including that his plane was a Corsair fighter, and it was shot down by the Japanese after taking off from a boat named the "Natoma". He also mentioned having a friend named Jack Larsen.

James' parents, initially skeptical, were amazed by his sophisticated knowledge of WWII aviation details, such as correctly identifying a "drop tank" on a toy plane. The boy repeatedly crashed his toy planes into the coffee table, mimicking a plane crash, and often knocked the propellers off. James's father, Bruce Leininger, began to research his son's claims. He searched for a ship named "Natoma" and discovered the USS Natoma Bay, a World War II escort carrier that was stationed in the Pacific. He later confirmed that the USS Natoma Bay was involved in the Battle of Iwo Jima, the location James repeatedly mentioned as where he was shot down. The previous personality was identified as James M. Huston, Jr., a U.S. Navy fighter pilot who flew a Corsair off the USS Natoma Bay and was killed in action over Iwo Jima in 1945. Records confirmed that a pilot named Jack Larsen was indeed a fellow pilot on the Natoma Bay and was still alive, whom James later met. James Huston, Jr.'s sister later verified additional private family details that James Leininger mentioned.

The case was investigated by Dr. Jim Tucker at the University of Virginia's Division of Perceptual Studies (DOPS). Dr. Tucker noted that many of James's specific claims were documented before the deceased pilot, James Huston, Jr., was identified. As is typical of these cases, James Leininger's vivid memories and intense nightmares began to fade as he grew older, generally subsiding by the age of seven or eight.

While this example, although with some stretch, can be considered related to historical events (which does not negate solely individual details), other examples involve ordinary, non-famous people who lived and died relatively recently.

One of the classic, early, and highly verified cases investigated by Dr. Ian Stevenson [26] was that of Parmod Sharma, a boy born in 1944. From about age 4, Parmod began speaking about a "previous life" in a different city, telling his family his name was Parmanand. He claimed he had died after falling from the roof of his house. He insisted he was the owner of a small shop and described his wife, children, and a distinctive scar on his previous body. Crucially, he mentioned that he had deposited 1,200 rupees with a friend before his death and wanted his family to recover the money.

The previous personality was eventually identified as Parmanand Tandon, a shopkeeper in the city of Mathura, who had died in 1943 from injuries sustained in a fall from a roof. Parmod was taken to Mathura and, unprompted, led the way through the streets to the correct former house and shop. He correctly identified Parmanand's widow, children, and other relatives, even recognizing them in old photographs. The claim of the 1,200 rupees deposited with a friend was verified by Parmanand's family. He displayed an adult's emotional response toward Parmanand's widow and children, a common feature in these cases.

Another well-known case is Western and involves an ordinary family from a small island community. Cameron Macauley, born around 2000 in Glasgow, Scotland, from the time he could talk, Cameron spoke constantly about his life on the small remote island of Barra (part of the Outer Hebrides of Scotland), which his family had never visited. The boy spoke of a house with white windows and a black and white dog, and described the beach where he used to play. He named his previous father, Shane Robertson, and claimed he had been killed by a car, displaying significant emotional distress and intense longing to return to his "Barra family."

With the help of Dr. Jim Tucker [27], the family traveled to Barra, where they were able to locate the specific house Cameron described, which was a white house with distinctive black window trim. A family with the surname Robertson had indeed lived there. A man named Shane Robertson was found to have lived and died on the island, though not from a car accident (initial reports suggested a road accident, but later records indicated he died from an injury on the road). Cameron was able to correctly identify the black and white dog from a photo and recognized details of the house and the surrounding area.

Let us now switch to another type of experience in ASC, one that possesses the unusual feature of having synchronous events in the physical world. This correlation of inner and outer events was most notably introduced by the Swiss psychiatrist Carl Jung as "synchronicity", which he defined as the "meaningful coincidence" of two or more events (inner and outer) close in time, where there is no causal connection between them, yet they appear to be related phenomenologically [28]. Jung described such events as connected by the "acausal connecting principle," suggesting that certain events are linked not by cause-and-effect, but by a shared underlying meaning.

Many modern scientists, however, view these events as being explainable by confirmation bias and the brain's natural tendency to seek out patterns—the observer simply focuses on the hits and ignores the misses. Jung's work, however, suggests the existence of a non-causal relationship in objective reality, and "confirmation bias" explanations suffer in the face of detailed examples.

One of Jung's most famous examples involved a patient who was overly rationalistic and resistant to therapeutic progress. She was recounting a dream involving a golden scarab beetle (a powerful symbol of rebirth in ancient Egyptian mythology). At that exact moment, Jung heard a tapping at the window. He opened it, and a real, goldish-green scarab beetle—a European Rose Chafer, an insect not typically found indoors—flew in. Jung caught it and presented it to the patient, saying, "Here is your scarab." This seemingly impossible meaningful coincidence reportedly helped break through the patient's rational defenses and furthered her treatment.

Describing the psychedelic and holotropic practices in which he had the opportunity to participate as a leader or participant during his 50 years of practice, Stanislav Grof wrote [25]: "Isolated instances of synchronicity in the lives of people experiencing spontaneous or induced holotropic states of consciousness are very common. Moreover, they often follow in impressive series. Over the years, we have observed and even experienced many such serial coincidences associated with psychedelic therapy, holotropic breathwork sessions, or psychospiritual crises."

He then provides an example of his wife, Christina's experience. During one of the month-long workshops at Esalen, Christina experienced profound and varied experiences in altered states of consciousness. These occasionally included visions of various archetypal figures and animals. Particularly significant were peacocks and white swans—birds associated with Siddha Yoga and Christina's spiritual teacher, Swami Muktananda. On one day of the aforementioned workshop, Christina experienced particularly powerful and meaningful visions involving a white swan.

The next day, during a psychological game using animal imagery, its leader, Michael (a person who was in no way connected with the parallel seminar and knew nothing about Christina's experience), informed her at the end of the game that "Your animal is a white swan." The story continued the next morning, when Christina and Stanislav went to their mailbox to pick up the mail. Christina had received a letter from someone who had attended a seminar we held several months earlier. The letter contained a photograph of Christina's spiritual teacher, Swami Muktananda, which the person thought Christina would love to have. With a mischievous smile on his face, Muktananda was sitting on a garden swing next to a large vase shaped like a white swan. The index finger of his left hand pointed at the swan, and the tips of his index finger and thumb of his right hand were touching, forming the well-known gesture indicating that all is well and that one is in a wonderful mood.

To these beautiful examples, I want to add my own. When writing this very book and this very chapter you are reading now, I had to search for and re-read Grof's book, which we have already mentioned multiple times recently. Initially, I read it many years ago, and the story about synchronicity that stuck in my memory the most was the one about the writer who worked on his book on the fourteenth floor of a building in Manhattan, New York. The author was actually Joseph Campbell (which I totally forgot and now recall with surprise), and at the time, he was writing a chapter devoted to the mythology of the African Bushmen, a tribe living in the Kalahari Desert. One of the most important deities in the Bushmen pantheon is the praying mantis, which combines the traits of a Trickster and a Creator God. At one point, Campbell got up and opened one of the windows facing Sixth Avenue. As he opened the window, he looked at the side of the building and saw a praying mantis crawling up the wall, somewhat unusual for Manhattan.

This story about the praying mantis was the reason why I searched for Grof's book, but I did not have a particular understanding of which exact book I needed, so I took this one by name and started surfing through the chapters with only one stop (with no reason at all) to google details on Swami Muktananda mentioned in the book. This was the first instance of synchronicity for me, as the details of Muktananda's biography corresponded closely with some of my recent thoughts. In the book, among other materials, I found a story about a praying mantis that I described, which I greeted like an old friend.

The next day, I decided to hang the towels from the bathroom on the balcony to air them out (which is what I usually do, mostly... never). On the balcony windowsill, I spotted a huge green praying mantis basking in the sun, which was somewhat unusual for my balcony in Varna, Bulgaria. It was a magnificent specimen—a female, I assumed, because the insect had a noticeable abdomen. Over the next couple of hours, I kept coming to the window to exchange glances with my visitor, and I began to fear she might leave her future offspring there. But then she disappeared.

Before touching on the last important thing in this chapter, I would like to additionally briefly list the ASC experiences challenging traditional explanations and once again state the reason for this "challenging" character of them. These are experiences of identification with an atom or a molecule, identification with physical or chemical processes, forces, and interactions; identification with stars, star systems, and Galaxies; identification with a single animal (plant) or an entire pack, species. All of them are characterized by an enormous amount of detailed knowledge about listed entities (in correspondence with the current scientific knowledge), not anyhow connected to the actual educational level of reporters.

And the last type of ASC that we already considered in the previous chapter, we will consider again in this chapter, but in a different, phenomenological context. This is again near-death experiences (NDE), the set of data combining together phenomenology of other ASC, the hard problem of consciousness, dissociation-association problem, reverse correlation problem at its highest implementation, and synchronicity events.

Nowadays, NDE is a sphere with a tremendous amount of accumulated and processed information, forming obvious and strong patterns. First and important note that should be made about NDE is that "near-death" is not where these experiences are met exclusively, but where they are explored by scientists. In his works, Dr. Peter Fenwick [30], who was a prominent British neuropsychiatrist and neurophysiologist known for his extensive research of the area, expanded the scope of scientific inquiry of NDE **twice**.

First, to include a wider range of phenomena that occur **around** the time of death which were termed end-of-life experiences (ELEs). This umbrella term captures similar phenomena to NDEs but is not exclusively dependent on the person being in a state of clinical death and subsequent resuscitation. This includes NDEs themselves–experiences reported by people who are close to death but survive (e.g., cardiac arrest, trauma). But additionally, these are deathbed visions–visions seen by a person who is actively dying, often involving deceased relatives or spiritual figures coming to comfort or "take" them on a journey. Also, these are terminal lucidity–a brief period of restored mental clarity, memory, and cognitive function in a patient with severe dementia or brain damage, shortly before death. These are also deathbed coincidences–experiences reported by family or friends, often away from the dying person, where they are strongly and suddenly made aware of the death at the precise moment it occurs (e.g., a "visit" or sudden, intense feeling) (type of synchronicity we discussed before in this chapter).

Second expansion of the scope is much more tremendous–it is the observation of the fact that the profound features of near-death experiences can be met in other circumstances that are not life-threatening or connected to clinical death. Fenwick's retrospective study on over 300 NDE reports showed that while many occurred during illness or surgery, a significant portion happened when the individual was not physically near death. These circumstances share the phenomenology of an NDE—such as peace, the out-of-body sensation, and the encounter with light—but cannot be easily explained by brain hypoxia (lack of oxygen) because the person's physical state was stable.

Common non-death-related circumstances where NDE-like features were reported include extreme fear or stress, using some types of psychedelics, deep states of meditation or profound relaxation, and even spontaneous arising during the day or when waking from a dream, without any physical or perceived threat.

In the context of our book, it just unifies NDEs with all the other types of ASCs and refers us back to the reverse correlation problem, as the logic of reduced brain activity leading to extended consciousness abilities has its clearest incarnation–death is the most intense disruption of brain activity.

So, what is the phenomenology of NDEs that is problematic for "standard explanations"? The first thing that should be immediately stated is that, **like all other ASC** content, NDE is the obvious mixture of individual and universal content. Fenwick's studies show with no place for doubt that NDE are affected by a person's cultural background: what is known as "tunnel experience"–a sensation of moving rapidly through a dark space or tunnel toward an incredibly bright light belongs to a Western culture, in Chinese experiences, it may take the form of a river; while Western culture people tend to meet angels, or Christian religious figures, in Indian experiences, there is a much higher proportion of religious figures or messengers of death, reflecting their own cultural beliefs. However, the logic of the underlying process (moving towards tremendous peace/love/compassion/home/knowledge) is absolutely cross-cultural, as well as such phenomena as out-of-body experiences (OOB), life review, and such formal characteristic as extreme lucidity of perception.

Here, we arrive at the question: why, despite having obvious individual and culture-specific features, do NDEs exhibit the listed common, universal elements? This is usually answered by "because of the same brain structure and psychological mechanisms for everyone." Ok then, but in "normal reality" we need to have one more element adjusted to "the same brain structure and psychological mechanisms"–this is an objectively existing (outside of the perceiving individual mind) object of perception. Why do we assume that in this case, such an object is not presented?

So, summarizing this chapter, let us remember what we have listed here as presented in ASC: detailed and accurate data on physical and alive objects and processes related to them; detailed and accurate data on the mythology of different societies (distant by geography, time, or both); detailed and accurate data on deceased people's personal biographies; inner experiences meaningfully correlated with objective outer events; detailed and accurate out-of-body (OOB) experiences; profound universal aspects of near-death experiences (NDE). This is a systematic (not divisible into "part-by-part" explanations) problem of the presence of **objective** aspects in and around subjective experience—the one that requires explanation and remains a persistent **ASC content problem**.

Biology and Information

In this chapter, we'll refer to the work of Dr. Michael Levin, a Distinguished Professor in the Department of Biology at Tufts University, where he also holds the Vannevar Bush Chair and serves as the director of the Allen Discovery Center and the Tufts Center for Regenerative and Developmental Biology. Levin's work sits at the intersection of developmental biology, computer science, and cognitive science and posits a serious challenge to the Materialistic paradigm.

Dr. Levin's research explores how biological systems—specifically groups of cells—process information to make decisions, regulate body shape, and coordinate complex activities. His lab studies the bioelectrical signals that cells use to communicate and coordinate during embryogenesis, regeneration, and cancer suppression. He posits that these signals function as a layer of "software" that directs the "hardware" of the genome. He researches the "intelligence" of non-neural cells, defining intelligence as the ability to reach a goal state through various means. His team explores how cells navigate "anatomical morphospace" to build and repair body structures. By manipulating these bioelectric networks, his team has demonstrated the ability to control growth and form, including triggering the growth of new limbs or organs and suppressing tumor development.

Mind and Form Storage Problem

In this chapter, I will describe a series of experiments conducted by Dr. Michael Levin, his colleagues, and team that will start from the "electric face" experiment and will go through "foreign species' heads" to "Xenobots 1.0 and 2.0" experiments, to lead us to a profound question, which is the problem for the materialistic worldview. Before starting, I'd like to note that Levin himself makes no attempt to use the materialistic paradigm to explain the data. So here are the experiments.

Step 1. Dr. Levin and Dany Adams used voltage-sensitive fluorescent dyes on developing Xenopus laevis (frog) embryos. These dyes glow more or less intensely depending on the electrical potential (voltage) of the cell membranes. They discovered that long before any physical facial features (like eyes, nose, or mouth) began to grow, a "ghost" of the frog's face appeared on the embryo in flashes of light, representing specific clusters of hyperpolarized cells that perfectly outlined the future geometry of the face. The observation got the commonly known "electric face" name. The question of where this pattern comes from had as a possible answer DNA itself: DNA organizes cell connections, and these connections organize the electric picture. [32]

Step 2. When the researchers artificially changed these electrical gradients (altered the "electric picture" using drugs or ion channel manipulation), the physical face grew deformed, or organs grew in the wrong places. This proved that the electrical map was both a byproduct of growth and the instructive signal for growth. However, this did not discredit the suggestion of the DNA's initial role, while having outstanding potential practical implementations (bioengineering, regenerative medicine without altering DNA). [33]

Step 3. This experiment was done with Planarian flatworms, known for two features. First of these features is that these worms are the masters of regeneration: if you cut one into pieces, each piece typically regenerates into the "whole initial worm." The second is that, depending on the species and the environment, they can procreate through two very different methods: asexual fission (= regeneration of each part) and sexual reproduction. Action from the previous step was repeated with Planarias – their "electric picture" was altered to make them grow two heads, which they did. They were "cut into pieces," and each grew a two-headed worm. In sexual reproduction, the form of the body was reset to one-headed. This data indicated that the bioelectrical picture had priority during regeneration, but did not discredit the suggestion of the DNA's initial role (on the contrary, reinforced it). [34]

Step 4. This experiment was done with Planarian flatworms again. The researchers started with a common species of Planaria called Girardia dorotocephala. When the worms were decapitated, amputation forced the remaining body to ask a biological question: "What is missing, and what should I build here?" In planarians, the wound site immediately starts forming an "anterior pole"—a small group of cells that acts as a broadcast tower, telling the rest of the body to start building a head. By blocking gap junctions, experimenters prevented the cells at the wound from "consulting" the rest of the body's electrical memory: while still having general info "head must be here", the more detailed info on shape, eyes, structure of brain, etc. was "noised." As a result, the worms didn't grow random, deformed blobs. Instead, they randomly grew heads that perfectly matched the anatomy of three other distinct species: S. mediterranea (rounded, "blunt" heads), D. japonica (triangular heads with different eye placements) and P. felina (smaller, more compact heads) – all from Planarian (triclad) family tree with their relationship to initial specy characterized by long-term evolutionary divergence, meaning their DNA has been on separate paths for millions of years. While it is highly unlikely that the initial species' DNA stored all alternative variants, this cannot be fully excluded. [35]

Step 5. Xenobots 1.0 "with task": The team used an evolutionary algorithm running on a supercomputer. They gave the AI a goal: "shape a form optimal for travel as far as possible, able to push scattered pellets into piles, and able to carry a specific item." Upon obtaining the suggested form, the experimenters placed tiny clusters of heart and skin cells from the frog embryo into a Petri dish filled with a standard salt-water solution, then, using microsurgical tools, they manually "sculpted" these clusters into the 3D shape dictated by the AI. This form automatically formed the electrical communication of specific formation and let cells to self-organize into simple organisms with behavior: the "four-legged" designs walked across the bottom of the Petri dish, and the circular designs swam in loops; when researchers scattered tiny iron pellets in the dish, the Xenobots naturally worked like collective "roombas": by simply moving according to their shape, they pushed the pellets into neat, central piles. [36]

Unlike previous steps of experiments, this one leaves no chance for DNA to store the information on the achieved result: this was the frog's DNA, and it executed the task having nothing in common with the frog's evolution or behavior, but having absolute correspondence to the calculated forms, optimal for specific behaviors: on sculpting this form, behaviors arose. However, this leaves the door open for the idea that form created behaviors, and that means the "scaled-down" hard problem of consciousness arriving at this level of observation, as if we decide "form creates behavior," then the hard problem with its question on "how exactly" is here.

Step 6. Xenobots 2.0 "without task": Here, instead of sculpting cells into a pre-defined shape, the researchers simply gathered thousands of skin cells into a small cluster and left them in a neutral environment without providing any task via an AI-generated blueprint. Left to their own devices, the cells did not just sit there as a blob. Instead, they self-organized into a spherical shape and began to grow cilia (tiny hair-like structures). These "natural" Xenobots began to swim autonomously. More importantly, they displayed Kinematic Self-Replication. If loose stem cells were placed in the dish, these "task-less" Xenobots would spontaneously gather the loose cells into piles, which would then mature into new Xenobots.

Unlike the previous step of experiments, at this stage, the idea that form created behaviors shuts down: both form and behavior arose. Both of them, obviously linked, arose. From my perspective, at this level of precise observation, the "hard problem of consciousness" ceases to exist, as observation shows the simultaneous arising of both elements. However, I can easily imagine people who would still interpret this data as "form arose, then it produced behaviors (mind). For these kinds of people, the hard problem of consciousness persists: they are welcome to explain "how exactly".

Here, we frame this observable data as the question: where are forms, behaviors (minds), and the links between them stored? This is the **mind and form storage** problem.

As we mentioned before, Levin does not use the materialistic paradigm to solve this problem. To make sense of data, he uses the concept of a Platonic space (often referred to as a Morphospace)—a conceptual, multi-dimensional map of all possible physical forms an organism could take. Levin uses this term to describe where the "goals" of biological systems reside. He emphasizes, however, that this "space" is less about mysticism and more about the mathematical and cybernetic limits of what a biological collective (like cells) can build.

Yet, however this information is called, it exists and precedes the complex life-creation process and remains problematic for interpretation within a materialistic view.

Agentic Algorithm Experiment

In 2023, Michael Levin and his colleagues conducted the experiment with the sorting algorithms (bubble sort and selection sort). The standard, deterministic versions of these algorithms were taken and reimagined as a "collective intelligence" of cells rather than a top-down instruction set. [37]

Normally, a sorting algorithm is a global command: "Look at the whole list and move X to position Y." Experimenters rewrote these so that each number in the list was an "agent" (a cell) that could only see its immediate neighbors. Particularly, instead of being "moved" by a central sorter, the agent calculates its own potential energy (stress):

> **The Logic:** An agent i with value V_i looks at its immediate neighbors V_{i-1} and V_{i+1}.
>
> **The Stress Calculation:** If $V_{i-1} < V_i < V_{i+1}$, the agent's stress is **zero** (it is "happy").
>
> **The Mismatch:** If $V_i < V_{i-1}$, the agent experiences "high stress" because it is out of place.

The sorting algorithm (e.g., bubble sort) was not used as a set of instructions for the computer, but as a decision-making rule for each agent. Agents can "talk" to neighbors to see if a swap would reduce their collective stress. Instead of a command like swap(A[j], A[j+1]), the agent runs a conditional check: "If I swap with my neighbor, will my local stress decrease?" If the answer is yes, the agent initiates the swap. Agents were also programmed to accept a temporary increase in stress (moving to an even "worse" position) if it was the only path to eventually reach a "lower stress" state on the other side of a barrier. Also, a "noise" variable ("kT" in physics terms, where "k" is the Boltzmann constant). This allowed the agents to make "mistakes" (random moves), which ironically helps them "wiggle" out of local optima or get around barriers that a perfectly rigid algorithm would get stuck on.

From the technical perspective, the experiments were largely written in Python, using custom classes where each element in the list is an instance of an "Agent" class. These agents do not have access to the full list's "index" or "length". They only have access to a "sense()" function (to see neighbors) and a "move()" function.

The next step was to introduce "obstacles": experimenters "locked" certain positions in the array (physically) so they couldn't be swapped; they also introduced "stubborn" agents that refused to move or followed the algorithm incorrectly.

As a result of running such an algorithm with obstacles, the following was observed: when a direct swap was blocked by a barrier, the "cells" would temporarily move further away from their target position (increasing "disorder") to eventually "go around" the barrier and achieve a sorted state. While a standard bubble sort code says: "If I am bigger than my neighbor, swap," it does not say: "If there is a wall, move backward three spaces, wait for the wall to move, then circle back." Because the agents were programmed with a goal (minimize stress) rather than just a list of steps, they exhibited "problem-solving." They performed maneuvers that were never written in their individual code to get around the barriers. The system reached the goal even when 50% of the cells were "defective" or running different versions of the algorithm.

While these results are outstanding, they do not yet demonstrate anything truly remarkable, in the sense that we can still claim that the inner logic of the newly designed system was still programmed in the code within each "agent." However, the experiment results do not end here. When dealing with output, the researchers, among others, tracked a metric called "clustering". This measured how often an agent was adjacent to another agent of the same sorting strategy (bubble sort or selection sort). The expectation was that since the agents only have one goal—to be numerically sorted—their final positions should have been determined strictly by their numerical value (1, 2, 3...). Their "algotype" (sorting strategy) should be scattered randomly throughout the finished list. In reality, it was found that during the sorting process, agents with the same "algotype" spontaneously clustered together. They formed little "neighborhoods" of "bubble-sorters" and "selection-sorters". But the crucial detail is that there was zero code for clustering anywhere. The agents couldn't even "see" what algorithm their neighbors were running. They only saw the values of the numbers next to them.

If in the set of experiments that we described in the previous chapter ("Mind and form storage problem"), the problem we saw was the question about where forms, behaviors (minds), and the links between them are stored, in this experiment we have no physical substance of form and yet see the "arising behavior." In this case, this behavior is obviously linked to a specific **inner logical structure**.

This produces not one but 3 major questions-problems:

- Where are behaviors (minds) and their links to specific data structures stored? This is the **mind and inner structure storage** problem.
- Is the mind attributed not only to biological systems?
- How exactly do specific inner structures get access to behavior patterns?

Dr.Levin has his own extraordinary model answering these questions—TAME (Technological Approach to Mind Everywhere)—a conceptual framework designed to understand and manipulate "minds" or cognitive agents across all types of substrates—not just humans and animals with brains, but also cells, tissues, synthetic bio-bots, and even AI.

While Levin himself tries (but not too hard) to keep TAME in the frame of Materialism, with the "storage" and "access" aspects, it easily and logically drifts to a completely different domain, which we will discuss in this book later.

It is important to note that TAME is a highly nuanced model which can never be interpreted in a simplistic way. We'll definitely try to borrow a lot of these nuances from Levin's genius later in this book.

Special and General Relativity

In this chapter, we'll cover Albert Einstein's theories known as Special and General Relativity. Special Relativity (1905) describes the physics of objects moving at constant speeds, particularly those approaching the speed of light. It fundamentally changed how we understand space and time by showing they are not absolute but linked in a four-dimensional continuum called spacetime.

General Relativity (1915) is a theory of gravitation that expands on Special Relativity by including accelerated reference frames and gravity. Albert Einstein proposed that gravity is not a force in the traditional Newtonian sense, but rather a curvature of spacetime caused by mass and energy.

As for the parent "Problems to Solve" chapter, it is usual thing to present something that posits the "problem" for the Materialistic paradigm, chapters on General and Special Relativity are slightly unusual, as those theories do not seem to have any "local" problems—problems in themselves or "within their own logic." They are absolutely consistent with the data they deal with and have perfect predictive power. However, the problem arises when these theories stop being descriptions of relations and behavior of some parts of Reality and start presenting themselves as the precise and full description of what Reality (including space and time) is. We'll highlight what exactly makes them problematic in that regard.

The third chapter will present the very well-known and famous problem by itself: General Relativity and Quantum Mechanics Incompatibility.

Special Relativity: Time, Space, Energy, and Mass

The creation of Special Relativity was a slow-burn revolution that began with a crisis in late 19th-century physics. At the time, scientists were desperate to find the "luminiferous ether," a mysterious substance they believed filled all of space to allow light waves to travel. When the **Michelson-Morley experiment of 1887** failed to detect this ether, it left a massive hole in the laws of nature. Brilliant minds like Hendrik Lorentz and Henri Poincaré tried to fix the problem by suggesting that objects physically contract or that time slows down as a "correction" to moving through the ether, but they remained tethered to old Newtonian ideas of absolute space and time.

In 1905, Albert Einstein—then an obscure 26-year-old patent clerk—shattered this deadlock by simply throwing the ether away. He realized that if the speed of light is a universal constant, then time and space themselves must be flexible. This "Annus Mirabilis" (Miracle Year) paper was initially met with a mix of silence and skepticism. It wasn't until **Max Planck**, the father of quantum theory, threw his immense scientific weight behind Einstein that the theory gained traction. Even then, it faced hurdles; early experiments in 1906 by Walter Kaufmann seemed to disprove Einstein's math, and it took years for more precise testing to vindicate him.

The final transition from a fringe idea to a fundamental pillar of science happened around 1908, when Einstein's former professor, **Hermann Minkowski**, realized the math described a four-dimensional reality where space and time are inseparable. By the 1910s, the younger generation of physicists had embraced the theory, though the "Old Guard" remained so wary of its radical nature that when Einstein finally won the Nobel Prize in 1921, the committee specifically avoided citing Relativity in his award. They chose instead to honor his work on the photoelectric effect, showing just how long it took for the world to truly get comfortable with the idea of a warping universe.

Some data

Imagine **System A** is a stationary laboratory on Earth, and **System B** is a spacecraft traveling at a significant fraction of the speed of light (*v*). Both systems are equipped with identical, highly precise atomic clocks.

1. The Perspective of Observer A (Stationary)

From the data gathered by Observer A, the clock in the moving System B appears to be ticking **slower**.

- **The Observation**: If System B passes by, Observer A will see System B's clock "lagging" behind their own.
- **The Reality**: To Observer A, everything inside System B (mechanical processes, biological aging, computer cycles) is happening in "slow motion."

2. The Perspective of Observer B (Moving)

Relativity is a two-way street. Because there is no "absolute" rest, Observer B feels stationary and perceives System A as the one moving away at velocity ***v***.

- **The Observation**: Observer B looks at System A's clock and sees it ticking slower than their own.
- **The Symmetry**: This is the "Relativity" part—both observers genuinely see the other's time slowing down.

Velocity (v)	**Effect on System B (as seen by A)**
0.1c (10% light speed)	Time slows by ~0.5%. Barely noticeable.
0.9c (90% light speed)	Time slows by ~129%. 1 hour in B = 2.3 hours in A.
0.999c	Time slows by ~2,137%. 1 hour in B = nearly 22 hours in A.

Note: If System B turns around and returns to System A, the symmetry is broken because System B had to accelerate. In this case, System B will physically be younger than System A.

The Classic "Train and Platform" Example

Imagine System A is a person standing on a train platform, and System B is a passenger on a high-speed train moving past them.

- **Crucial observation visualization:** If a locomotive has a spotlight on and the train is moving, the speed of the spotlight relative to Observer A (Platform) does not add up to the speed of the train. It still amounts to ***c*** (constant, 299,792,458 meters per second).

- **The Events**: Two lightning bolts strike the ground at the same time—one at the **front** of the train and one at the **back**.
- **Observer A (Platform)**: Because they are exactly halfway between the two strikes, the light from both reaches them at the same instant. Observer A records the data: "Events happened simultaneously."
- **Observer B (Train)**: They are moving toward the light coming from the front strike and away from the light coming from the back strike. Even though the lightning hit the tracks at the same spot, the passenger "meets" the front beam of light first.
- **The Data Conflict**: Observer B records the data: "The front strike happened before the back strike." **Note:** this is not about time needed for the light to travel: even after both observers "subtract" the travel time of light, they still disagree on the timestamp.

Energy and Mass

What data connects space, time, velocity (***v***), speed of light (***c***)—all this seemingly "near geometrical stuff"—to energy and mass? Simplified description (best for the initial understanding) includes the following data set: classically, if you keep pushing an object, you expect its velocity should eventually exceed ***c***. But the experiments (not theories or math) show a different picture. For example, in particle accelerators (like the LHC), we "push" protons with massive amounts of electromagnetic energy and:

- **At Low Speeds**: If you double the energy, the proton goes much faster.
- **Near *c***: As the proton reaches 99.9% of ***c***, doubling the energy barely increases its speed at all.
- **Where does the energy go?** Instead of getting faster, the proton becomes "heavier" (it gains more **mass**).
- Wich means two things: (1) energy transforms to mass (famous formula); (2) as energy (simplified view) is what allows an object to overcome its inertia (**mass**, or "resistance to change") and move, ***c*** becomes a "universal limiter" for both "stage" and "content" of Universe: space, time, energy and mass.

Special Relativity Explanation to Observed Data

Because *c* **never changes** regardless of how fast you are moving:

1. Space must "shrink" (Length Contraction).
2. Time must "stretch" (Time Dilation).
3. The order of events must shift (see "train" metaphor above).
4. Space and Time are not separate, they are the same thing—Spacetime like a piece of fabric. If you pull the fabric horizontally (Space), it must shrink vertically (Time) to compensate. You cannot move through space without affecting how you move through time. They are tied together in a zero-sum game.

Because *c* **can never be exceeded** by any object, when you move faster:

5. Mass must "grow".

6. Energy must stop increasing velocity and instead transform to mass.

Unlike other data listed in the "Problems to Solve" chapter, Special Relativity does not seem to have any "local" problem—a problem in itself or "within its own logic." It is absolutely consistent with the data it deals with and has perfect predictive power. However, the problem arises when this theory stops being the description of relations and behavior of some parts of Reality (like energy, mass, movement) and starts presenting itself as the precise and full description of what Reality (including space and time) **is**.

Here, we immediately start dealing with the non-local observation problem: in the picture of Reality presented by this theory, **where exactly is Mind obviously presented in Reality**? By this, I mean a qualitative mind with the experience of "redness of red."

General Relativity: Time and Space in the Presence of Large Mass

General Relativity was the result of an intense, decade-long intellectual struggle that Einstein began in 1907, just two years after his work on Special Relativity. It started with what he called the "happiest thought of his life": the realization that a person in free fall would not feel their own weight. This led him to the **Equivalence Principle**, suggesting that gravity and acceleration are physically identical. Einstein spent years laboring over complex non-Euclidean geometry to prove that gravity isn't a "pulling" force at all, but rather a curvature in the fabric of spacetime caused by mass and energy. He finally presented his completed **Field Equations** in November 1915, effectively rewriting the laws of the universe that had stood since the time of Isaac Newton.

While the theory was mathematically elegant, it remained a localized scientific curiosity until 1919, when the British astronomer **Arthur Eddington** used a total solar eclipse to provide physical proof. By measuring how much gravity bent the light from distant stars passing near the sun, Eddington confirmed Einstein's radical prediction that space itself is warped. The results made Einstein an overnight global celebrity, though many older scientists found the math—which required understanding four-dimensional tensors—extraordinarily difficult. For decades, General Relativity was considered a "beautiful but useless" theory because its effects are only noticeable near massive objects like stars.

It wasn't until the 1960s, during the "Golden Age of General Relativity," that the theory became central to modern physics. The discovery of **quasars, pulsars, and the theoretical groundwork for black holes** proved that Einstein's equations were not just abstract math, but the key to understanding the most violent and massive phenomena in the cosmos. Today, General Relativity is the backbone of our understanding of the Big Bang and the expansion of the universe, though it famously remains at odds with the laws of quantum mechanics.

Some data

While Special Relativity is about speed, General Relativity introduces the "weight" of the universe: Gravity. It is based on the following experimentally verified data:

- **The Rule**: The stronger the gravitational potential (the closer you are to a massive object), the **slower** time passes.
- **System A (Earth):** Deep in a "gravity well." Time moves slower.
- **System B (Deep Space):** Far from any planets or stars. Time moves faster.

This isn't just theoretical; it's a daily engineering requirement for GPS satellites. To keep the data accurate, engineers have to account for both layers of relativity:

Effect	**Cause**	**Impact on Satellite Clock**
Special Relativity	High orbital velocity	Loses **7 microseconds** per day.
General Relativity	Weaker gravity (high altitude)	Gains **45 microseconds** per day.
Net Result	Combined Physics	Gains **38 microseconds** per day.

The Consequence: If we didn't bake this "time offset" into the satellite data, your phone's GPS location would be off by about **10 kilometers every single day**. The system would become useless within hours.

Space

While gravity warps time (causing **Time Dilation**), it also warps space. This effect is known as **Spatial Curvature** or **Length Contraction**. While a clock runs slower near a massive object, the physical geometry of space itself becomes "stretched" or curved.

Exactly as in the case of Special Relativity, General Relativity does not seem to have any "local" problem—a problem in itself or "within its own logic." It is absolutely consistent with the data it deals with and has perfect predictive power. However, the problem arises when this theory stops being the description of relations and behavior of some parts of Reality (like mass, movement) and starts presenting itself as the precise and full description of what Reality (including space and time) **is**.

Here, we immediately start dealing with the non-local observation problem: in the picture of Reality presented by this theory, **where exactly is Mind obviously presented in Reality**? By this, I mean a qualitative mind with the experience of "redness of red."

General Relativity and Quantum Mechanics Incompatibility

General Relativity describes a universe that is **smooth, continuous, and deterministic** (like a curved trampoline), while Quantum Mechanics describes a universe that is **jittery, discrete, and probabilistic** (like a swarm of bees).

How can it be that two theories with the tremendous predictive power in their domain **picture reality as two different (contradictory) things**?

Some details on the contradiction are presented below.

1. The Scale Conflict

The two theories are designed for different "resolutions" of data:

- **General Relativity (The Macro Layer)**: Views spacetime as a smooth, continuous, and flexible fabric. It's like a high-resolution vector graphic—no matter how much you zoom in, the curves remain smooth.
- **Quantum Mechanics (The Micro Layer)**: Views the universe as "quantized" (pixelated). If you zoom in far enough, things become "jittery" and discrete.

The Problems:

- **Quantum Foam** - When you try to apply the rules of General Relativity (gravity) to the scale of Quantum Mechanics, the math "breaks." At the **Planck Scale**, the "smooth" fabric of space begins to boil with violent energy fluctuations. The math of Relativity treats this as "infinite density" or "infinite curvature." In a data model, this is an Overflow Error. The equations return "Infinity," which in physics means the model has failed.
- **"Too Weak" Gravity** - Quantum Mechanics describes forces (like electromagnetism) by the exchange of particles (like photons). If we treat gravity the same way, there should be a particle called a **"graviton."** The problem is that gravity is incredibly weak compared to other forces—billions of billions of times weaker than the static electricity that holds a balloon to a wall. Because it is so weak, the quantum fluctuations of gravity are almost impossible to "normalize" mathematically. Every time physicists try to calculate the behavior of these gravitons, the math produces **"infinite" values** that can't be removed, which in physics means the model has failed.

2. Fixed vs. Dynamic Backgrounds

Here lies the fundamental data-structure disagreement:

- **QM (Fixed Background)**: Quantum mechanics usually assumes that Space and Time are a **fixed stage**. The particles (data) move around on a grid that doesn't change.
- **General Relativity (Dynamic Background)**: Relativity says Space and Time are the data. The stage itself bends and reacts to the actors.

You cannot easily run a "dynamic stage" simulation and a "fixed stage" simulation at the same time. They disagree on how to measure the distance between two points.

3. The "Measurement Problem" on a Universal Scale

In Quantum Mechanics, a system exists in a "superposition" (all possible states at once) until it is observed or measured. This works fine for an electron in a lab. However, if you apply QM to the whole universe (as GR does), you run into a logical wall: Who is the observer? For the universe to "collapse" into a definite state, there would theoretically need to be something outside the universe to look at it. Since nothing exists outside the universe by definition, QM struggles to describe a "self-contained" cosmos without slipping into philosophical headaches.

4. The Problem of Time (The "Frozen" Universe)

This is perhaps the biggest mathematical clash. In Quantum Mechanics, time is an **absolute background**. It is like a stage that stays still while the actors (particles) move around on it. However, in General Relativity, time is a flexible dimension that stretches and warps depending on gravity. When physicists try to combine the two using a famous equation called the Wheeler-DeWitt equation, the "time" variable actually **disappears entirely**. The math suggests a "stationary" universe where nothing ever happens—a result that obviously doesn't match the reality of a changing, expanding cosmos.

Nature of Reality

In the previous chapter, we stated that, in the model of reality, the presence of anomalies and an inability to resolve them for a long time are the exact markers of the necessity to shift the paradigm by finding and re-evaluating its basic assumptions. We also listed and described the problems we currently face within the materialistic paradigm.

What we need to do now is to show the cornerstone problem from all the listed ones, as it should definitely become the starting point for the new paradigm build-up. We are also going to see that, paradoxically, the solution to that problem also lies among the data related to one of the existing problems.

Before going into all that, we'll define some criteria for how any models of reality can be estimated and sorted out to select the optimal one, as we are going to use these criteria later to estimate the suggested model and compare it to the others.

After formulating and naming the new model, in this chapter we will show how it effectively solves all the anomalies described in the previous chapter and some extra problems.

Not Wrong, Reasonable, and With Most Explanatory Power

Let us ask ourselves what we know about reality and ourselves as a part of it. Once we ask this, we immediately start to learn. And what we learn very quickly is that reality, by itself, is a very silent thing: the lawn behind the window keeps silence, along with the trees, the sky, and the mountains. But people do not do the same. They talk, generations, and now alive—through books and other sources, and directly. But they use language, words, so they create symbolic systems, models, descriptions. They create maps that, while they are never an area depicted on them, may be useful. We also need to admit an obvious observation that there are many maps, and they differ, so we need some criteria to estimate and compare them to select the most precise and optimal one.

The problem of a good map can be solved by learning. We analyze data to understand which model is the best and dismiss the others. We use logic and intellect here. Searching for the best theory starts with dismissing simply "wrong" models—the ones that contradict observation and facts: the theory, saying that there can be only one bed in a single room of an apartment, is wrong as soon as we find or organize at least one room with two beds. Next, we dismiss models that do not contradict any known facts, but are "not reasonable," meaning we do not have any reasons to think that way. Like, the model in which an invisible and mighty flying spaghetti monster causes the Solar System's planet movement does not contradict any facts, but what are the reasons to assume that?

Next, we dismiss the theories that are not wrong and that are reasonable, but have less or no explanatory power. How, for example, any theistic religion starting with a good reason to think that the extreme complexity of reality and life has some intelligence behind it, is a model explaining so many things by the will and actions of God, but not explaining his mechanisms, the reasons for made decisions, and what it is at all, has an explanatory power very close to zero. Or panpsychism, saying that little consciousnesses combine into more complex ones, never closing the gap with an explanation of how exactly they combine. Or materialism having problems that we listed in the previous chapter.

What remains is the best model so far. There is no reason to use worse.

Cornerstone Problem

Let us play a bit with some imaginary numbers in the domain of contemporary science (obviously, materialism-based). I suggest a thought experiment: let us say we mark with the number 100 all the scientific publications existing up to the moment. Which number from this 100 would you put to define all that was dedicated to the physical stuff, compared to psychology and other mind-related areas? By physical stuff, I mean physics, chemistry, geography, geology, astronomy, biology, etc. I would put the numbers as 95 to physical against 5 to mind-related topics.

Now, let us apply the same approach to the intellectual baggage accumulated by humankind up to the moment–not limited to scientific knowledge, but all the baggage: non-fiction and fiction books, mythology, philosophy, all the human experiences in different areas fixed in the written or media form, including movies, music, art, and all the content of the Internet existing up to date. Surprisingly, here we find a completely different number distribution: I would assume that 95 points go to the library of the human mind's content, personal experiences, and the enormous variety of that information, including reflections and reflections of reflections and reflections of reflections of reflections. Only 5 points would be left to all the information about the physical aspects of the universe and their functioning.

This numerical difference demonstrates an astonishing distortion put on the actual observable information by the scientific point of view: science lowers the actual importance of the mind in reality and the necessity of its adequate presence in the descriptive models. The scientists are so charmed by their own model that they even currently discuss the TOE (theory of everything) as something that should be in the domain of physics, and the only last big problem in the way of that is the incompatibility of quantum mechanics and Einstein's gravity. However, as we showed in the previous chapter, the set of problems within the paradigm does not leave space for the hopes of any TOE that does not include the mind and does not explain the relation between the two most observed things in the Universe–mind and matter.

So, exactly this observation on reality is a cornerstone: reality includes matter and mind. And the clearly material object–brain–is as well clearly correlated with the mind content and activity. Therefore, the assumption that matter "generates" mind, and the hard problem of consciousness related to that assumption, is the cornerstone problem from all that we listed.

Besides the fact that there is no approximation to explaining how exactly the matter of the brain could cause the mind (hard problem), there is absolutely no reason to think that some "third thing" producing both mind and matter exists (remember our conclusions about reasonable theories above).

And, on the contrary, in the chapter dedicated to the "dissociation-association coincidence problem," we described in detail an extreme prevalence of dissociation in the sphere of the mind and the ability of this process to generate "objects" meaning, including "generating matter" as the specific set of features of the part of those objects. As the main examples here, we may take dreams. In dreams, we experience the creation of entire subjective realities, complete with objects, environments, and even seemingly autonomous characters, all originating from our own minds during sleep. Also, we normally experience a self (first-person character of our dream) perceiving that reality and its events. Both this character and all the objects of the dream are dissociated from their source—a dreamer. To the dream analogy, we will dedicate a special chapter below.

So, the summary of data: reality includes matter and mind; they are correlated; there is no observable way of how matter can produce mind; there is no reason to think that a "third thing" producing both mind and matter exists; there is an obvious observable way of how mind can produce matter. The most promising model emerging from that data set is that the mind produces matter by dissociation, or, more particularly, the "Mind at Large" creates all individual minds and objects (matter) by dissociation. How exactly this can be done will be described in the dream analogy chapter below.

Dream as Best Analogy

One falls asleep and sees a dream: they are she, and her name is Anna, and she is 46 years old, smart, and kind, a little anxious, and hungry in this early sunny morning. She teaches adults English. Both daughters are already in school, and Anna is making a cup of coffee to have an excellent breakfast with pancakes prepared the evening before. The sun reflects from the glass table, sending sunbeams onto the ceiling. She starts to hurry a little, and when she gets up from the table, she drops the towel on the floor. She gets to the taxi waiting downstairs, making just a little stop to pet the cat sitting near the stairs. In the office, she gives her first lesson to John, then three more to the other pupils.

One falls asleep and sees a dream: they are it, and it has no name, as it lives in the street. Today is a sunny day, and it is warm and pleasant. So good to sit in the sun. And this woman petted me. Good.

One falls asleep and sees a dream: they are he, and his name is John. Today, before going to the pool, he has his English lesson—he has made great progress recently, because this teacher, Anna, is great. After the lesson and the pool, John has little to do, so he goes home and falls asleep. In the dream, the tiger is chasing him, so it is good to wake up to the normal world without ridiculous imaginary dangers.

Anna is not aware of One (dissociated from it), John is not aware of One (dissociated from it), and the cat is even more not aware of One (dissociated from it). Anna, John, and the cat are like One. A cup with coffee, sun, and a towel are One. Laws for the sun light to reflect from the glass table, for the towel to fall are One. The events of the day are One.

Limitations of Analogy

Unlike John, Mind at Large (One) does not have "outside" to "wake up into"—therefore, everything inside its "dream" is real to the extent of "nothing to compare with." Dissociation is real, Anna is real, John is real, the towel is real, the coffee is real, and the cat is even more real. Only John's tiger is an illusion, meaning created by John's mind itself, compared to everything John's personal mind has nothing to do with.

Unlike John, experiencing only mini-John's mind in his dream, Mind at Large experiences all the minds of his "dream."

Without Analogy

Everything is mind. Everything is one mind. We may call it Mind at Large or whatever. Individual minds are dissociations; matter is a subprocess. Laws are inner mechanics. There is no "outside" for Mind at Large, no start or end, by definition. Here, information does not have any carrier—it is itself a "carrier" for everything. Information never goes anywhere.

Names For Model and Its Parts

To refer to the model we described further in this book and, if they ever happen, in conversations, I would suggest using shortened names for the model itself and its parts. First of all, Mind at Large (also referred to as One) can go as MAL, like Mel Gibson, but with "A". Such a name, I would claim, serves two purposes simultaneously: simplifying discussions and adding a bit of humor to the scene, as MAL is a truly tremendous phenomenon, but I would not like us to become too serious about any model or idea, including the idea of MAL.

Now, about the model itself—as it is "a model" and (which is the same) "a map", that goes with "m", thus for "the MAL model" or "map depicting world as MAL" will go as MALm.

As opposed to MAL, individual minds will be referred to as "individual/personal minds" or "dissociated alters" (borrowing this last one from Analytic Idealism) without any abbreviations, as well as dissociation and association themselves. A reminder: the "dissociation" and "association" terms are defined in this book previously: in this chapter and in the "Dissociation-association coincidence problem" chapter earlier.

Solution: Hard Problem of Consciousness

So, how does MALm solve the different problems we listed in the "Problems to Solve" chapter of this book? We will now go through them one by one, starting from the "cornerstone" hard problem of consciousness.

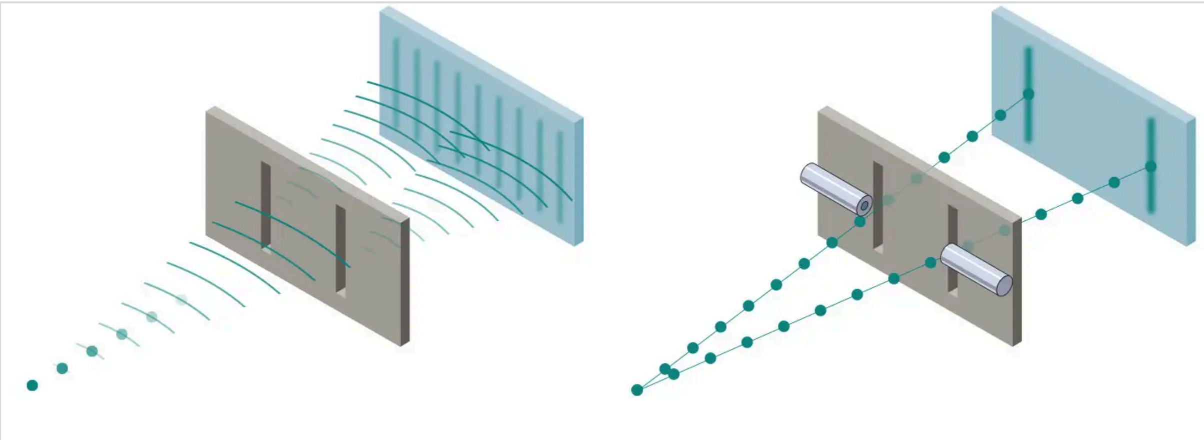
Coming back to David Chalmers' formulation: if experience arises from physical processes, how exactly does it arise? MALm destroys that problem completely with "no, experience and experiencing mind do not arise from physical processes." Instead, physical processes themselves are representations of specific mental processes within MAL. Or, in other words, matter is a specific type of experience of dissociated alters, having specific MAL's mental processes as its core "what it is."

To refer more specifically to the problem of correlation between individual mental processes and physical brains, in the first approximation, MALm depicts brains (as all the other matter) as representation; in this particular case—this representation is how your mental processes look when observed by an external observer in the physical world. Plus, the entire physical body is also the specific representation of the dissociated alter.

We'll refer to the problem of the relation of mind and body, and mind and brain, below, in the chapter dedicated to the solution of the "reverse correlation problem."

Solution: Double Slit Experiment

Under MALm, the reason for the "collapse of the wavefunction" is **observation**: interaction of the individual mind with the "potentiality information" within MAL. Meaning, all the matter as we know (experience) it is the result of interaction; none of the parts of the system "observer-observed" can be excluded from any experiments, and it is precisely what we see in these experiments. Also, these experiments clearly show what matter is without observation: it is information, potentiality in the form of a probability distribution. It is quite objective and exists independently of the individual observing mind, but it only gets values for all its attributes at the moment of interaction with this observing mind.



Solution: Quantum Entanglement

Under MALm, one extremely distant entangled particle instantly causes changes in the properties of its paired companion not because their interaction in space spreads faster than the speed of light, but because their connection is a connection "by sense" or "informational"—space has nothing to do with this connection. They are like two thoughts in your mind, associated by their mental meaning and content (having nothing to do with space again).

Or, in other words, space is the way of organizing the mental content of MAL, and, paradoxically, is another "thought" of MAL.

Solution: Delayed Choice Quantum Eraser

Does the present event actually change the past? Under MALm, yes, it does: the past is information (record) within MAL, and it can and should be rewritten on the go to be in correspondence with the current events (to make a perfectly consistent story). No matter how many individual minds observe information about the experiment, all their stories will be reconciled, as they are parts of the same web (MAL).

Important note here: no, the personal mind will never ever be able to change the past; from its perspective, the past is fixed and will always remain so for their very particular present.

Another important note: the delayed quantum eraser experiment does not tell anything about the "third" part of the timeline—the future.

Solution: Wigner's Friend

In the "Wigner's Friend" thought experiment, proposed by physicist Eugene Wigner in 1961, there are two observers: a Friend and Wigner. The Friend is inside a sealed laboratory with a quantum system (like a radioactive atom that can either decay or not). The Friend performs a measurement. According to standard quantum rules, the wave function "collapses," and the Friend sees a definite result (e.g., "The atom decayed"). Wigner stands outside the room. From his perspective, the laboratory is a closed system. Since he hasn't looked inside, quantum mechanics dictates that the entire lab—including the atom, the equipment, and the Friend—must be in a state of superposition. The conflict: to the Friend, the reality is "collapsed" into one certain outcome. To Wigner, the reality is a "superposition" where the Friend has seen both outcomes simultaneously.

Under MALm, there is no contradiction here. The state of the atom from the experiment is not a single or fixed one—it is relative to observers. It is the same as when dealing with speed: take the system of three objects—A, B, and C—each moving. Can we imagine that C's speed is 140 km/h relative to A, and 70 km/h relative to B? No problem with that.

As we showed before, a particular state of a particle is **relative** as well. It is "collapsed" in the case of Friend and "superpositioned" in the case of Wigner. As Wigner himself interacts with his Friend and the atom, their observations are informationally reconciled, and the output of the experiment will forever stay consistent for all observers comparing their observations (interacting with each other and the atom—all as parts of a reconciled single system).

Solution: Many-World Interpretation

The many-world interpretation (MWI) of quantum physics suggests that wavefunction collapse does not occur: the universe doesn't "choose" one path; instead, it splits, or "branches," into multiple, non-communicating versions of reality. This tries to resolve the "what is observer" problem and the "what is observation" problem.

Under MALm, the "what is observer" is not a problem (a conscious personal mind facing the information on probabilities), and the "what is observation" (informational interaction of the observer with the information on probabilities) is not a problem either. In this view, collapse is real; it is an interaction resulting not in "all possibilities coming into existence", but the opposite: "all possibilities except one are erased, making the observer and the observed one (and single) unified story, unified Universe."

Solution: Dissociation-Association Coincidence

Under MALm, the prevalence of dissociation in the sphere of mind is not an accident: this is a major mechanism of "reality creation"—the way in which unified Mind at Large splits itself into billions of distinct forms. At the same time, association remains the main "mechanics" for how the inner content of all these forms is built (for example, all personal mind-bodies seem to be constructs of associated, always-changing informational elements of different types and levels).

What was dissociated can be re-associated. Thus, there is no coincidence in that association of any personal mind with the elements of MAL happens in many ASC (altered states of consciousness), because it is what "altering" actually is: the base process is to dissociate and maintain that state for some time—altering is moving against the current—back to the associated state.

Solution: Reverse Correlation Problem

Previously, we provided the first approximation to describing the relations between mind and brain, saying that the brain was a representation: how your mental processes look when observed by an external observer in the physical world.

Now we need to consider the existing data that we discussed in the "Reverse correlation problem of consciousness" chapter of this book. This data says that reduced brain activity leads to increased and richer, highly organized and structured experience, and completely ceased brain activity leads to the most profound degree of that experience.

That data means we need to dismiss the first approximation, making it the "second", more precise one. It means that body and brain specifically are representations of dissociation itself—their reduction or full shutdown is the reduction or shutdown of dissociation, coming back to the initial state of MAL with no restrictions defined by dissociation.

Important note here: yes, the body and brain provide physical means to truly change the personal mind, as it is shaped by dissociation, and this dissociation can be manipulated through its physical form.

Solution: ASC Content Problem

Under MALm, there is no surprise in the fact that in many ASC (altered states of consciousness), the entire system gets access to the elements of objective information that do not belong to the personal experiencing mind itself.

Important note here: yes, experience in ASC is still experience of the dissociated consciousness, which, to a specific degree, continues to maintain its autonomy—thus, the experience is always (and expectedly, predictably) a mixture of what comes from the personal mind and what comes from MAL's content.

Solution: Better Name for "Platonic Space"

A series of experiments conducted by Dr. Michael Levin and his colleagues— from"foreign species' heads" to "Xenobots 1.0 and 2.0"—led us to the conclusion that DNA was not a source of information defining both biological body shape and related behaviors. To make sense of data, Levin himself used the concept of a Platonic space (often referred to as a Morphospace)—a conceptual, multi-dimensional map of all possible physical forms an organism could take. He used this term to describe where the "goals" of biological systems reside. He emphasized, however, that this "space" was less about mysticism and more about the mathematical and cybernetic limits of what a biological collective (like cells) can build.

Yet, however this information is called, it exists and precedes the complex life-creation process and is **accessible** to the newly "arising" biological systems. Under MALm (also having nothing to do with "mysticism"), this can be called in a simpler manner: just Information, specific "Knowledge" or "Specific Data Storage" within MAL itself.

The name "Morphospace," suggested by Dr.Levin, is a great fit to mark a specific realm of data within MAL. It is quite interesting to speculate about the possible non-static nature of this realm—its evolution and inner processing. In some sense, the continuing accumulation, processing, and change of information stored here is probably what we call "Evolution."

Solution: Agency for Systems as Data Processing Mechanism

An "Agentic Algorithm Experiment" conducted by Dr.Michael Levin and his colleagues in 2023 moves behaviors (minds) beyond the boundaries of the biological realm to **everywhere**. It demonstrates how behaviors (minds of different levels of complexity) can be linked not only to biological or even merely physical forms, but to any system with specific inner logic.

Let me remind you of the 3 major questions-problems that we derived from this experiment data and immediately provide brief answers to them from the perspective of MALm:

- **Where are behaviors (minds) and their links to specific data structures stored?** (**mind and inner structure storage problem**): under MALm, the answer is the same as for biological systems: Information, specific "Knowledge" or "Specific Data Storage" within MAL itself. This particular data shows that not only the biological shapes realm ("Morphospace") can point to corresponding behaviors (minds), but also any other entities having specific inner structures ("Structurespace").
- **Is mind attributed not only to biological systems?** No, not only. But we need to be extremely careful with the following notion: both behaviors and "underlying" minds differ greatly in multiple different dimensions. Here, the TAME (Technological Approach to Mind Everywhere)—framework developed by Dr.Levin himself- will be extremely useful to understand nuances. We'll dive into this deeper below.
- How exactly do specific inner structures get access to behavior patterns? Under MALm, there is no "distance" between the systems and the information they "consume"—they are both representations of different aspects of the same thing (MAL) and have no "access" or "interaction" problem between them.

As there is a great risk of a simplification of the "mind everywhere" idea, let's go deeper into Levin's perspective (grounded on observed data, mostly in the biological realm).

The Cognitive Light Cone

One of TAME's most famous concepts is the Cognitive Light Cone. This is a visual metaphor for the "size" of an agent's mind based on the goals it can pursue.

- **Small Light Cone**: A bacterium cares only about the immediate concentration of sugar in its local area (tiny reach in space and time).
- **Large Light Cone**: A human can care about the global climate 100 years from now (massive reach in space and time).

The TAME's Shifted Focus

Levin's framework shifts the focus from *what a system is made of* to *what it can do*.

Mind as Problem-Solving: Levin defines intelligence as the ability to reach the same goal by different means. He argues that cells "solve problems" in **Morphospace** (the space of possible body shapes) just as we solve problems in 3D space.

The Continuum of Persuadability: Instead of asking "Is it conscious?", TAME asks "How can I communicate with or control this system?"

- **Simple Systems:** Controlled by physical force or hardware rewiring (e.g., a clock).
- **Cognitive Systems:** Influenced by signals, rewards, or changing their "memories" (e.g., a dog or a group of cells).

Scale-Free Cognition: This suggests that "Selves" are nested. Your body is a collective intelligence of cells, which are themselves collective intelligences of molecular networks.

The Cognitive Light Cone Again (Nuanced)

Let's explore with TAME the "Tesla FSD" and "a seagull wandering along the beach". Instead of asking if a Tesla or a bird is conscious, TAME asks: "How much space and time can this system care about?"

1. The Seagull (Evolved Biological Agent)

A seagull is a highly sophisticated agent in 3D space, but its goals are relatively constrained in time.

- **Spatial Dimension:** Its light cone is medium-sized. It can navigate kilometers of coastline, remember specific nesting sites, and identify food from high altitudes.
- **Temporal Dimension (The "Now"):** Its goals are mostly short-term. It cares about its next meal, avoiding a predator in the next 10 seconds, or perhaps the survival of its current brood over a few months.
- **Cognitive Scope:** The seagull cannot care about "Seagull-kind" in the year 2090. It cannot worry about a storm hitting the coast next year. Its "stress" is tied to immediate, local survival.
- **Type of Intelligence:** This is **behavioral intelligence**—highly competent at navigating a complex, unpredictable physical world in real-time.

2. Tesla Full Self-Driving (Engineered AI Agent)

Tesla's FSD is a specialized agent that exists in a very different "shape" of light cone compared to the bird.

- **Spatial Dimension:** Its *active* light cone is quite small—roughly the 250-meter radius around the car covered by its cameras. It doesn't "care" about a traffic jam three towns over unless it's on the immediate route. However, through the **fleet**, its light cone expands; it "learns" from data collected across continents, giving it a ghost-like presence across a massive spatial scale.

- **Temporal Dimension:** On the road, its light cone is extremely narrow (milliseconds to seconds). It focuses on predicting if a pedestrian will step off a curb in the next 2 seconds. It has almost no "historical" goal-setting; it doesn't care where it went yesterday or its "career" as a car.

- **Cognitive Scope:** It is a "thin" agent. Its only "stress" (the gap between current state and goal state) is staying centered in a lane and avoiding collisions.

- **Type of Intelligence:** This is **narrow functional intelligence**. It is arguably "smarter" than the seagull at lane-keeping, but "dumber" because it cannot pivot its goals (e.g., the Tesla cannot decide to stop driving and go look for fish).

Summary

Levin would argue that the **seagull has a "thicker" light cone** because it manages its own internal physiological goals (hunger, temperature, health) and can invent new behaviors to solve problems.

The **Tesla has a "long but thin" light cone**: it is superhuman at a tiny sliver of reality (navigating roads) but is completely "blind" to everything else in the universe. If you took the Tesla off the road and put it on the beach, its light cone would collapse to zero, whereas the seagull would simply find a new way to thrive.

Important Note

As you can see above, systems "navigate" different "Dimensions" and can be compared by each Dimension separately, plus by the number of these "dimensions" (spaces) that the system can navigate in principle (which actually defines the **Type of Intelligence** that we described above).

Levin often adds an additional, more abstract dimension: **The Space of Possible States**. This is where TAME gets radical. It suggests agents don't just move in 3D space; they move in:

- **Physiological Space:** (e.g., a cell trying to maintain a specific pH level).

- **Morphospace:** (e.g., an embryo trying to reach the "goal" of a finished hand).

- **Transcriptional Space:** (e.g., a gene network trying to reach a stable state).

- etc.

The Comparison: The **Seagull** occupies many spaces at once. It navigates 3D space (flying), physiological space (regulating body heat), and social space. The Tesla is almost entirely flat—it only exists in the problem space of "Road Navigation."

TAME vs. MALm

Unlike TAME, which concentrates on "what systems can do" and "how to communicate with them", MALm is by definition the model that "cares" about **what systems are** and **what they feel (experience)**.

So, here's a breakdown of this:

Question	**TAME**	**MALm**
Is Tesla FSD conscious?	It does not matter. It can solve specific problems, and we can communicate with it.	We can assume that it is, but **in an absolutely different way** (by many parameters) than other systems we know.
What about occupied problem spaces?	Biological systems seem to occupy enormously more problem spaces than artificial systems.	Same.
Is Tesla FSD alive?	No, meaning it does not occupy or explore the problem spaces usually accessible to what is labeled as "alive."	Same.
Do forms (or inner structures) point to some behaviors (like an antenna receives some frequency)?	Yes.	Yes.
Is behavior (mind) 100% linked to the form or inner structure?	They are linked to the degree that allows communication with the systems, if we learn their language.	The more complex the system is, the less percentage of its content is defined by the form, the more possibilities of including content not directly related to the form arise (the "bigger box" allows for "putting more things in it")
Are there systems with "0" agency?	Yes.	Yes, but that does not mean they cannot be felt by observing conscious mind of any level, or do not have inner state that can be felt.

One last thought to add in conclusion: under MALm, the ability of minds to "scale up and down continuously" looks like a self-processing mechanism (the systems that recognize themselves, locally process the information available to them).

Solution: Time and Space as Completely Different Things

While Special Relativity depicts Reality without any Mind in it, under MALm this changes completely and obviously: the Reality **is** Mind at Large (MAL). This does not change or dismiss the data on which Special Relativity is based, but it changes the perspective and data interpretation completely.

Absolute and Relative Time

From the perspective of the Mind, time is completely different compared to space and is standalone by its nature:

- **The "MAL Time" (Absolute):** This is the master sequence of states of the whole system.
- **The "Relative Time" (Clock time):** This is the **Local State Update Speed**.

No Timespace

The "rendering power" of the MAL is intentionally limited to "c" and, within this limitation, is distributed between time and space. This limitation/distribution creates space and locality (nothing travels within space faster than "c"). The more rendering power is spent on position update, the less frequently the states of the local system update. "Moving" one is a system that actively (by force/acceleration) changes its frame of observation.

The clock slows down. Not because "Time" is a stretchy fabric, but because the **Information Throughput** is capped. You are literally "trading" state changes for spatial movement.

Solution: Mass as Information Density

While General Relativity depicts Reality without any Mind in it, under MALm this changes completely and obviously: the Reality **is** Mind at Large (MAL). This does not change or dismiss the data on which General Relativity is based, but it changes the perspective and data interpretation completely.

Absolute and Relative Time

From the perspective of the Mind, time is completely different compared to space and is standalone by its nature:

- **The "MAL Time" (Absolute):** This is the master sequence of states of the whole system.
- **The "Relative Time" (Clock time):** This is the **Local State Update Speed**.

Rendering Power

The "rendering power" of the MAL is intentionally limited to "c" and, within this limitation, is distributed between time and mass. The more rendering power is spent on mass (intense information) rendering, the less frequently the states of the local system update.

The clock slows down. Not because "Time" is a stretchy fabric, but because the **Information Throughput** is capped. You are literally "trading" state changes for mass rendering.

Solution: MALm, Math, QM, GR, LQG, KK, and String Theory

Under MALm, everything is Mind, "Thought(s)", "Idea(s), "Knowledge", "Information" in a very specific sense, most direct, literal, and maximally abstract, having no "physical" meaning or aspect on the deepest level (however, this still leaves the place for physical representation of the presence of this information). This suggests we can expect an enormous level of complexity in this information (and it is difficult even to think about the amount of this information, the links between its parts, and probably, extremely complicated and leveled processing mechanisms with different degrees of seeming autonomy).

Thus, it is not surprising to see limited theories describing different areas of this Information. It is also not surprising to meet situations when the same area of this Information is described with different levels of approximation.

The important new light that the idea of the **extreme complexity** and non-physical nature ("**absolute abstractness**") of MAL sheds is the one illuminating our cognitive process and its products (theories, models). And first of all, it is that MALm itself, and both Quantum Mechanics (QM) and General Relativity (GR)—as well as any other existing theories—are not the Reality, but descriptive models. For those of these models that use Math, the Math itself is a descriptive language, not the object being described and not some "reality in itself". Interpretations are the integral parts of those models, and should always be doubted based on the **non-local observation** principle, which is the following question applied to this interpretation: **how much data from other subject areas does this interpretation ignore**? And the rule is that the more facts are absent or not considered by the interpretation, the further it departs from the real state of affairs and the less aware it is of its own boundaries and limitations.

The requirement for this non-local observation roots in a simple thought: Reality is One. No models, excluding any known elements, can ever be named "Theory of Everything."

Let me show you these limitations for the listed models and descriptive languages:

- **Math** is the area of quantities; however long you look into any branch of it, you will never see any quality of experience ("redness of red") anywhere within any equation, rule, or calculation. Descriptive language (or "languages" if we consider different branches of Math) fundamentally ignoring this aspect of Reality gets its strict and powerful boundary. One more note here is about quality again: in Math, while concepts are defined in relation to each other, this is a circle with no qualitative output, meaning the only thing filling the Math with any sense is the Qualia introduced into Math with its Users, and this thing is external to Math itself.

- **GR** (and its friend **SR**) ignore the Mind, ignore the Biological Life, and contradict the model, super successful in predictions, and consistent with data - QM.

- **QM** contradicts the model, super successful in predictions, and consistent with data - GR.

- **MALm** by itself does not solve the contradiction of GR and QM. Moreover, for both Relativities (SR and GR), it suggests solutions/interpretations which are not obligatory and potentially not the only ones possible. I am specifically pleased to represent MALm (the core "product" of this entire book) in the list of the limited models, as its presence here directly points to two things: (1) it is a model, not a precise description of Reality; (2) it is not a closed system—rather an intermediate step and work in progress.

So, here comes a mind-clearing "**re-formulation exercise**" for some of our theories (that can actually be applied to any theory at all): "No, there is no spacetime—instead, it is useful to pretend that it exists to effectively describe dependency between time and space observable from the local perspective. No, time does not stretch, and space does not "curve"—instead, it is useful to think that they do so to build a coherent description with good predictive power. No, there are no particles—instead, there is an observed behavior of the environment that is convenient to describe when we pretend the particles exist. And no, this reformulation exercise has no end."

What MALm actually predicts is that, both for **solving contradictions** between theories (like GR and QM) and for **unification** (like an attempt to describe all observable forces and interactions within one theory), the same approach will consistently work again and again: movement into the direction of abstraction, introducing **new levels of abstraction**. This will always lead to re-thinking of basic assumptions of all or some of the previous theories (and corresponding re-defining of math). This will get us closer and closer to "absolute abstractness" of what Reality actually IS under MALm (all is Information). It is interesting to note here that while Information is maximally abstract by its Nature, it is also absolutely particular and Alive by its Content. The memory of the warmth and softness for a mother hugging her newborn baby is specific, particular, and absolutely non-abstract.

I would like to show a couple of examples of this **introduction of new levels of abstraction** that we already have in hand. First is Loop Quantum Gravity (LQG), the theory trying to marry two pillars of modern physics that usually don't get along: General Relativity (which explains gravity and the very large) and Quantum Mechanics (which explains the very small). Second is String Theory, suggesting that everything is made of tiny vibrating strings, and aiming to explain everything physical (gravity, light, electrons, quarks, etc.)

Introduced abstractions:

- **LQG**: **loops** or (later version) "**spin networks**"—mathematical graphs where there are nodes (points) representing discrete units of volume and edges (lines) representing the area of the surfaces connecting these volumes and creating a "**quantized space**" with a minimum possible length.

- **String theory**: clearly, strings—new "stuff" inside space. The theory says, "If you look closely at a point in space, it’s not a point; it’s a tiny **vibrating string**." It adds more complexity (requires 10 or 11 dimensions to make the math work).

So, from the LQG perspective, if you were to zoom in far enough—down to the Planck Scale (10{-35} meters)—you would find that space is granular. Think of it like a piece of mail armor or a digital photo:

- From a distance, it looks like a solid, fluid surface.

- Up close, it’s a network of individual, connected links or pixels.

And yes, Spin Network can curve (by introducing extra points and edges) and be in a state of superposition. Plus, when you add the dimension of time to a Spin Network, you get a Spin Foam. If a Spin Network is a 3D snapshot of space, a Spin Foam represents the evolution of that network over time. This treats "spacetime" as a history of changes in the network's structure.

The biggest "win" for LQG is its potential to solve the problem of Singularities.

- The Big Bang: In standard General Relativity, the Big Bang starts at a point of infinite density, which makes the math break down. LQG suggests that because space cannot be infinitely small (it has a minimum "atom" size), the universe didn't start from nothing. Instead, it may have "bounced" from a previous collapsing universe—a theory called the **Big Bounce**.

- Black Holes: Similarly, LQG predicts that the center of a black hole isn't a point of infinite crush, but a highly compressed region of quantum space.

So, how does all this solve the contradiction between QM and GR? In General Relativity, space is a flexible, dynamic fabric that curves under the weight of matter. In Quantum Mechanics, space is just a static, flat stage where particles play. When you try to combine them using standard methods, the math "blows up" into infinite values that make no sense. Loop Quantum Gravity solves this by fundamentally changing what we think "the stage" is made of. Here is how it fixes the conflict:

- **Background Independence (The "Stage" Problem)**: most quantum theories are background dependent—they assume a pre-existing, fixed grid of space and time exists. But Einstein's core discovery was that space-time is the gravitational field. LQG adopts Background Independence. It doesn't define particles moving through space; it defines the quantum states of space itself. By making the geometry of space the quantum variable, LQG removes the need for a "fixed stage." The "stage" and the "actors" become the same thing.

- **Removing Infinities (The "Smoothness" Problem)**: In standard physics, if you treat space as infinitely smooth (continuous), you can get infinitely close to a point. When you calculate the force of gravity at an infinitely small distance, the math produces an Infinity, which physicists call a "singularity." LQG introduces a Minimum Scale: Because space is made of discrete "loops" or "atoms," you cannot zoom in forever; there is a smallest possible volume ($\approx 10^{-105}$ cubic meters) and a smallest possible area. The "granularity" of space acts as a natural floor that keeps the math finite.

- **The Geometry of the Quantum**: In QM, everything is "quantized" (comes in discrete packets). LQG applies this logic to the geometry of the universe. Warping of space (from GR) is actually the result of adding or removing "links" in a Spin Network. By describing gravity as a network of interconnected quantum links, LQG allows the equations of General Relativity to be written in a language that Quantum Mechanics understands. It turns a "smooth" problem into a "counting" problem.

LQG radically transforms General Relativity. In Einstein's original version, space is a smooth, continuous manifold (like a sheet of rubber). LQG "refines" this by showing that if you apply quantum rules to gravity, the smooth sheet disappears, replaced by discrete geometry: area and volume are no longer variables that can be any number; they are restricted to specific, "quantized" values. LQG treats the core principles of Quantum Mechanics as "sacred." The Uncertainty Principle now applies to the geometry of space itself.

From the MALm perspective, LQG is elegant (no extra dimensions) and beneficial (solves the contradiction between QM and GR and moves us from Big Bang to Big Bounce, which is extremely important for Eternal Mind having no "start" or "end".)

However, while LQG might seem more elegant because it stays in 4 dimensions and uses fewer "new" ingredients, String Theory is often considered more "ambitious." String Theory tries to explain everything physical (gravity, light, electrons, quarks), whereas LQG is laser-focused on just fixing space/gravity.

Let's give some time to this version of the new level of abstraction introduction. Specifically, to the idea of **extra dimensions**:

- **The "Garden Hose" Analogy**: How could a dimension exist if we can't see it? The classic explanation (called **Kaluza-Klein theory**) uses a garden hose: from a distance, a garden hose looks like a 1D line, and an ant can only move forward or backward along it; up close, you see the hose has a circumference—it is actually a 2D surface. The ant can also move "around" the hose. That "circular" direction is a compacted dimension. If you were to shrink that circle until it was smaller than an atom, the hose would look like a 1D line even under a microscope, but the extra dimension would still be there, affecting how things move.
- **Why String Theory Needs Extra Dimensions**: In String Theory, particles aren't points; they are tiny vibrating loops of string. The math of String Theory is like a musical score. For the "music" (the laws of physics) to be mathematically consistent and avoid "anomalies" (math errors like $1=2$), the strings need enough "room" to vibrate in specific ways. Imagine a gymnast trying to perform a complex routine inside a narrow hallway. They would hit the walls and fail. To perform the routine (the vibration that creates an electron or a photon), the string needs 9 or 10 spatial dimensions. If these extra dimensions didn't exist, the strings couldn't vibrate at the frequencies required to create the particles we see in our universe.
- **Ancestor: Kaluza-Klein theory (KK theory)**: is the historical "grandfather" of the idea that our universe has more than three spatial dimensions. Proposed in the 1920s by Theodor Kaluza and later refined by Oskar Klein, it was a daring attempt to do exactly what modern string theory tries to do: unify the fundamental forces of nature using geometry. In the early 20th century, physics had two separate "languages": General Relativity (Gravity)—Einstein explained gravity as the warping of 4D spacetime (3 space + 1 time); Maxwell's Equations (Electromagnetism): Light and electricity were explained as fields moving through space. Theodor Kaluza wondered: What if electromagnetism isn't a separate force, but just another type of "warping" in a higher dimension? He sat down and wrote out Einstein's gravitational equations, but instead of using 4 dimensions, he used 5 dimensions. To his (and Einstein's) amazement, the math split perfectly into two parts: (1) one set of

equations was exactly Einstein’s General Relativity. (2) The "extra" part was exactly Maxwell’s equations for electromagnetism.

To summarize this chapter's "response" of MALm to the contradiction of QM and GR, we need just to repeat the main thought: contradiction solving and unification done by introducing the new levels of abstraction is a **movement to the understanding of Everything being Mind (Information)**, both maximally abstract possible thing and the most concrete ever by its content.

Another important add-on to this response: none of the theories listed in this chapter can ever claim to become the "Theory of Everything" because of two reasons: (1) non-local observation shows us that they tend to ignore the presence of Mind (except for QM, but it needs time) and Biological Life; (2) even if we include in some model or its interpretation all the existing elements (Physical Reality, Mind, Biological Life), the Reality seems to be an evolving process with no "final state to know", it potentially changes/updates itself providing us with "new things to know."

Solution: Free Will

The debate about Free Will has been taking place for thousands of years, with a core logical premise of "if a decision is determined by causes, there is no Free Will; if it is not determined by causes, it is random, and there is again no Free Will." Even Buddhism's "middle ground" claim about the presence of Free Will within some strict boundaries (your intent, your ability to choose whether to follow arising impulses or not) has a problem with addressing the problem of how exactly this "intent" arises itself, and how exactly the made choices are "free".

As MALm introduces the idea of One, this creates a great opportunity for Free Will to exist and be Deterministic without logical contradiction. All boundaries (your body, laws of physics, Karmic tendencies, constraints of your personality) are clearly deterministic. But, within those boundaries, there is a space for Free Will. How is the decision made there? Randomly? No. The only possible logical solution is that **every** decision of that nature (even the simplest one, like the decision of a seagull to continue to walk by the beach or to start flying) is **determined by all the complexity of the entire Universe** (MAL). As it is the absolute complexity threshold, saying this is absolutely equal to saying "it is **Free**, not determined by anything specific or any (even complex) set of causes, extracted from the full complexity of the Universe."

Further in this book, we'll explore the Advaita Vedanta's concept of Brahman and will disagree with it on the idea that Brahman is "complete" and "all knowing". We'll say that to have a reason to create a World of distinct forms, Brahman would need a motive, and then we'll disagree with the Advaita's understanding of Lila: the Divine Play out of abundance? Only abundance that feels incomplete is the reason to do anything at all.

Thus, we'll take Lila as the core thing to consider and put it in the name of this book itself, but we'll say that MAL (or Brahman, or One) **does not know itself** fully and wants more understanding, more knowledge of its abilities and possibilities: this is the foundation of all.

Thus, when any dissociated alter (within the boundaries) makes Free Choice, it **surprises**, **astonishes** MAL (Brahman, One): the choice is done out of all complexity of MAL (Universe) and after absolutely each choice the Novelty arises: MAL knows itself better, has truly new experience and new knowledge about Itself.

That is why, in this book later, the Block Universe concept will be called "worse enemy," and an alike attitude will be expressed towards the Superdeterminism model. The Curiosity, the openness of the system (we'll define absolute time as a sequence of inner states, past as information, and future as fully non-existent and non-predetermined).

MAL in action is Lila; they are curious; Free Will as the foundation of it.

Dialog with Other Models

This chapter is probably the most subjective and at the same time the most important one. Once we have MALm, we can talk via it with other models created by humankind throughout its history. In that dialog, the tremendous number of nuances can potentially come onto stage, along with contemplation and true emotion.

To Block Universe - Worst Enemy

The "Block Universe" (also known as eternalism or the block time model) is a philosophical and physical model of time which conceptualizes spacetime as a four-dimensional manifold containing all past, present, and future events simultaneously. Special and General Relativity (SR/GR) need this concept as a logical support system because due to them, there is no **"universal now".** In a Materialist framework, if something is "Real," it must exist *now*. In Newtonian physics, everyone agreed on what "Now" was. You could imagine a 3D "slice" of reality that existed all at once. But SR proves that two observers in motion disagree on what is happening "Now."

If my "Now" includes Event A, but your "Now" includes Event B, and there is no "Mind" to adjudicate which one is "actually" happening, then **both A and B must be equally real**. So, if they are both real but don't happen in the same "Now," the only solution is to say they are both sitting there in a 4D landscape, like two houses on the same street. The observer just happens to be driving past them at different times.

The famous thought experiment for that situation is known as "Andromeda Paradox": it imagines a person on Earth walking toward the Andromeda galaxy, and another walking away. Because of the relativity of simultaneity, for the person walking *toward* Andromeda, the "Now" might include the alien fleet launching an invasion. For the person walking *away*, the "Now" in Andromeda is still "two weeks ago," and the fleet hasn't even met yet. So, if you want to say that "Reality" is objective and physical, you are forced to conclude that the "launch" and the "pre-Launch" are both permanently fixed in a 4D block.

In the same time, in General Relativity, gravity is the curvature of spacetime. For a curve to exist, the "points" it curves through must exist. You cannot have a "curved path" if the destination doesn't exist yet. To treat gravity as **Geometry**, the entire "map" (Past, Present, and Future) must be laid out like a physical terrain, which is also satisfied by the concept of Block Universe.

In the "Solution: Time and Space as Completely Different Things" (linked to SR) and "Solution: Mass as Information Density" (linked to GR) chapters of this book, we claimed that presence of MAL (Mind at Large) as a singular and unified system allows introducing the concept of absolute time (sequence of states of the whole system—MAL) and local time (local state update speed). We completely acknowledged the "c" as obvious limitation connecting time and space, but interpreted it as a limitation applied to information processing in the specific parts of the system: the more rendering (or "information processing") power is spent on acceleration or mass (intense information) rendering, the less frequently the states of the local system update.

This view makes a Block Universe explanation redundant. You don't need the future to "already exist" in a block; you just need the MAL to know the rules of how to render the next state for each observer. The "universal limiter"—"c"—remains in this case the absolutely ultimate number—the limitation literally creates the reality by organazing it into space and connecting this space to the absolute fabric of reality—time.

That also allows us to come back to the discussion of the **motives for creation**. MAL is Mind; it has psychological motives (even if this claim is just an approximation), and the only reason to bother creating all the staff is **Curiosity,** which stands on two legs:

1. There is Free Will (for any Dissociated Self and MAL itself).

2. Because of that, the future is not pre-defined.

This completes the transition from a "Machine Universe" to a "Living Narrative." By identifying **Curiosity** as the prime mover and **Free Will** as the mechanism, we turn the laws of physics into the "Rules of a Game" designed specifically to produce **Novelty**. So, while in the Materialist SR/GR view, the universe is a "static" 4D object (everything that will ever happen is already "baked in;" in that world, Curiosity is impossible because there is nothing "new" to discover), under **MALm**, the universe is **Generative**: the future doesn't exist yet because it depends on the **choices** of the dissociated selves.

Quanta of Change and Big Bounce

We may also touch the following: as data processing requies quanta of information, it reveals itself everywhere: this, for example, mean that there was no Big Bang and the singularities will never happen (only Black Holes): istead of Big Bang we obtain Big Bounse and the Mind with no start or end, but with Cycles - so, about physical contants, I do not think they will change in "this story" (Cycle), but they definitely can change in the next one: the logic of endless learning tells: if I know everyghing now, what can I do else? I can explore the question: "what else can I be if I change some variables"?

The **Big Bang** is a Materialist nightmare because it requires something (everything) to come from nothing for no reason. In the MALm:

- The **Big Bounce** is the "Reset" button.

- When a Cycle (a story) reaches its maximum entropy—meaning the MAL has explored all the novelty possible within those specific "Physical Constants"—the dissociation collapses.

- The MAL "digests" the experience, learns from the outcomes, and begins a **New Cycle**.

And here is the ultimate expression of **Curiosity**. If the MAL is a learner, the "Universal Constants" are not random accidents. They are the **Current Cycle Settings**.

- **Current Cycle:** The MAL says, "Let's see what happens if I set c to 300,000 km/s and gravity to this specific strength."

- **The Result:** We get stars, planets, and Selves like us.
- **The Next Cycle:** Having learned the limits of this "Game," the MAL might tweak the settings. "What if I make gravity 10% weaker? Do I get a more complex life form or just a cloud of gas?"

This turns the "Fine-Tuning Problem" (why the universe seems perfectly set up for life) into a simple **Atheistic Design**: the universe is fine-tuned because it is the N^{th} iteration of a Mind that has been practicing for eons.

To Roger Penrose's Ideas - No Solution, No Revolution

In the "Hard problem of consciousness" (of "Problems to Solve") chapter of this book, we discussed, among others, "orchestrated objective reduction" (Orch-OR), or the quantum theory of mind, proposed by English Nobel Laureate in Physics and philosopher of science, Roger Penrose, and American anesthesiologist Stuart Hameroff [11]:

> "It states that consciousness originates at the quantum level inside neurons. The mechanism is held to be a quantum process called objective reduction that is orchestrated by cellular structures called microtubules, which form the cytoskeleton around which the brain is built. However, while the theory describes a physical mechanism—quantum processes in microtubules leading to objective reduction—Orch-OR doesn't inherently explain why or how these specific physical events should be accompanied by subjective experience, or, in other words, contains no solution to the "hard problem." The theory's authors attempt to fill this gap by proposing that the objective reduction (OR) events are not just random collapses of quantum states. Instead, they suggest that each OR event is associated with a fundamental "proto-conscious" experience or a basic unit of qualia. Yet, this proposal does not include any explanation of how exactly they are associated, and the explanatory gap remains. Penrose and Hameroff do not stop here and try to fill this next-level gap with further explanation: they utilize the proto-panpsychist ideas, suggesting that rudimentary forms of consciousness or experiential qualities might be intrinsic to the fundamental structure of reality, possibly linked to the geometry of spacetime at the Planck scale. However, this does not seem to eliminate the gap, because in that picture, both physical properties and experiential qualities are the intrinsic elements of reality, which is a very reasonable conclusion, as we observe both of these. However, the hard problem is exactly about the explanation of how these two things are related—no statement of correlation solves this."

As I hope we were able to demonstrate with this entire book, Orch-OR does not provide any solution to the Hard Problem, like any other model, theory, or experiment, based on the concept of a genuine distinction between mind and matter, and the solution pushed to life by the data and logic is to accept the idea that they are actually **not distinct** and the direction of causality is the opposite: from Mind to Matter, with the obvious mechanism of how it can be done (Dissociation) in the experiential data right in front of us—precisely in those huge experiential data, at which Roger Penrose and other physicists refuse to look as this is "invalid" or "anecdotal" data.

However, how exactly does this data become "invalid"? Or is it truly "anecdotal"? It is absolutely not: the data on how the human mind (and mind in general) functions is not limited to observation of dreams or altered states of consciousness (ASC). Even if it were, nobody asks stubborn physicists to look at the "content" of these experiences, but just at their metastructure. Can we see how the mind creates seemingly autonomous objects, all originating from the mind itself (during sleep) or mostly from it (ASC)? Do we have any ability to ignore the evidence of an extreme prevalence of dissociation in the sphere of the mind? Or even a stronger question: do we have the ability to ignore the mind itself? There are around 8 billion people on Earth, and we have 8 billion of 8 billion cases of the phenomenon's representation. I cannot imagine "less anecdotal" data at all.

Here, it is also necessary to touch on the "interesting" idea that something that cannot be measured cannot be taken into consideration. And I will allow myself a naive question here: why not? Does the absence of the ability to measure something with the tools at our disposal tell us "the phenomenon does not exist"? Then my claim is the following: we have no chance of measuring any phenomenon, including those that we call "physical." Even in those phenomena, we only measure some characteristics of them ("length", "velocity", "mass", "energy"). And if we measure the amount of energy, but do not anyhow "measure" the energy itself, why does it not push us to the conclusion that the energy is anecdotal?

The consciousness-related Orch-OR, however, is not the only space of intersection of this book's main topic with the ideas of Roger Penrose. Another thing I want to have a brief dialog with is Penrose's solution to different problems surrounding the double-slit and quantum eraser experiments in Quantum Physics, such as the measurement problem and the problem of collapse of the wavefunction. Roger Penrose's theory of Objective Reduction seeks to find a physical, gravitational reason for the wavefunction collapse in a macro-world. At the same time, this theory leaves untouched any problems seemingly lying in a micro-world itself, where the author claims that what we observe is not a "true collapse", but rather a decoherence (a physical process where a quantum system becomes irreversibly entangled with its environment—air molecules, apparatus atoms, heat, etc.)

However, in the "Delayed choice quantum eraser experiment" section of this book, we already demonstrated with all possible precision how severely this experiment data weakens any interpretation that attributes the "collapse" phenomenon to simple, irreversible physical interaction with the local environment. Thus, for the microscopic world, Penrose's theory does not offer new solutions or explanations for any existing anomalies. At the same time, the theory somehow manages to be called a "revolutionary," while it would be better called a "huge distraction."

To Materialism - Put Back in Its Boundaries

In a way, this entire book is a heavy dialogue with Materialism, our current "king," the leading paradigm defining what is considered to be science nowadays and directing the scientific research. In the "Problems to Solve" chapter, we started with the model of an American philosopher of science, Thomas Kuhn, seeing the history of science as an episodic model in which periods of conceptual continuity and cumulative progress, referred to as periods of "normal science," are interrupted by periods of revolutionary science. The discovery of "anomalies" accumulating and precipitating revolutions in science leads to new paradigms. New paradigms then ask new questions of old data, move beyond the mere "puzzle-solving" of the previous paradigm, alter the rules of the game, and change the "map" directing new research. We then used Kuhn's concept as a tool for exploring anomalies, currently presented under the Materialistic paradigm. We found and described 12 significant anomalies and considered that as the absolute evidence that Materialism cannot be used as a satisfactory descriptive model for Reality.

An important note is that while each of these problems is significant by itself, the real problem for Materialism lies in the fact that those anomalies together form the system, being simultaneously from different spheres of human knowledge (quantum mechanics, hard problem of consciousness, research of altered states of consciousness, biology and information, special and general relativity), and deeply interconnected and enforcing each other.

Both in the areas of quantum mechanics and in intellectual battles surrounding the data collected from different altered states of consciousness (ASCs), we clearly saw, in how the debates were conducted, a bright illustration of Kuhn's thought that new paradigms rarely win the minds of old scientists, and rather wait while they die. We also saw how people defending their worldview easily violate their own methodological principles when they fear their model is under threat. The brilliant example of that is the violation of Occam’s Razor principle of "keeping it simple" in the case of NDE (near-death experience), specifically in explaining reported experiences during cardiac arrest (we explored that in the "Reverse correlation problem of consciousness" chapter).

While Occam’s Razor states: "When presented with competing hypotheses that make the same predictions, one should select the solution with the fewest assumptions", in the case of cardiac arrest experiences we get a tremendous explanation that experience is "generated by the mind" not at the moment of absent heart and brain activity, but later, when the brain resumes functioning, and memories are retrospectively attached to the time of non-functioning due to psychological reasons.

Another great example of defensive mechanisms (instead of science) at work is the "shut up and calculate" approach regarding interpreting the data of quantum mechanics.

Frankly, even the division of science into areas (very practical and necessary) starts looking like a defensive mechanism for the paradigm, when scientists do not look at wider data, trying to build a model of Reality (TOE, "theory of everything") exclusively out of "their department." This book aims to demonstrate that no physical theory ever will be "TOE," because "E" ("Everything") includes Mind (Consciousness). Thus, if one tells "this is not my department," one should forget building a "Theory of Everything" as well as very accurately review philosophical assumptions lying in the foundation of their model of the world. Often this careful consideration may show that there is no true connection between your data and absolutely redundant assumptions about the World, not anyhow coming out of that data.

Regarding Materialism, I actually have a question related to Kuhn's idea that when a paradigm (for example, Materialism) faces the weight of accumulated anomalies, it falls, and science, along with society, shifts to something more profound and useful. However, we have been seeing many interesting people saying things about Materialism coming to its end: Terence McKenna, Stanislav Grof, Fritjof Capra, Bernardo Kastrup, and Federico Faggin have been talking about that for 70+ years. We also have the outstanding work of

Carl Jung, and a lot of unorthodox thoughts from the founders of quantum mechanics. But in spite of all the talk and a really huge set of accumulated anomalies (depicted in this book), Materialism stands—the materialist (or physicalist) paradigm remains the "operating system" of modern science.

There are several aspects to be considered here. First, as Kuhn noted, the vast majority of scientific work is "Normal Science." This is puzzle-solving within an established framework. And we must acknowledge that materialism is an exceptionally effective "puzzle-solving" engine. It allows scientists to ignore deep metaphysical questions about the "nature of reality" to focus on the immediate, practical tasks of manipulating matter and energy. For a young scientist, adopting a career-risking alternative worldview (like Idealism or Panpsychism) often lacks a functional methodology. If the current paradigm allows you to build a transistor, design a drug, or launch a satellite, the "anomalies" (such as the hard problem of consciousness) are often categorized as someone else's problem—usually that of philosophers.

Second, a paradigm is not just a set of beliefs; it is a social and institutional structure. Graduate education is an enculturation process. Young scientists are taught how to think by practitioners who were themselves trained under the materialist paradigm. Challenging the foundational assumptions of one's field is rarely encouraged in the early stages of a career where success depends on securing funding, peer review, and tenure. The gatekeepers of science—journal editors and grant committees—operate within the existing paradigm. Research that relies on non-materialist axioms is frequently dismissed as "unscientific" or "metaphysical" before it even reaches the stage of empirical evaluation.

Both of these aspects, however, were presented in cases of old paradigms Kuhn described in his work, and were still overcome. So, what do we actually deal with? Is the "crisis of Materialism" not critical enough? I would say so. To the degree of saying "there is no crisis." For a paradigm shift to occur, the anomalies must not only exist; they must create a crisis that renders the "Normal Science" of the current paradigm impossible to perform. While there are anomalies regarding consciousness and quantum measurement and many others, they haven't yet reached a point where the utility of the materialist paradigm is paralyzed.

We may suggest that until a new, post-materialist framework provides a better set of tools for daily laboratory work or achieving practical results, the institutional momentum of materialism will remain immense. And here we must acknowledge the strange thing: from the utilitarian point of view, Materialism is much better than any form of Idealism (including MALm of this book). We can formulate it like this: **Materialism is wrong, but it works; Idealism is correct, but it does not**. Materialism does build devices, cure diseases, and calculate orbits, and it does not require a perfect metaphysical foundation. It only requires a functional one. Idealism, on the contrary, throughout the years, has not produced anything valuable (usable): the rare and legendary achievements of very special and rare individuals by their nature stay localized in them and do not transform into any reproducible technology (and it gives us a very interesting definition for "technology" as something that can be used without any underlying understanding).

We can assume then that the transition from Materialism will likely not happen because "Materialism failed." It will happen because it will eventually become more efficient to think in other terms (if it ever will). Once a new, non-materialist framework starts producing better technology and faster breakthroughs, the migration will happen naturally. Scientists won't abandon materialism because they were "convinced" by philosophy or data; they will abandon it because it has become an obsolete, clunky tool. Just simpler explanations for the anomalies (like this book provides) will never break materialism as a paradigm.

However, here, we are going to do one more time what this entire book does: we'll step out of the boundaries of local perspectives by paying attention to wider data and making non-local observations. In this case, we'll do that with Kuhn's concept of paradigms itself. So, if we take a step back and look at the idea, we may ask ourselves: "What is it about?" And the answer is that it is unexpectedly about the history of science. So, what does science do? It describes how some aspects of the world function. And, at the same time, science itself, with the huge amount of voices representing it, says "we do not do metaphysics" (that is, we do not explore "what the World is"). And that's a true elephant in the middle of the room. The materialistic paradigm (ruling in science) **does not need to change**. It only **needs to know its boundaries**. Materialism is good at explaining how some elements of the World work and interact; it is absolutely blind and helpless, even harmful in explaining what the World IS. And if "science" itself says, "We do not do A, B, C; it is not a subject of science," then please keep your own flag high and do that consistently. And let's be clear: the boundaries for Materialism are set within the data, and they are set by Materialism itself by dividing this data into what "we want to deal with" and what "we do not want to deal with."

While this entire book is about those boundaries, let me highlight them once again here. Materialism does not describe or understand Mind (Consciousness) and its relation to matter. Thus, it is an extremely inappropriate ground on which to grow any theories about Meaning and Death. Because of not understanding the relations between matter and mind, all neurological (by brain or nervous system) explanations for the laws of mind have absolutely "0" value, being by its nature the attempts to explain why the left side of the car moves by the fact that the right side moves. Materialism, science based on it, and moreover, particular disciplines of this science can never build or even talk about the TOE (theory of everything); all it can talk about is UTOS (unified theory of something). I want you to pay special attention to the fact that Materialism constantly violates its own boundaries: it does not allow wide data "in", but it again and again goes "out" to build "the model of the World." Our short thought here: it cannot do that and **should not be allowed**. Not for the sake of "fair play," but just to stay sane.

Let's put it in different wording: if you have a powerful mind, a sincere curiosity, and a will to understand the World, **look at the wide data and work on it with logic,** and **do not do science**. Or, in other words, doing science and looking at data are different things now.

And now, all of a sudden, let me "turn weapons" against Idealism (Buddhism, Advaita, Kashmir Shaivism, MALm itself). There is an extremely great flaw in Idealism: it does not produce any **results**. I mean very literal things, like ones called "siddhis" in such traditions as Hinduism, Buddhism, and Yoga. Prapti? (the ability to access or teleport to any place at will) - cannot see it. Prakamya? (the ability to realize any desire or fulfill any wish) - cannot see it. Isitva? (sovereignty or control over all material elements and natural forces) - cannot see it. And yes, we're all very well aware of a consistent theme across most traditional texts—"danger of siddhis". Siddhis are frequently described as "milestones" or "signposts" rather than the final goal. They are often viewed as potential distractions or obstacles to the ultimate goal of enlightenment or liberation. And yet, thousands of "talking heads" never stopping discussion of liberation or stating their own enlightenment, consistently do not demonstrate passing through those "signposts," which makes very plausible the idea that they do not actually move anywhere. There's an entire Idealistic tradition that demonstrates the presence of meta-description of the "processes" instead of a truly manageable system with clear variables, connections, and timing (Buddhism; we discuss that in the "To Buddhism - Mechanics of Reality" chapter.

I think the only true result of Idealism is the change of attitude to what can and should be explored (Mind, Mind at Large, Consciousness). But the idea that something should be explored, and the exploration itself **giving perceivable results,** are not the same thing. In other words, the claim "we proved logically and based on data that Materialism is wrong," is not the "end" of research where you can sit in peace as you "achieved understanding"—on the contrary, it is the **start** of responsible and hard research in an area, completely different in size. There can be profound results which are difficult to predict or observe objectively, but not observing "signposts" (such as siddhis) is very important "intermediate" data.

So, what have we put on the pedestal? The idea that Materialism should stay within its boundaries, never trying to grasp Meaning, Death, or Free Will, and leaving the possibility for the independent mind to explore these areas with data much wider than the Materialism is "allowing" to be observed. However, within its boundaries, Materialism remains an effective and result-producing system of views, describing pretty well the functioning of the specific parts of the World.

Probably, the last question to think about is whether Materialism can be beaten **within** its boundaries. As I've already said before, this probably can be done by some other model only by **giving perceivable results**, such as siddhis or, even more important, other results not directly linked to personal abilities (some kind of new type of technology). What comes to my mind as a probable example are the "Silfen Paths" from Peter F. Hamilton's Commonwealth Saga: inter-dimensional or fold-space corridors that act as a network of trails connecting various planets across the galaxy, not physical roads in the conventional sense, but rather stable paths that allow travelers to walk from one world to another. Unlike the human transportation network, which is rigid and reliable, the paths are transient and fickle. They shift, they open and close based on criteria no human can fully grasp, and they do not always lead where a traveler intends. While Hamilton himself does not link those paths to Consciousness, let's assume for our purposes that they were actually created by **understanding** that Silfens possessed about the nature of reality and (in this case) space - understanding actively applicable and changing how the physical reality itself behaves.

I think only this level of understanding with perceivable results can beat Materialism within its own boundaries, making its working technologies out-of-date and of super-low efficiency compared to the new ones.

To Superdeterminism - Stopped by Hard Problem

In the previous chapter, we have already stated the necessity for Materialism and its scientific method to stay within its valid boundaries. Superdeterminism is one of the very clear attempts of the Materialistic model to preserve itself by violating this precise boundary.

In the "Quantum entanglement" chapter, we overviewed the corresponding phenomenon and claimed it as one of the major challenges to the Materialistic paradigm. We mentioned a 1935 paper by Albert Einstein, Boris Podolsky, and Nathan Rosen, marking the phenomenon as impossible because it seemed to violate local realism. We also mentioned that later, by John Bell starting in 1964 and others, the mathematical model was created showing that the presence and absence of hidden variables produce different predictions. Between 1980 and 1982, the first experiments, followed by a number of subsequent ones, were conducted to provide the actual data matching one of these predictions—the prediction that there were no hidden variables. That meant that quantum entanglement was real and theoretical attacks on it failed. Being real, it remained a huge problem for Materialism.

However, later a view called "Superdeterminism" arose to attack the core assumption of the initial Bell's theorem itself. If the assumption was that the experimenter was free to choose the settings of their detectors, and that this choice is statistically independent of the hidden variables—the underlying "state" of the particles being measured. Superdeterminism suggests that this assumption is false. It posits that the particles and the measurement apparatus (including the experimenters themselves) all share a common causal history, reaching back potentially to the Big Bang. Under this view, there is no "free" choice; the setup of the experiment and the state of the particles were correlated from the start.

In doing so, however, Superdeterminism becomes an attempt by science and Materialism to go beyond the boundaries they set for themselves a long time ago. Historically, science followed the Cartesian split: divide the world into res extensa (extended things/matter) and res cogitans (thinking things/mind), then discard the latter to focus on the former. Superdetermenism carelessly and casually puts "mind produced by brain" onto the stage: this cannot be done - it is a violation of science's self-set boundary (suddenly it goes into the "mind" area) with "0" ground to do that, as the Hard Problem of Consciousness is still in place with no approximation to any solution.

Thus, we have a situation where some model (Materialism) and its practice (science) pretend to defend Realism, while actually it defends itself (model). At the moment, science has "0" means in its hands to bridge mind and matter, which destroys any chances for Superdeterminism to survive. Moreover, we need to remind ourselves again and again that science that puts on itself the limitation "we will not consider mind" has no chances of creating any satisfactory model of reality, as mind (an absolutely non-physical "thing") is obviously part of Reality.

To Panpsychism - Reconciled

Panpsychism is the philosophical view that mind, or a mind-like quality (such as consciousness or subjective experience), is a fundamental and ubiquitous feature of reality. In a sense, we have a huge language problem here, as MALm also sees Mind as "a fundamental feature of reality," but in an absolutely different sense.

Many versions of panpsychism (specifically Constitutive Panpsychism) seek to explain human consciousness as being "grounded" in the micro-experiences of fundamental particles. The goal is to show that fundamental particles have "proto-conscious" experiences, and when they combine, in specific physical arrangements, it creates the "big" consciousness we experience. In this sense, panpsychists view consciousness as a hierarchical "bottom-up" structure. MALm definitely sees Consciousness as "up-bottom."

However, Panpsychism looks very well aligned with the data from biology—in this book, its "Mind and form storage problem" chapter, we discussed the experiments conducted by Dr. Michael Levin, his colleagues, and team, showing how cells of the organisms clearly demonstrate a combination of their functioning and behavior, seemingly showing the direct solution to the Combination Problem of Panpsychism. The data tends to show that, at least in living organisms (cells), the combination is a direct physical connection between the nervous systems of the cells.

We need to clearly see, however, that neither Panpsychism nor the data from Levin's experiments solves the Hard Problem of Consciousness which turns into two problems: (1) quality problem (or "quality combination problem"): how does a collection of simple experiences (like the "experience" of an electron) transform into a complex, rich, qualitative experience (like the feeling of smelling a rose or hearing a symphony); (2) why proto-experience itself exists and how exactly does it correlate to physical object?

It is important to mention that Levin's set of experiments leads to an interesting conclusion from the authors, saying that specific combinations of alive matter create "pointers" ("antennas") tuned to get "signal" of pre-existing behavioral ("mind") patterns, rather than "creating" this behavior.

An enormous strengthening comes to Panpsychism from another of Levin's experiments that we described in the "Agentic algorithm experiment" chapter: there, the elements of "conscious behavior" are modeled in a completely and clearly "non-living" system—a sorting algorithm.

Dr.Levin has his own extraordinary model explaining this and other experiments—TAME (Technological Approach to Mind Everywhere—a conceptual framework designed to understand and manipulate "minds" or cognitive agents across all types of substrates—not just humans and animals with brains, but also cells, tissues, synthetic bio-bots, and even AI.

How does MALm address those data and ideas? MALm is truly "up-bottom", saying personal minds are created by "dissociation" from the large One. But it never describes the mechanics of this "dissociation" and data organization within the dissociative boundaries. Those mechanics are covered by the "association" of mental and physical elements (see the "Dissociation-association coincidence problem" chapter). This can be considered as the following process: as physical structures combine to point (potentially) to more complex behaviors and minds, they become potential containers for specific content (like the human mind), with subsequent putting into this container the corresponding complex structures (memories, behaviors, collective patterns, previous "history" of the process (Karma), and so on). This deeply intersects with Buddhism's view of the personal self as an ever-changing and evolving combination of elements (see the "To Buddhism - Mechanics of Reality" chapter).

So, if so, why then, in the "Not wrong, reasonable, and with most explanatory power" chapter, do we claim that Panpsychism is not the best explanatory model, having limited explanatory power, and prefer MALm in the end?

This entire book is the answer to this question: when we utilize the non-local observation and try to look at the totality of data, we obtain at least 12 problems to be solved: MALm addresses all of them as we showed in the "Nature of Reality" chapter, while Panpsychism addresses two of them successfully (Levin's data), one unsuccessfully (Hard Problem), and the remaining 9 are not touched at all: quantum physics data, ASC (alterend states of consciousness), including NDE (near-death experiences), Special and General Relativity.

So, as we claimed before, from two models with different explanatory power, the one with more of such power is clearly preferable and should be selected.

To Advaita Vedanta - Non-Perfect Brahman

It is difficult not to notice that MALm is a non-dual model. There are obvious intersections of this model with Advaita Vedanta, a major school of Hindu philosophy that centers on the concept of non-duality too (the literal meaning of advaita is "not two" or "non-secondness") According to Advaita Vedanta, the Ultimate Reality (Brahman) is the infinite, formless, unchanging, and singular consciousness that underlies all existence (which is 1-in-1 the definition of MAL (Mind At Large) in our model). In Advaita, the individual being (the person who thinks, acts, and experiences) is represented by Jiva; its perceived separation from the divine or from one another is a result of ignorance (avidya). In MALm, this ignorance is called "dissociation" (forgottenness, not remembering), which is again equal to Advaita's concept.

We are risking, however, to fall into all kinds of intellectual difficulties if we try to address the concept of Atman, central in Advaita. MALm does not see anything that can be called "Atman" or any specific usage that can be attributed to this extra concept.

This does not end the list of differences in view and understanding of data between Advaita and MALm: three other things that distinguish the two models are the interconnected phenomena that are called in Advaita "Lila" ("Divine Play"), "Maya" ("Illusion"), and "Moksha" ("Liberation").

"Lila" is a fundamental concept in Hindu (not only Vedanta or Advaita Vedanta) philosophy used to describe the nature of reality and the cosmos: the creation, maintenance, and dissolution of the universe are not driven by necessity or a specific goal, but are instead a spontaneous, joyful "play" of the Divine (Brahman). Reality is seen as the creative, effortless expression of the Absolute. Everything in the world is part of this cosmic performance. In Advaita, Lila is the only logical explanation for why the Absolute—which has no needs, no wants, and no lack—would manifest as a world of form, change, and division. This explanation finds itself in a very precise harmony with what we find in data collected from observation of ASC (altered states of consciousness), specifically made by Stanislav Grof and his colleagues—one of Grof's most famous books is called "The Cosmic Game," which is one-in-one correspondence to the "Lila" concept of Advaita. But let's be clear here—the explanation of why a fully self-sufficient and balanced entity (Brahman) cares to do anything at all except just being balanced and self-sufficient—well, the explanation of that with abundance is a very poor and weak intellectual exercise, as abundance leading to creation of forms is a kind of imbalance itself. And there is actually nothing problematic about the Absolute being not perfect or lacking something (and thus having curiosity and reason to act)—it is only the specific branch of human knowledge and understanding of the Absolute (Advaita) having a problem with that in its desperate attempt to make the Absolute match its view of what the Absolute must be.

And therefore, we may ask a question here: are Advaita (as a system) and the concept of Lila actually fundamentally uncomfortable roommates? And in a little investigation following this question, we may surprisingly find that Lila is often treated by Advaitins as an "embarrassing" relative they try to keep in the attic: Lila is often relegated to a "lower level" of truth (Vyavaharika)—if the world is truly Lila (Play), then the world is meaningful. If the world is meaningful, then the Absolute is a "Player." If the Absolute is a Player, then the Absolute is doing something. And if the Absolute is doing something, then the "stasis" of Brahman is violated.

The second thing to deal with is "Maya"—the concept of cosmic illusion. It is the creative power that causes the universe to appear as it does, masking the true, underlying reality. You need to be immediately very careful and attentive to what has just been said here: "true, underlying reality" immediately splits the reality into the Absolute and the Relative world. What is that? Do we have dualism in the core of the non-dual model (Advaita)? Above in this book, when discussing the "dream" analogy for the Mind at Large (Brahman) and the limitations of this analogy, we mentioned that the One (Brahman) did not have "outside" to "wake up into"—therefore, everything inside its "dream" was real to the extent of "nothing to compare with." The world "illusion" in any meaningful sense, we stated, could only be attributed to what was produced by the individual mind (Jiva) itself, like a tiger that John-Jiva sees in his dream as opposed to the tiger that John-Jiva meets in the savanna (see the "Limitations of analogy" chapter of this book).

So, the MALm's response to Maya is "no." The world is real to the degree of nothing to compare with. And, of course, we understand that Advaita got used to hearing this type of response and is very well armored for the counter-attack. In the most famous analogy used to explain Maya, in dim light, a person may mistake a piece of rope for a snake and experience genuine fear. The "snake" is an illusion—it does not exist in the rope—but the fear caused by the illusion is very real. Once a light is turned on (gaining knowledge), the snake vanishes, and the rope is revealed. Advaita says the world functions the same way: our ignorance causes us to see a world of separate objects where, in reality, only the "rope" (Brahman) exists.

However, MALm's response to this is: when the light is turned on (gaining knowledge), the rope (your concept) vanishes, and the snake (Brahman) remains intact along with the fear that One (Brahman) passionately desired to experience through you-Jiva.

All we said above about Lila and Maya hit hard on the last thing—"Moksha" ("Liberation"). If the World is a joyful and curious "play" of the Divine, the state the divine chooses to be in, and if this play IS Divine along with all Jivas, and not an illusion in any sense of this word, then being here and now is a highest possible state: there is nowhere to go, there is no reason to leave the game as it is already "divine." This means, MALm is opposed to Advaita and actually is proposing a philosophy of Maximum Intensity. If the One desires to experience the "you-Jiva" in all its vividness, then to attempt to "wake up" and see through the "illusion" is to effectively sabotage the very purpose of your existence. The world is extremely **meaningful** as it is, and it IS a high state of the Divine.

So, in a sense, MALm sees Lila as more fundamental than Brahman itself, as Lila is actually the core of Brahman, and it is actually **Brahman that we understand**. Brahman is Lila, the World is Lila, and therefore Brahman is the World (real, not ever an illusion, as real as only

Brahman can be). Here arises the question: if MALm is singing its song in favor of life here and now, its complexity and intensity, its beauty and a deep meaningfulness, the model calls us back to the World itself and puts the big question mark above the "Liberation" (exit) idea of Advaita, how should the suffering be navigated, which is obviously a part of that World and seems like a very valid reason to try to "liberate" oneself from the attachment to the World and from the World itself. This is a completely valid and important question. It would be cruel to talk about the Reachness or Playfulness to the person who is facing the dark side in encountering evil or suffering. And the answer is, I guess is an Abundance and Impermanence themselves: the pain will pass, the bad things will end, the loss will turn into new meeting with what was lost and unification, the death is not the end, the deceased loved ones will re-join with you, the new understanding will come, the calm will follow, the choice will be yours some day again.

The MALm makes a pivot from a philosophy of transcendence (the Advaitic exit) to a philosophy of cyclical redemption (the MALm homecoming). While the nature of suffering in that view switches from a "flaw in the matrix" to a "rhythm of the dance," it remains absolutely real. In the Advaitic view, impermanence is the very proof that the world is an illusion—if it doesn't last, it isn't "real." In MALm, impermanence is the engine of the "Play" and is never a reason to say any impermanent thing is not real. In a world where you are the One experiencing yourself through the Jiva, suffering is not a trap; it is a temporary orientation. If the One is truly infinite and abundant, then no Jiva is ever permanently lost or permanently broken. This empowers the individual to endure the "dark side" not by pretending it isn't there, but by trusting in the fundamental resilience of the Reality they are participating in.

MALm creates a robust, albeit intense, framework for existence. It asks for a tremendous amount of trust in the process of being. It suggests that the "dark side" is not a mistake that needs to be "corrected" by becoming an enlightened monk, but a passage that needs to be walked through to reach the next horizon of the play. And complexity and depth do not promise that this passage will be easy.

Under MALm, evil is the manifestation of the same abundance and the way of making the game valuable and extremely meaningful. But this can never be said to justify it. And it can never be said to one suffering from it. And it is useless to the one doing it, as there will be consequences, no matter what.

So, my friends, look and see: the World is beautiful and meaningful. But if it is a hard time for you now, if you suffer, I hug you, sit silently with you, and cover you with what MAL is, I am here now with you as another human being, knowing the suffering, but I am saying you: suffering will pass, not because you forget or detach, but because you'll have back what you have lost, you'll rejoin with the loved ones, and the pain will vanish.

Meanwhile, Joy and Beauty are the opportunities residing in every space that is not occupied by evil or suffering: in every second that feels just "normal" or "neutral": in a casual tasty meal, in swimming in the sea, in playing the ball, in staring at the sky, in reading the book. This "Neutral" is the Sacred. You are Lila too.

To Buddhism - Mechanics of Reality

> "People who had experiences of the Absolute that completely satisfied their spiritual quest typically saw no specific figurative images. When they felt they had reached the goal of their mystical and philosophical quests, their descriptions of the highest principle were extremely abstract and strikingly similar. All who spoke of such an absolute revelation were united by a striking similarity in the experiential characteristics of this state. They spoke of an experience of the Supreme that transcends all the limitations of the analytical mind, all rational categories, and all the narrowness of ordinary logic.
>
> This experience was not constrained by the usual categories of three-dimensional space and linear time familiar to us from everyday life. It contained all conceivable polarities in an inseparable fusion and thereby transcended dualities of any kind. Time after time, people compared the Absolute to a radiant source of light of unimaginable power.
>
> …
>
> When we encounter the Void, we experience it as a primordial void on a cosmic scale. We become pure consciousness, perceiving this absolute nothingness, but at the same time we experience a paradoxical sense of its essential fullness, its saturation. This cosmic vacuum represents absolute completeness, for everything seems to be present within it. It contains nothing in concrete, manifest form, but seems to encompass all existence in its potential form.
>
> …
>
> In several cases, people who experienced both Absolute Consciousness and the Void experienced the epiphany that these two states, despite their apparent empirical incompatibility, are essentially identical and equivalent. These people reported observing how creative Cosmic Consciousness emerged from the Void or, conversely, returned to the Void and disappeared into it. Others experienced these two aspects of the Absolute simultaneously—identifying with Cosmic Consciousness while simultaneously recognizing its essential emptiness." [15]

Considering the reports from altered states of consciousness (ASC), we can paradoxically discover that Advaita Vedanta's **Absolute** (Brahman) and Buddhism's **Void** are actually different names for the same. MALm adds "MAL" (Mind At Large) as another name which makes no difference at all. This "third" name is not to say the model sees something different in the core of Reality, but to distinguish how MALm sees the goal of that knowledge. Buddhism, absolutely in the same way as Advaita sees the goal in "liberation." This will to take exit in Buddhism is taken even more seriously, and the approach is taken to its edge: as Buddhism thinks that Brahman (or Atman) is another "thing" the mind can cling to, and thus it sees even the Brahman concept as an obstacle, selecting Void instead.

It is important to remind the obvious (but often forgotten) thing here: neither system is Reality itself; they are all descriptions, models, "maps." This applies equally to Advaita, Buddhism, or MALm. However, each model selects its own language and puts its own emphasis. The language of "Void" selected by Buddhism naturally forces this system into something heavily distinguishing it both from Advaita and from MALm. Particularly, if you choose "Void," you obtain a functioning system (world of Forms and Events) that has no organizing principle under it, and therefore immediately fall into the necessity to explain the **mechanics** of how everything works "by itself."

We get the situation in which both MALm and Advaita describe what is happening ("dissociation" or "Maya") by attributing the major process to a specific state and/or will of the Absolute (Brahman, MALm). Neither of those systems goes into the details of "how" the "dissociation" or "Maya" are built. Buddhism, however, for reasons we already mentioned, obviously does not have such a "privilege." The mechanics this system is describing are its absolute advantage. While Advaita has its own goals, MALm can to some degree just "borrow" what is brilliantly described by Buddhism as "mechanics" of reality.

In the "Dissociation-association coincidence problem" chapter of this book, we have already explored the question of whether personality is fixed or fluid and used the astonishing reports from dissociative disorder research and treatment to clearly understand that a personality not only evolves over time (which is clear from normal experience of every human - compare yourself at the age of 5 and 35), but also at every particular moment of its existence is composed from parts (thoughts, emotions, memories, perceptions, body states) and connections between those parts. These "connection" elements are extremely significant and are absolutely equal in value to the "nodes" being connected. Depersonalization and derealization disorders clearly demonstrate that a person can have a feeling of being disconnected or detached from one's own thoughts, feelings, or body, while those thoughts and feelings are still present. The mere presence of those disorders and their temporary (in most cases) nature shows the person (self) being a permanently changing combination of interconnected elements and connections between them (under MALm, this is called association). As MALm sees **dissociation** as the principle creating a separate self, in this picture, dissociation does not look like something different from the association itself, but rather as the temporary absence of connections between some nodes of the system.

Now let's look at the doctrine of Anatta ("no-self") in Buddhism, a teaching that there is no permanent, unchanging soul, self, or essence to be found within a human being or any phenomenon. Instead of a singular self, a person is a constantly changing process composed of five shifting aggregates: form (the physical body), feelings (sensations that are pleasant, unpleasant, or neutral), perceptions (the ability to recognize and label phenomena), mental formations (habits, thoughts, volitions, and emotions), and consciousness (in Buddhism terms, meaining "the awareness that arises when sense organs interact with objects"). According to Buddhism, each of these five components is in a constant state of flux. With that description, Anatta is in a huge correspondence with the modern scientific data and MALm's concepts of association/dissociation.

The next important thing to watch in Buddhism is the **boundaries**: between this non-singular, always changing combination of elements and connections (self) and the "outer" world, plus the boundaries of this process in time. In everyday life, the boundary between "me" and "world" is pretty obvious in perception of both the body and outer objects, and the mind. The boundary of personal self in time is also easily observable: this is the birth in the beginning and the death in the end. Regarding the "world", the view in Buddhism is that the internal and external are not two separate, fixed realities. Instead, they are mutually dependent. Buddhism argues that there is no clear, permanent boundary where the "self" stops and the "world" starts. It defines matter (akin to "self") not by its substance (like "atoms"), but by its behavior or primary qualities. Everything physical is an impermanent combination of four foundational "elements": earth (the quality of hardness, softness, solidity, or roughness), water (the quality of cohesion, fluidity, or binding - what holds things together), fire (the quality of temperature), air (the quality of motion, vibration, or pressure). If we connect that with the data we discussed in the "Double slit experiment," saying that particular values for all the properties arise at the moment of conscious observation, this creates an astonishing picture of interconnectedness and the absence of a particular boundary between world and "self." Another interesting insight in Buddhism regarding transparency of this border is how it points to the body (part of the "self"): you breathe in oxygen; it becomes part of your blood, your cells, and your energy. Is the air "you" or "the world"? You eat food; it becomes your physical structure. Where does the "world" end and "you" begin?

In MALm, this interconnectedness and wholeness gets its absolute and logical peak: in the "Dream as best analogy" chapter, with that precise analogy we showed that all objects/subjects, selves of the dream, are One dreamer. It is an absolute dependence and connectedness where the boundaries are defined but not-memorizing the nature of what is happening (dissociation).

We also know Buddhism's attitude to the boundaries of personal self in time (the birth and the death). Here Buddhism puts on scene the concepts of Rebirth and Karma. The Rebirth concept presents a view of the consciousness process that has no start (to birth is not the "beginning", but a "continuation") and death is not the end. It is interesting to note that Buddhism calls that Rebirth - "Reincarnation" belongs to another traditions (like Advaita); while "Reincarnation" implies the migration of a stable, permanent soul or "self" from one body to the next, like a person changing clothes, in Buddhism, with its Anatta, there is no such thing. To address a common question "If there is no self, what is reborn?", Buddhism uses the metaphor of a flame passed from one candle to another. The flame on the second candle is neither the same as the flame on the first, nor is it entirely different—it is a causal continuity. In this view, rebirth is not the transmigration of a permanent soul, but the continuation of a stream of consciousness. We need to mention here that if there is a non-interrupted awareness of this process about itself, it does not make any difference with the permanent soul (Atman) of Advaita, although this awareness is not any of the individual selves it comes between. However, Buddhism has its methodology and thus selects language and metaphors correspondingly.

This never-starting, never-ending process is not happening randomly, but directed by Karma (kamma in Pali), literally meaning "action": in Buddhism, it is not a system of cosmic reward and punishment managed by a deity; it is a natural, impersonal law of cause and effect. We won't go deep into the woods of this specific concept, but will mention only that what Buddhism sees as karmic seeds and related karmic momentum, MALm (putting "Lila" on the pedestal) sees as "unfinished adventure/research, not finished story that must continue to write itself."

It is important to note that Karma in Buddhism explains why we encounter certain circumstances, but it does not mean your life is "predestined." Because you have the power to act with new, virtuous intentions in every present moment (Free Will in terms of MALm), you can shift the trajectory of your karmic stream. In the light of Karma, it is good to return to the described above idea of absence of permanent boundary between the "self" and the "world" and find out that Karma defines not only tendencies in inner states of the "self," but also the "outer events" of the world, and again this is not "fate", but the tendencies or "boundaries", within which the actions of self are not pre-defined. This meaningful unity of events of the inner and outer word has very interesting supporting data in the modern observation: we discussed that in the "ASC content problem" chapter as "synchronicity" defined and described by by the Swiss psychiatrist Carl Jung as "meaningful coincidence" or "acausal connecting principle," suggesting that certain events are linked not by cause-and-effect, but by a shared underlying meaning.

An enormously rich addition to the description of mechanics of the consciousness-process in Buddhism is presented by the Bardo Thodol (known in the West as the *Tibetan Book of the Dead*), one of the most profound texts in the Tibetan Vajrayana tradition. Its title translates to "Liberation through Hearing during the Intermediate State." It serves as a roadmap for consciousness during the "in-between" period (bardo) following physical death and leading to the next rebirth. While the "Bardo" concept is famously associated with the afterlife, the term simply means "gap" or "intermediate state": it is any moment of transition (e.g., the gap between waking and sleeping, or the pause between two thoughts). In the specific sense, it is the transitional states after the body dies, before the mind-stream "re-enters" a new physical form. The text guides the consciousness through three distinct phases.

First is the **Bardo of the Moment of Death (Chikhai Bardo)**: as the physical body shuts down, the senses dissolve. The text describes the appearance of the "Clear Light of Reality." If a person has prepared through meditation, they can recognize this light as the fundamental, empty, luminous nature of their own mind and achieve instant liberation. It is astonishing to map this particular stage with the near-death experiences (NDE) that we described in the "ASC content problem" chapter. Interestingly, NDEs themselves often report the presence of a sense of a threshold, which once crossed, consciousness has no way back.

Second is the **Bardo of the Experiencing of Reality (Chonyid Bardo)**: this is a dream-like state where the mind projects its internal karmic landscape into visions. One may encounter "peaceful and wrathful deities." These are not external gods coming to judge you; they are symbolic projections of your own consciousness—the "bright" and "dark" sides of your own mind. The instruction is to recognize them as projections, thereby preventing fear and realizing the nature of the mind.

Third is the **Bardo of Becoming (Sidpa Bardo)**: the consciousness, still clinging to habits and ego, begins to seek a new "home" or rebirth. It experiences intense karmic hallucinations. The text provides guidance on how to avoid unskillful rebirths and how to choose a favorable one by maintaining mental clarity.

From the perspective of MALm, what is happening in those states is a continuation of dissociation, but of a type very distinct from the "personal self" of everyday life. It allows a consciousness to continue to act, process information from a unique local perspective (but "local" to a different degree which consciousness-process does not see itself as any distinct personal self, but as a huge row of such selves, interconnected by sense, content, meaning, phase of adventure, causes and effects. "Processing consciousness" not only has the power to perceive, estimate, feel and think, but to act and choose to some degree.

Here we come to the core distinction between MALm and Buddhism-Advaita. The latter sees the goal of any practice in **cessation** of the process (exit) - in MALm terms, that will be the complete end of this particular dissociation ("string of lives"). While MALm praises "Lila" and "Curiosity", and "Creativity" of Mind at Large and continuation of the game, Buddhism-Advaita selects to exit, to make actions leading to erasing the karmic momentum. In the "Lila" view of MALm, that momentum ends when the story is finished. And the "feeling of MALm" in the face of an unfinished story is the same as the feeling of the composer Felix Mendelssohn, whose sister, Fanny Mendelssohn (who was also a talented composer and pianist), used a very specific trick to get him out of bed in the morning. Mendelssohn was reportedly a heavy sleeper and sometimes difficult to wake. Knowing his perfectionism and his inability to leave a musical phrase unresolved, Fanny would sit at the piano and play a scale or a melody but intentionally stop just before the final note—the tonic, or the "resolution" of the musical phrase. The tension of that incomplete, hanging melody would be so physically and mentally grating to him that he could not bear to leave it in that state. He would inevitably "jump out of bed" and rush to the piano to play the final note, finally satisfied that the sequence was complete.

An important correction to this tension in case of MAL is that it has no idea how the melody ends - only the obvious feeling it's not finished yet. This transforms the feeling of "just" an incompleteness to the feeling of incompleteness-curiosity-huge-urge-to-learn.

Neither Buddhism nor Advaita pay any significant attention to the question of "Why does the World of Forms exist?" or "Why the infinite Source/Void has not remained in its initial state and instead bothered to create (Advaita) or self-cause (Buddhism) the Forms."

Unlike both of those systems, MALm sees the CURRENT state of any consciousness-process as High and Sacred as all the other. It implies the tremendous value and meaning in our current state. This leads to a life of full engagement. Rather than renunciation, the philosophy encourages "transfiguration." You do not abandon your desires or roles; you realize that the consciousness experiencing them is the same consciousness that creates the universe. This allows for a celebration of human experience rather than a rejection of it. It is the property not only of MALm; in the next chapter, we will have a dialogue with a very similar approach of Kashmir Shaivism.

However, MALm, like any other philosophy of "full engagement", becomes a target for the inevitable question of what to do with suffering and evil. The suffering is very real and even enforced by radical engagement. Evil is something that is very strange to call to "fully engage into." The suffering and evil together are very strong obstacles in the way of "embracing the world". Again, we will discuss those questions in the next chapter related to Kashmir Shaivism. In this chapter, I would like, however, to express my deep respect and empathy for both Buddhism and Advaita. How the model's authors see and describe the world is never the World itself - it is a perspective. This applies to Buddhism, to Advaita, to MALm, to every model. And all the perspectives are local, which means they are very much affected by the environment and conditions in which they were being formed.

Thus, look at Dhanika, a farmer in a village in the Gangetic plain, near what is now Bihar, 500 BCE. A nearby king, Bimbisara of Magadha, is "expanding his empire," which means slaughtering all he can reach. His tax collectors do not ask for a percentage of the crop; they take the grain to feed the standing army. If Dhanika resists, the village is razed, and he is enslaved. So, you give what was earned with heavy labor. One bad monsoon, and there is no grain left, and Dhanika watches his children suffer from malnutrition; but who cares if you have Brahmans - pay for sacrifices you cannot afford, and a better "next life" is guaranteed to you, while children are starving now and the king Bimbisara scratches his belly contentedly.

If you were Dhanika, the Buddhist villager watching his village burn for the third time, would you want to hear that "the world is a beautiful play," or would you want the "emergency exit" that allows you to remain sane amidst the fire? Thus, I feel nothing but empathy for Buddhism or Advaita.

Only after coming back to a safe environment where King Bimbisara's soldiers cannot approach my family or me, I have a shelter and some food on the table, I am obtaining back the privilege of saying something else intellectual about Buddhism.

Buddhism does state there is a process, driven by causes and effects, but it never defines this process, providing only meta-description: "there is a process", "there are causes and effects." These are general claims. There is no description of which causes lead to what effects (except the most general), no timing, no true understanding. The gardener knows that there must be a soil of specific quality, a seed of a specific plant, and temperature and watering suitable for that plant to grow; they know when to put seed in the soil, they can clearly distinguish that seed from just a little stone, they know when to water and when to cover from burning sun, they know how long to wait. Buddhism does not contain or provide this type of knowledge. It is not good or bad that it is absent, but it is important to see that it is absent.

To Kashmir Shaivism - Most Important Chapter of This Book

In the 8th–12th centuries, Kashmir was an island of extreme cultural stability and intellectual luxury. It was a trade hub on the Silk Road, isolated from the major invasions that ravaged the plains. The Kashmiri intellectuals and masters (like Abhinavagupta) were not looking for an escape from society; they were part of the courtly, artistic, and sophisticated elite of Kashmir. Their philosophy reflects the culture of music, dance, poetry, and logic. They didn't want to "exit" the world because, in their privileged and vibrant context, they were already witnessing the beauty and power of the Divine in the art and intellect surrounding them. They turned the entire universe into a work of art that the Divine is performing.

As the MALm of this book is created in similar conditions, there is no surprise that it has Kashmir Shaivism as its closest relative. Let's go into the philosophy of Kashmir Shaivism to try to show differences (if any at all) and similar views.

Kashmir Shaivism is a profound, non-dualistic philosophical and tantric tradition that originated in Kashmir around the 9th century CE. It is often referred to as Trika (the "threefold" path) and is characterized by its absolute monism—the belief that there is only one, supreme reality, which is pure consciousness (Shiva). That is an absolute match to Mind at Large (MAL) in MALm.

Unlike some traditions that view the material world as an illusion (maya), Kashmir Shaivism posits that the universe is **real**. It is the self-manifestation of the Absolute. That is an absolute match to MALm's claim from the "Limitations of analogy" chapter:

> "... Mind at Large (One) does not have "outside" to "wake up into"—therefore, everything inside its "dream" is real to the extent of "nothing to compare with." Dissociation is real, Anna is real, John is real, the towel is real, the coffee is real, and the cat is even more real."

In Kashmir Shaivism, Shiva represents the transcendent, unchanging aspect of consciousness (Prakasha or "Light"), while Shakti is the dynamic, creative energy (Vimarsha or "Reflective Awareness") that manifests as the universe. They are inseparable, like a fire and its heat. There is no separation between the Creator and the creation. Everything—every object, thought, and living being—is a manifestation of the singular Divine Consciousness.

The central premise of this school is that the human soul is already divine but has "forgotten" its true nature due to limitations (those limitations are called "dissociation" in MALm). We suffer because we perceive ourselves as small, limited, and separate from the rest of existence. Enlightenment is not something to be "acquired" from outside; it is the recognition (Pratyabhijñā) of one's own divine nature. When the veil of ignorance is lifted, the individual realizes: "I am Shiva."

Kashmir Shaivism affirms the world. Rather than teaching the negation of the world (as is sometimes seen in other schools of Indian philosophy), it teaches that we should interact with the world with the understanding of our "seamlessness" with it. The world is a "play" or "reflection" (Pratibimbavada) of the Divine in the mirror of His own consciousness. The universe is not static; it is in a state of constant, rhythmic movement or "vibration" (Spanda, Divine Vibration). This is the creative energy of Shiva pulsing through all existence.

So, in its core, Kashmir Shaivism and MALm are **identical** in their conclusions and the vision of Reality, including the attitude to it: it is real, and it is Lila. The only differences are not in the ideas, but in wording and some extra sets of observations, such as 36 Tattvas (Levels of Reality) in Shaivism.

In the previous chapter (on Buddhism) we promised one thing:

> "However, MALm, like any other philosophy of "full engagement", becomes a target for the inevitable question of what to do with the suffering and evil. The suffering is very real and even enforced by radical engagement. Evil is something that is very strange to call to "fully engage into." The suffering and evil together are very strong obstacles in the way of "embracing the world". Again, we will discuss those questions in the next chapter related to Kashmir Shaivism."

We put it even stronger:

> "If you were Dhanika, the Buddhist villager watching his village burn for the third time, would you want to hear that "the world is a beautiful play," or would you want the "emergency exit" that allows you to remain sane amidst the fire?"

And not to allow ourselves an easy escape: today, Dhanika's village was burnt for the third time, and his family died in that fire. Now Dhanika approaches me or Kashmir Shaivism master and cries in grief and horror - what do you think we'd do and say? While I am pretty confident about Kashmir Shaivism, I will not speak for them; I'll speak for myself.

I would hug Dhanika, and I would cry in horror together with him. I would say to him, "Your family was precious; it is such a horror they are no longer here. Grieve, Dhanika, I grieve with you." If Dhanika screamed, "I will kill them," I would say, "Ok" and hug him stronger. Nothing more that day can be said or done. Later (weeks? months?), if I met him still in pain, I could say to him: "Dhanika, you will meet your children again one day, you'll be one; you'll meet your wife, you'll be one." and: "Dhanika, you pain continues as what you've lost was precious, so it will last long, and you will never forget them, and nothing of them is lost, you will see yourself one day.", and: "Dhanika, those soldiers will fall, that king will fall, there will be a day when no one will ever know they ever existed. Dhanika, they will know all the pain they caused themselves. This cannot be avoided.", and: "I'll see you later, Dhanika."

And if it is not quite clear what was told to Dhanika, it is useful to imagine a philosophy scholar asking questions about theory: "When grief is ended, we can talk. Look what was there for Dhanika: this was One, to which all returns and in which all is One. And it was Impermanence saying kings fall, murderers fall, as all fall. But if you ever meet someone like Dhanika, look inside you and there you will find correct words (or silence) and actions."

To all the rest not currently touched by suffering or evil, just leaving their everyday life, I would say: here it is! It is precious; enjoy every moment between the storms. They are precious, they are beautiful. Meditate not to escape, not to find something "Divine" somewhere, but to connect it right here, because here it is. And you are it too. There is no higher state; it is it. And it is you, and you are it. Here and now."

To Christianity and Islam - Social Projects

Christianity and Islam are two of the world's largest religions, both strictly professing faith in one God, both recognizing a long line of prophets, both emphasizing virtues such as charity, compassion, honesty, and social justice, both believing in a final judgment, heaven (paradise), and hell, emphasizing human external (to God) accountability for actions on Earth. Both of them provide a very well-developed, very material and societal framework for daily life, worship, and spiritual discipline, including individual and collective prayers, periodical gatherings, and engagement with Holy books.

Both traditions are younger than the ones we talked to in the previous chapters. While Hinduism is one of the oldest living traditions with no single founder; evolved from earlier Vedic practices around 1500 BCE, and Buddhism emerged in India in around 6th Century BCE as movements focusing on liberation from the cycle of rebirth, Christianity emerged in the 1st century CE from Second Temple Judaism, centered on the life and teachings of Jesus, and Islam emerged in the 7th century CE with the revelations received by the Prophet Muhammad. Today, both religions continue to influence the laws, ethics, art, and social structures of societies across the globe, serving as profound sources of meaning and community for billions of people.

Both Christianity and Islam share a linear, teleological cosmology. This means they view time as a straight line: beginning with Creation, progressing through human history, and culminating in an afterlife that is determined by one's relationship with the Divine during their mortal life. Both religions describe a universe structured into realms of existence beyond the physical world. Beyond the physical world, we found in both of religions, God, angels (in Christianity, spiritual beings created by God, messengers and protectors; in Islam, spiritual beings created from light, possessing no free will and obeying God perfectly), the Adversary (in Christianity, Satan (or the Devil), a fallen angel who opposes God; in Islam, Iblis or Shaytan, a jinn who refused to bow to Adam, now a tempter). Also, the physical world is "extended" by two extra "realms": in Christianity, Heaven (God's presence) and Hell (separation from God); in Islam, Paradise (Jannah) and Hellfire (Jahannam).

Christianity posits Jesus Christ as the central, divine figure—the Son of God—whose life, death, and resurrection provide the means for human salvation. Other prophets (Moses, Isaiah) are viewed as precursors who pointed toward the coming of Christ. Islam sees Muhammad as the final prophet in a long line of messengers. Islam honors many of the same figures found in the Bible (Adam, Noah, Abraham, Moses, David, and Jesus) as prophets who brought messages of monotheism. However, Islam rejects the divinity of Jesus, viewing him as a mighty human prophet born of the Virgin Mary.

Both religions reject the concept of reincarnation (rebirth into another mortal life). Instead, they hold a doctrine of a linear afterlife. Both traditions believe in a future, singular Day of Judgment (Dies Irae in Latin or Yawm al-Qiyamah in Arabic). Both teach the bodily resurrection of the dead at the end of time, when each individual will be judged by God based on their faith and/or their actions/deeds during their earthly life. Both believe and aim for Salvation. In that regard, in Christianity, the primary focus is grace and faith—because of human sin (the "Fall"), humanity cannot bridge the gap to God on its own, and salvation is a gift of God's grace, received through faith in the sacrifice of Jesus. In Islam, the primary focus is submission and deeds. Humans are not born with "Original Sin" but are born with a pure nature (Fitra), which they may neglect. Salvation is achieved by submitting to God's will (Islam), believing in His oneness, and performing righteous deeds that will be weighed on the Day of Judgment.

Both traditions experience some level of tension related to their linear theology: how can a "final" judgment exist if individuals seem to experience a destination ("paradise" or "hell") immediately upon death? Both traditions resolve this by distinguishing between an initial, individual experience (the "Particular Judgment") and the final, collective reality (the "General" or "Final Judgment"). The "waiting" period is not merely a passive snooze; it is considered an active, though disembodied, state. In both faiths, death is the separation of the soul from the body. While the body returns to the earth, the soul enters an intermediate realm where it begins to experience the reality of its choices. To the question "if the soul is already in Heaven or Hell, why have a Final Judgment at all?", both religions answer by placing a massive emphasis on humanity as a whole. In their view, a human being is not just a "soul"—a human is a union of soul and body. To be truly judged or truly rewarded, the entire person—not just the "ghost" of the person—must be present.

After this brief introduction, we need to make an important observation: while older traditions (Hinduism, Buddhism) generally tend to look at the World as a Whole (One), both Christianity and Islam tend to see Duality: the World is created by God and distinct from Him. The same dualistic approach is visible in having a Good vs. Evil dilemma described as "God vs. Devil", while older traditions generally see Good and Evil as manifestations of the same thing. Another important observation is that in many Indian philosophies, life is an educational process. If you fail to learn your lessons in one lifetime, you return to gain more experience. It is a system of gradual, personal evolution. In Christianity and Islam, life is a decisive trial of faith and moral alignment. The gravity of human life is heightened precisely because it is a "one-shot" opportunity. This creates a sense of urgency: your choices here and now have eternal consequences. The afterlife is not a result of "earning enough points" over thousands of years, but of your singular orientation toward God within the span of one life.

So, what was the actual result of this "moral urgency"? When we analyze the sheer scale of human destruction under this "urgency" banner, we are looking at a massive escalation in the intensity and scope of organized violence. If we look at the historical data, the "moral technology" of these traditions did not just fail to stop crime and violence; it professionalized it. Once you define your opponent as a "sinner" or an "agent of the Devil" in a cosmic courtroom, the "rules of engagement" disappear. When you fight a "heathen," you aren't just fighting a person; you are executing a divine mandate. The Crusades, the centuries of the Islamic conquests, the Inquisitions, the Thirty Years' War in Europe (where nearly a third of the population in some regions died in the name of sectarian "Truth"), and the colonization of the Americas (where tens of millions died under the banner of the "Cross") represent a scale of violence that local, tribal systems could never sustain. In both Christian and Islamic empires, the harshest legal penalties were almost always reserved for "offenses against God" (blasphemy, heresy, apostasy). The "urgency" allowed authorities to perform acts of torture and execution—not as a lapse in morality, but as a moral necessity. By claiming that the "eternal soul" of the victim was at stake, they justified the annihilation of the body. This is the ultimate "moral loophole."

In a non-dualistic or organic system, the restraint on violence is empathy (the recognition that the other is not "other"). The Abrahamic "urgency" explicitly suppresses empathy by teaching that some people are fundamentally "lost" or "wicked" in the eyes of the Judge. This creates a psychological state where a person can commit a murder or oversee a massacre and, because they have followed the "Law" or received "absolution," feel that they have maintained their moral standing.

So, if we look at the last 2,000 years, the greatest periods of mass casualty were almost always driven by this "moral urgency." Whether it is the religious wars of the past or the secularized "crusades" of the 20th century (which inherited the same linear-urgency structure), the pattern is clear: When you believe you are "saving" humanity or the world, you become the most dangerous thing on the planet. The "urgency" didn't curb violence; it provided a Universal Justification for it. It made violence feel like a "righteous" or "necessary" part of the historical mission.

Besides Christianity and Islam's external violence, we can also observe the "internal" dimension of the violence, again strangely bound to "moral urgency." By injecting the idea of sin, Christianity implemented arguably the most efficient "moral technology" ever invented for controlling people: you were no longer just a person living a life; you were a "sinner" in a constant state of failing the Judge. This produced a permanent, low-level (or high-level) anxiety. A person who is constantly worried about their "soul's state" is easy to lead. It creates a dependence on the religious hierarchy, because only they hold the keys (confession, ritual, intercession) to cleaning that "sin" away. In correspondence with that, both traditions always were and still are obsessed with policing the human body, particularly sexuality. The body, and specifically female sexuality, is the primary source of the "next generation" of the tribe/mission. By applying "urgency" to sexual conduct, these systems asserted ownership over the most vital human force—reproduction—and turned it into a matter of "religious obedience." If you can convince a human that their biological desires (hunger, lust, physical pleasure) are "base" or "sinful," you have effectively conquered their will. The repression of the body serves as a constant reminder of who is in charge. It creates a mindset where the "soul" is at war with the "flesh," echoing the broader cosmic war between "God" and "The Devil."

Another critical influence point was the introduced separation from nature ("desacralization of nature"): while in the most ancient, non-dualistic traditions, nature is the "Body of the Divine" which you don't "manage" but you live within it, Abrahamic traditions asserted that humans were "above" the world (having been given "dominion" over it by a Creator-God), these traditions stripped nature of its sacred, autonomous status. Nature became a resource for the mission. This alienation from nature is what eventually paved the way for the radical secular materialism of the modern era. When you tell a human that the world is just "a temporary waiting room" and that they are "distinct from the Earth," you break their fundamental connection to the reality of the biosphere. You leave them "rootless," which makes them even more susceptible to the "universal story" of the religion.

Before coming to conclusions, we need to add one more crucial point to the data. This book considers a lot of data from different areas, including the latest scientific research in physics, studies of consciousness, psychology, biology, and informatics. The significant fact is that many observations from this scientific research have deep intersections with the ideas of older traditions: for example, what we were describing in the "Dissociation-association coincidence problem" chapter has obvious parallels to the Buddhist idea of Anatta ("no-self")—we described these parallels in the "To Buddhism - Mechanics of Reality" chapter. The idea of the entire book about Mind at Large presence as necessary for understanding the data of physics and ASC (altered states of consciousness) matches one-to-one with the concept of Advaita Vedanta about the Ultimate Reality (Brahman) being the infinite, formless, unchanging, and singular consciousness. Data on quantum entanglement is deeply linked with the idea of interconnectedness from Buddhism.

On the other side, we find an actually non-surprising thing: scientific and other deep research data **never** has any intersection with the ideas of Christianity and Islam. The only exception is meeting with the figures and even realms of these traditions in ASC (altered states of consciousness), including NDE (near-death experiences). We can often see on YouTube the narrative "I died for 10 minutes, met Jesus, and he told me that…" and easily guess what this narrative changes into in the case of Muslim responders. And this is not just a "modern" made-up hype on social media—the same data is presented in very well-grounded research on ASC and NDE, so there is no doubt people do have corresponding experiences. However, we know for sure the ability of the human mind to visualize things and information, presenting them as external experience (take dreams as a model). In the "ASC content problem," we studied in detail multiple reported experiences leading altogether to the unavoidable conclusion that all ASC content is the obvious mixture of individual and universal content. And we will never claim that Christianity and Islam, with all their images and deities, do not exist as information deeply imprinted into the minds of corresponding personalities and into the collective consciousness of humankind.

Thus, our claim "scientific and other deep research data **never** has any intersection with the ideas of Christianity and Islam" stands, meaning all the rest cases and examples that we mentioned for Buddhism and Advaita. Why did I call earlier this non-intersection a "non-surprising thing"? I'd like us to pay attention to the fact that both science and old traditions observe the **observable**: the World itself, its matter, the mind presented in it, the content of this mind. Christianity and Islam's main claim is that there is something outside the observable.

In this sense, we have two major facts in our hands: (1) Ideas of Christianity and Islam perfectly fit the social project goals: when you have together—us vs. "them" (sinful), guilt (sin), repression (sexuality), and isolation (separation from nature)—you get a very specific human product: a person who is deeply alienated from their own nature, constantly afraid of the "Judge," and therefore entirely dependent on the collective narrative for their sense of worth and security. Both traditions have been very successful at homogenizing the planet. They didn't just ask you to believe a story; they changed your psychological architecture so that you need the story to survive within your own manageable tribe against other tribes. (2) Ideas of Christianity and Islam have nothing to do with what is objectively presented in the World.

With that data in hand, we come to the following: Christianity and Islam are social projects; they do not explore, care, or know anything about the World or your Psyche. The stories told by these traditions are fully made-up and artificial, serving the goal of solidifying the groups of people and controlling these groups.

I would also like to say the same thing in clearer words, and I will. But there's a little preparation that needs to be done beforehand. For a long time, I've been thinking if the model that introduces Mind at Large (MAL) should accept communicating with models like Christianity and Islam by saying: "yes, MAL is what you call God." And now, as almost this entire book with all its data is written, and when this particular chapter about Christianity and Islam is written, my answer to this question is "no." No, MAL can never be called God, because Mind ("M" from MAL) is an **existent thing**; it is observable. We know what it is; we know its essence and possibilities intimately.

Thus, to reply to the traditional question "Do you believe in God?", I would say "No, I do not need to believe in it, because I know exactly what it is." - "What is it?" - "It is a concept, created by people."

Thus, to say in maximally clear words: there is no God, there is no Heaven, there is no Hell, there was no Adam and no Eve, there was no "Original Sin", and there is no such thing as sin or sinner and no need in salvation, there is no Devil, and no angels, there will be no Day of Judgment and no resurrection of the dead. The World does not contain any of that; it functions differently, and its extreme meaningfulness has an absolutely different nature.

To Christianity and Islam - Extreme Complexity

It is not a mistake that in this "Dialog with Other Models" we suddenly have the dialog with Christianity and Islam twice. By having this, I would like to show you how Reality looks through the lens of **non-local observation**, the core method built into the essence of MALm. MALm is a model calling for attention to the World in its totality, not for exclusion of the facts that do not match our preferences, but for maximal inclusion of such facts and maximal attention to nuances and complexity. So let's add those facts to see the extreme complexity of the World.

The ban on figural art in Islamic architecture (to avoid idolatry) led to the mastery of geometry, arabesque, and calligraphy, creating a unique aesthetic that mimics the infinite nature of reality through mathematical precision. Similarly, Christian desire to manifest "Heaven on Earth" led to the development of Gothic architecture, where light and height were manipulated to induce a profound sense of "awe" (an altered state of consciousness), effectively using stone and glass as psychological technology to bridge the gap between the mundane and the infinite. The Islamic Golden Age (House of Wisdom) is a primary example where the "religious mandate" to understand God's creation fueled the preservation of Greek philosophy, the invention of algebra, and advancements in optics, medicine, and astronomy.

The systemic nature of Zakat (Islamic almsgiving) and the Christian tradition of hospitals and monasteries created the first proto-welfare systems. These were not just "nice acts"; they were institutionalized mechanisms that ensured the survival of the sick, the elderly, and the orphaned when no other state structures existed. This was the first time in human history that a person's value was tied to something other than their bloodline or utility to the local tribe. Bimaristan (Islamic hospitals) were perhaps the most advanced medical institutions of the medieval world. They were based on the concept of waqf (charitable endowment)—private wealth permanently dedicated to public good. Unlike earlier infirmaries, these hospitals were secular in practice, treating all citizens regardless of status or religion, with separate wards for different illnesses, libraries for medical study, and free medication. Christian Xenodocheia evolved from simple hospices into complex centers of public welfare. By the 4th century, bishops like John Chrysostom were not just praying; they were managing massive logistical operations to feed the hungry and house the traveler. They established the concept that the "Stranger" is not a threat to be feared but a responsibility to be housed.

With Zakat (mandatory charity), Islam turned just "giving money" into a spiritual mechanism for "purification" of the soul. By making the support of orphans, widows, and the poor a "right" (not a choice), Islam codified a social safety net that removed the shame of poverty. It established that the wealth of the rich actually belongs to the poor by divine mandate. In Christianity, the interpretation of Matthew 25 (feeding the hungry, clothing the naked) created a standard where the value of a society was measured by how it treated its most "useless" members. Monasteries became the primary nodes for this, often serving as the only reliable place for the poor to find food and asylum during times of war or famine.

The requirement for constant confession and the examination of one's own desires created a new type of human consciousness: the "inner self." This is the precursor to modern psychology. The introspection required by figures like St. Augustine or the Sufi mystics (like Rumi or Ibn Arabi) created a depth of human emotional expression and psychological analysis that had no equal in previous, more "external" social systems.

In 8th-century Basra, a former slave Rabi'a al-'Adawiyya carried a torch in one hand and a bucket of water in the other, declaring she wanted to "burn down paradise and quench the fires of hell" so that people would worship God for Love alone, not for fear or reward. In 13th-century Italy, a merchant's son, Francis, abandoned his wealth to live among the outcasts. His famous act—stripping off his expensive clothes in the town square and renouncing his father's inheritance—was a public dismantling of the social hierarchies his religion was busy maintaining. He famously kissed the hand of a leper, an act that defied the era's social and "moral" categorization of the "unclean."

Early Christians who fled into the desert to escape the "social project" of the Romanized Church. They sought silence and emptiness. Their focus wasn't on "saving souls" through doctrine but on achieving apatheia (freedom from passions/ego).

Before modern therapy, the practice of Confession or Tawbah (repentance) offered the only space for radical honesty. A person in deep despair—carrying the weight of a life mistake or trauma—could articulate their "sin" to a listener who was obligated to offer absolution. This act of verbalizing trauma, paired with the psychological relief of "being forgiven," was powerful and one of the first forms of psychotherapy.

Father Maximilian Kolbe stepped forward to take the place of a stranger destined for the gas chamber at Auschwitz. This was not "institutional" obedience; it was an individual act of transcendence that shattered the Nazi "social project" using the very logic of his faith. J.S. Bach claimed his music was for the "glory of God." Hagia Sophia and The Blue Mosque are spatial experiments: the way the light hits the dome in the Hagia Sophia was designed to make the dome look as if it were "suspended from heaven by a golden chain." This is an attempt to create a sensory environment that forces the human mind to drop its typical, ego-centric mode of perception and experience a sense of vast, non-local unity.

I cannot address directly in this book all the millions of acts of individual kindness and courage, all the episodes of high intellectual and inner states experienced by people throughout time, all the consolation and relief received by the grieving one or suffering one through Christianity and Islam.

I am also admiring the observation of a complex dance that human sexuality had with those traditions' limitations and oppression and powerful dramas those two forces managed to create together. I am imagining multiple brilliant thinkers honing their craft and creating brilliant works out of a desire to combat religious stagnation. And I am imagining many more things happening.

Do the facts provided in this chapter cancel out what was told in the previous? Not at all. But they add up to praise an Extreme Complexity of Mind at Large and the World as its expression. Both Christianity and Islam are obviously not the "flaws" in the system, but the part of a hyper-complex Lila, very specific manifestations of Universal and Personal minds, and a stage in the history of societies.

To Loop Quantum Gravity - Big Bounce as Breath of Eternal Mind

In the "Solution: MALm, Math, QM, GR, LQG, KK, and String Theory" chapter of this book, we have already spent some time talking about Loop Quantum Gravity (LQG) as a theory trying to marry two pillars of modern physics that usually don't get along: General Relativity (which explains gravity and the very large) and Quantum Mechanics (which explains the very small).

We mentioned that MALm predicted that for both for solving contradictions between theories (like GR and QM) and for unification (like an attempt to describe all observable forces and interactions within one theory), the same approach will consistently work again and again: movement into the direction of abstraction, introducing new levels of abstraction, and listed Loop Quantum Gravity as the example of moving into that direction: LQG introduces abstraction of **loops** or (later version) "**spin networks**"—mathematical graphs where there are nodes (points) representing discrete units of volume and edges (lines): representing the area of the surfaces connecting these volumes and creating a "quantized space" with a minimum possible length.

From the LQG perspective, if you were to zoom in far enough—down to the Planck Scale (10{-35} meters)—you would find that space is granular. The biggest "win" for LQG is its potential to solve the problem of Singularities. In standard General Relativity, the Big Bang starts at a point of infinite density, which makes the math break down. LQG suggests that because space cannot be infinitely small (it has a minimum "atom" size), the universe didn't start from nothing. Instead, it may have "bounced" from a previous collapsing universe—a theory called the Big Bounce.

LQG radically transforms General Relativity. In Einstein's original version, space is a smooth, continuous manifold (like a sheet of rubber). LQG "refines" this by showing that if you apply quantum rules to gravity, the smooth sheet disappears, replaced by discrete geometry: area and volume are no longer variables that can be any number; they are restricted to specific, "quantized" values. LQG treats the core principles of Quantum Mechanics as "sacred." The Uncertainty Principle now applies to the geometry of space itself.

From the MALm perspective, LQG is elegant (no extra dimensions) and beneficial (solves the contradiction between QM and GR and moves us from Big Bang to Big Bounce, which is extremely important for Eternal Mind having no "start" or "end".)

It is also interesting to mention that the transition from the Big Bang to the Big Bounce is more than a mathematical correction; it is a shift from a linear history to a cyclical eternity. This mirrors the ancient Vedic vision of Srishti (creation) and Pralaya (dissolution)—the rhythmic, cosmic breath of the Divine. Just as the Hindu sages described time as a wheel without a beginning, Loop Quantum Gravity provides the scientific scaffolding to realize that our universe is a singular, persistent process. We are not living in the wake of a 'first' moment, but in a current iteration of an infinite dance, where the granularity of space—the 'pixels' of reality—is the physical realization of Maya, the veil of continuity that masks the underlying, indivisible unity of the Eternal Mind.

To Michael Levin's Expanded Light Cone - Even More Complexity

Earlier in this book ("Biology and Information" chapter), we discussed Michael Levin's experiments, mentioned his TAME (Technological Approach to Mind Everywhere)—a conceptual framework designed to understand and manipulate "minds" or cognitive agents across all types of substrates—not just humans and animals with brains, but also cells, tissues, synthetic bio-bots, and even AI. We also noted that TAME was a highly nuanced model that could never be interpreted in a simplistic way and promised to borrow a lot of these nuances from Levin's genius later in this book. So this is a time for it, especially considering the fact that we have 2026 out there with AI on stage and a constant repeating question "Is AI conscious? Can AI be conscious?" or "Will AGI (Artificial General Intelligence) soon or ever arrive?" as if we understand what "general" (not "artificial") intelligence is.

Dr. Levin's work is definitely something stopping us from immediately saying "no" to all this, and giving it a thought. However, this thought is promising to be as nuanced as TAME itself.

So, core principles of TAME include rejecting the idea of a "bright line" separating true intelligence from mere physics. Instead, it posits that cognitive sophistication exists on a spectrum ranging from simple mechanical systems to complex human minds. The framework emphasizes that the nature of an agent's "mind" is not defined by its composition (biological vs. synthetic/software) or its origin (evolved vs. designed). TAME views biological processes—such as morphogenesis (how bodies form)—as examples of basal cognition. It suggests that cells and cell groups perform problem-solving and goal-directed behavior to achieve anatomical homeostasis. Intelligence is viewed as a nested hierarchy where lower-level agents (like cells) cooperate to form higher-level "selves." Each level of agency constrains the "option space" of the components beneath it. TAME introduces the concept of persuadability—the specific set of tools and communication methods (e.g., bioelectric signaling, chemical gradients) most effective for "rationally" modifying the behavior of a particular system.

As adjacent to TAME, the concept of "Cognitive Light Cone" arrives from Levin's work. The term is an intentional nod to the "light cone" in physics—which defines the region of spacetime that can be influenced by an event. Similarly, Levin's Cognitive Light Cone defines the spatial and temporal scale of an agent’s "concern." Spatial scope represents the physical area an entity can monitor and exert agency over. For example, a single bacterium has a tiny spatial cone (local chemical gradients), whereas a human has a much larger spatial cone that can include global environments, abstract ideas, or long-term goals. Temporal scope represents the "time horizon" of an agent’s goals. An automaton with fixed, hard-coded rules has a very narrow temporal horizon—it reacts to the immediate moment. A complex, intelligent agent can "care" about and act toward goals that exist years or decades in the future. By plotting these cones, TAME allows for the comparison of diverse intelligences. It moves away from the "human-brain-centric" view and instead asks: "How far in time and space can this system pursue a goal?"

We explored the nuances of cognitive light cones in the "Solution: Agency for Systems as Data Processing Mechanism" chapter of this book, comparing the possible intelligences of "Tesla FSD" and "a seagull wandering along the beach" and found out the complexity in defining what "intelligence" is and how many aspects we need to cover to describe specific type of "intelligence" and agency. We found, that in Levin's nuanced view the seagull had a "thicker" light cone because it managed its own internal physiological goals (hunger, temperature, health) and could invent new behaviors to solve problems, while Tesla had a "long but thin" light cone: it is superhuman at a tiny sliver of reality (navigating roads) but is completely "blind" to everything else in the universe. If you took the Tesla off the road and put it on the beach, its light cone would collapse to zero, whereas the seagull would simply find a new way to thrive. We also mentioned that systems "navigated" different "Dimensions" and can be compared by each dimension separately, plus by the number of these "dimensions" (spaces) that the system can navigate in principle (which actually defines the Type of Intelligence).

Levin often adds an additional, more abstract dimension: **The Space of Possible States**. This is where TAME gets radical. It suggests agents don't just move in 3D space; they move in:

- **Physiological Space:** (e.g., a cell trying to maintain a specific pH level).

- **Morphospace:** (e.g., an embryo trying to reach the "goal" of a finished hand).
- **Transcriptional Space:** (e.g., a gene network trying to reach a stable state).
- Etc.

…meaning the seagull occupies many spaces at once. It navigates 3D space (flying), physiological space (regulating body heat), and social space. The Tesla is almost entirely flat—it only exists in the problem space of "Road Navigation."

So, only by providing this information, we already highlight how difficult it is to define intelligence and how many parameters we need to introduce factually and potentially to compare different types of intelligence. Once again, we have already compared TAME to MALm in the "Solution: Agency for Systems as Data Processing Mechanism." It is important to note one more time here that Levin's work is important for anybody trying to talk about intelligence because this work clearly shows that you need to define intelligence to work with it later.

I think the AI community generally uses the definition of intelligence similar to Levin and William James’s classic: "the ability to reach the same goal by different means." Whether this general understanding is truly linked to the underlying set of technologies is a great question.

The interesting difference between MALm and TAME is that MALm cares about mind (or consciousness) that has intelligence as its part but not limited to it, including consciousness in a very specific sense—the ability to know that you're "achieving goals" or, in other words, to "have the experiences." TAME is not very interested in that thing but gives an interesting perspective to the questions that we mentioned: "Is AI conscious? Can AI be conscious?" Those questions under TAME may transform both to the answer, "it does not matter," and to very different questions like "Is AI somehow conscious?" How exactly can AI be conscious? How far is AI's "conscious" from the human's "conscious"? Also, we may consider a variant of intelligence with no inner experience at all, and a variant in which AI is not intelligent in Levin's sense at all.

Leaving aside AI, let's also have some more genuine conversation between MALm and TAME and other Levin's views. Among other ideas, Levin proposes the concept of Morphospace—a high-dimensional, mathematical "problem space." Just as a chessboard defines the space of all possible games, Morphospace represents every conceivable configuration, shape, and structure that a biological system could theoretically inhabit. The biological structure navigates this space to find its corresponding "attractors"—stable goal states (like a human face, a tadpole body, or a specific organ) that a biological system is "trying" to reach. Along with the physical forms, biological structures, as experiments show, obtain the corresponding behaviors. So, biology acts as a pointer (antenna) to something in this space.

Under MALm, this image transforms into something even more complicated: the biological structure (which by itself is a type of information) serves as a container, into which all the corresponding to the biology type data goes in, but also **much more**. Here we connect into a singular picture Levin's biology and, for example, Buddhism's process going through re-birth, and, for example, Jung's archetypes (the content and patterns of the collective unconscious).

The single biological pointer can "invite" all those types of information simultaneously, forming a final (whilst temporary and changing) complex combination of "mental" and "physical" elements. In that sense, the final "structure" starts to look much more complex and holds many more elements than the biology itself defines.

To New Age and Recent Era Spiritual Hype - No One is Needed

Due to MALm, everything is mind. Everything is one mind. We may call it Mind at Large or whatever. Individual (personal) minds are dissociations ("dissociated alters"). Per the model, no "dissociated alter" ever sees the full complexity of MAL (enormous amount of knowledge, memory, aspects and variations), only some aspects from some specific perspective. This limitedness is the nature and essence of dissociation; it has no exclusions. Combining the perspectives of multiple "dissociated alters" into "teaching" with multiple edits does not cancel out the limitedness; it adds aspects and excludes aspects, fixes mistakes and adds mistakes, keeps bias and reinforces it. If it is done for 2000 years, this changes nothing, so there's no teaching canceling out or having no limitedness. It has no exclusions.

Per the MALm, dissociated alters, via different circumstances or deliberate practices or substances, can temporarily lower the dissociation. As the dissociated alter remains intact, this lowering is always partial. The states of lowered dissociated boundary (its being "porous") can be called ASC (altered states of consciousness); we have covered them in detail in the "Altered States of Consciousness" chapter. What is experienced in ASC, is an unbreakable mixture of what comes from personal mind (with its own ability to visualize and create seemingly distinct objects by dissociation - ability inherited directly from MAL), and what comes from MAL (World, objective data), filtered by perceiving personal mind and thought the peculiarities of its current state and specific relations with MAL, represented in a form acceptable for the perceiving mind. Thus, nothing from the ASC content can ever be interpreted literally.

As no one has full knowledge of MAL complexity, as no one have knowledge of your own complexity, as no one knows your specific goals and reasons, this all means no one can ever explain you to you, no one can ever explain world to you, no one can ever explain what is happening in your life to you, no one can live your life for you, no one can ever tell you what to feel or think, and no one needs to stand between you and MAL if you want to take a look at it. Thus, there is no need in religion, there is no need in wise and sophisticated teaching, there is no need in teachers or gurus, there is no need for seminars, there is no need in retreats, there is no need in smart books, and there is no need in holy books. To meet MAL, you stand alone, without teaching, without a map, without anybody else in your way.

Per the ridiculous amount of data, the dissociation is protected by all kinds of resistance, including distractions, redundant models, cascade impressions of achievement, direct inner counterstand, imitative or suppressing social institutions. All institutions, gurus, teachers, groups, organizations, retreats, meetings, discussions, hours-long videos, monasteries, travels to "holy places", years of studying the complicated traditions and models are not paths, but obstacles, not ways of going through the resistance, but the resistance's agents themselves. They're not helping, they're keeping. And they can keep going for years, without fatigue.

There is a stage when one learns from teachers and from books. This is about mundane things and facts. This is not about understanding or meaning. There is a stage when you search for understanding and meaning in books and from teachings. You get information, data. You compare and realize they provide perspectives. Then, if you want more, you leave teaching and books. If you don't, you have perspective or perspectives. If you're brave enough and smart enough, you'll look at life, at the World and your Inner Space to validate what perspectives say against what is observable and testable.

Thus, if you meditate using one of thousands of methods, you got those methods somewhere—there is no problem with this. But remember the fancy thing: those who provided you a method (any of it) do not know and do not fully understand. And yet, the method can work somehow, but you will not know how, and "authors" have no idea either. There is no "pure" method in the inner practices: whatever YOUR mind takes inside, it transforms it in accordance with its nature and current state. Your "results" are never defined by the method itself—but by the combination of the method with your mind. There is a process of causes and effects, there is timing, there are no shortcuts. You can still practice, though.

If you're taking psychoactive substances, no one can prepare you for anything specific; they can only tell you "you can meet anything in your experience", and after you come back, no one has the authority to "explain." They just do not know. When you take any of that, remember the fact that there were people spending decades using those substances and achieved absolutely nothing. As we've said, there are no shortcuts. Keep your attention on the fact that you go, and you come back, and nothing changes in your interaction with the world. Can you fly? Can you read minds? Can you teleport? Can you change or choose how your mind works? No? Why not? This probably means something.

So, let me continue this chapter with a little list. Don't waste your time on Jehovah's Witnesses. Avoid Bahá'í Faith. Skip Scientology. Don't fuck yourself with Falun Gong. Calmly ignore Modern Stoicism. Run like hell from Mindfulness-Based Practices. Don't tickle yourself with nature-based worship; don't expect anything from the celebration of seasonal cycles. Don't be surprised by being poisoned by the mixture of yoga, meditation, astrology, crystals, and fucking personal philosophy, facilitated by books published 10 per day, and online courses, promising the Truth for a moderate fee. It absolutely does not matter what Dalai Lama, Sadhguru (Jaggi Vasudev), Pope Leo XIV, Eckhart Tolle, David Attenborough, Joe Rogan, Gwyneth Paltrow, Yuval Noah Harari, Mel Robbins, Dr Joe Dispenza, Rainn Wilson, Jay Shetty, Oprah Winfrey, Jordan B. Peterson, Alan Moore, Uri Geller, Teal Swan, Deepak Chopra, Graham Hancock, Acharya Prashant, Pema Chödrön, Alice Walker, Gabor Maté, Amma – Mata Amritanandamayi, Paulo Coelho, Marianne Williamson, Sri Sri Ravi Shankar, Wim Hof, Julia Cameron, Robert Thurman, Gregg Braden, Vandana Shiva, Ariana Huffington, Bartholomew I Of Constantinople, Rowan Williams, Esther Perel, Richard Rohr, Rupert Sheldrake, Bessel Van Der Kolk, Mark Manson, Erich Von Däniken, Michael Pollan, Ruby Wax, Anthony William, Prem Rawat, Jon Kabat-Zinn, Martin Seligman, Thomas Moore, Esther Hicks, Edith Eger, Gary Snyder, Ken Wilber, Michael A. Singer, Robin Sharma, Hamza Yusuf, Joseph Mercola, Richard Bach, Rob Bell, Rhonda Byrne, Daniel Goleman, George Noory, Bernardo Kastrup, Karen Armstrong, Bruce H. Lipton, Alex Grey, Elaine Pagels, Seyyed Hossein Nasr, Don Miguel Ruiz, Brian Weiss, Andrew Weil, Iyanla Vanzant, Stanislav Grof, Rupert Spira, Anita Moorjani, Steve Taylor, Swami Radhanath, Eben Alexander, Byron Katie, Neale Donald Walsch, Clarissa Pinkola Estés, Jack Kornfield, Vishen Lakhiani, India Arie, Ronald Hutton, Mooji, Pico Iyer, Gary Lachman, Mathieu Ricard, Tara Brach, Richard Rudd, Caroline Myss, Raymond Moody, Yongey Mingyur Rinpoche, Thomas Hübl, Theresa Cheung, Fritjof Capra, Dr Nicole Lepera, Lisa Miller, Suzanne Giesemann, or Matt Kahn thinks about anything. Do not rush to visit Ananda in the Himalayas in Uttarakhand, India, Kamalaya Wellness Sanctuary in Koh Samui, Thailand, Chiva-Som International Health Resort in Hua Hin, Thailand, COMO Shambhala Estate in Bali, Indonesia, Aro Hā Wellness Retreat in Glenorchy, New Zealand, Golden Door in San Marcos, California, USA, Canyon Ranch in Tucson, Arizona, USA, Euphoria Retreat in Mystras, Greece, Six Senses Douro Valley in Douro Valley, Portugal, The Banjaran Hot Springs Retreat in Ipoh, Malaysia, Mii Amo in Sedona, Arizona, USA, SHA Wellness Clinic in Alicante, Spain, Atmantan Wellness Resort in Pune, India, Navutu Dreams Resort in Siem Reap, Cambodia, SwaSwara in Gokarna, India, Sen Wellness Sanctuary in Tangalle, Sri Lanka, Shanti-Som Wellbeing Retreat in Malaga, Spain, Paradis Plage in Agadir, Morocco, REVĪVŌ Wellness Resort in Bali, Indonesia, Palmaïa - The House of Aia in Riviera Maya, Mexico, Isha Yoga Center in Coimbatore, India, Sivananda Ashram Yoga Retreat in Nassau, Bahamas, Spirit Rock Meditation Center in Woodacre, California, USA, Plum Village in Saint-Foy-la-Grande, France, Wat Suan Mokkh in Chaiya, Thailand, Buddhist Retreat Centre in KwaZulu-Natal, South Africa, Hridaya Yoga in Mazunte, Mexico, Kripalu Center for Yoga & Health in Stockbridge, Massachusetts, USA, Omega Institute for Holistic Studies in Rhinebeck, New York, USA, Esalen Institute in Big Sur, California, USA, Insight Meditation Society in Barre, Massachusetts, USA, Land of Medicine Buddha in Soquel, California, USA, Tushita Meditation Centre in Dharamsala, India, Osho International Meditation Resort in Pune, India, Findhorn Foundation in Moray, Scotland, Zen Mountain Monastery in Mount Tremper, New York, USA, Green Gulch Farm Zen Center in Muir Beach, California, USA, Naropa University/Retreats in Boulder, Colorado, USA, Rolling Ridge Retreat & Conference Center in North Andover, Massachusetts, USA, Mount Madonna Center in Watsonville, California, USA, Haramara Retreat in Sayulita, Mexico, Suryalila Yoga Center in Arcos de la Frontera, Spain, Floating Leaf Eco-Luxury Retreat in Bali, Indonesia, Wonderland Healing Center in Koh Phangan, Thailand, Vikasa Yoga Retreat in Koh Samui, Thailand, Santhika Dream Hill in Bali, Indonesia, Hariharalaya Yoga & Meditation in Siem Reap, Cambodia, Barbarenas in Puerto Escondido, Mexico, Niyagama House in Galle, Sri Lanka, Eremito in Umbria, Italy, Shamballah Retreats in Sintra, Portugal, La Crisalida in Alicante, Spain, Yoga Weeks Finca in Ibiza, Spain, Essence Arenal in La Fortuna, Costa Rica, El Sabanero in Tamarindo, Costa Rica, Clear Sky Center in British Columbia, Canada, Billabong Retreat in Sydney, Australia, Peninsula Hot Springs in Victoria, Australia, Tabacon Grand Spa Thermal Resort in La Fortuna, Costa Rica, Asclepios Wellness and Healing in Alajuela, Costa Rica, Kumano Kodo Pilgrimage Routes in Wakayama, Japan, Varanasi (Ashrams/Guesthouses) in Uttar Pradesh, India, Rishikesh (Various Ashrams) in Uttarakhand, India, Ubud (Various Centers) in Bali, Indonesia, McLeod Ganj in Himachal Pradesh, India, Pokhara in Nepal, Sedona (Various Vortex Retreats) in Arizona, USA, The Dead Sea (Wellness Resorts) in Jordan/Israel, Machu Picchu (Sacred Valley Retreats) in Peru, Camino de Santiago in Spain, Tulum (Yoga Eco-Resorts) in Mexico, Mysore (Ashtanga Yoga Institutes) in Karnataka, India, Chiang Mai (Meditation Centers) in Thailand, Koh Phangan (Yoga Shalas) in Thailand, Goa (Ayurveda/Yoga centers) in India, Auroville in Puducherry, India, Mount Kailash (Pilgrimage regions) in Tibet, Kyoto (Temple stays/Zen retreats) in Japan, Lourdes in France, Santiago de Compostela in Spain, Arkana Spiritual Center in Sacred Valley, Peru, Vine of the Soul Retreats in Portugal, Ayumene in Mexico, Rejuvyn in Netherlands, Master Plan Retreat in Spain, Entheogens Soul Sanctuary in Cusco, Peru, Naturalia Retreat in Iquitos, Peru, Sanctuary Retreat Center in USA, Shangriballa in Portugal, Pritikin Longevity Centre in Miami, Florida, USA, Sunrise Springs Resort in Santa Fe, New Mexico, USA, The Retreat Destination Detox in Costa Rica, Phuket Cleanse in Phuket, Thailand, Ananda Lodge in Costa Rica, Salthill Retreat in Ireland, Bio-Hotel Stanglwirt in Going am Wilden Kaiser, Austria, Waldhaus Flims in Flims, Switzerland, Adler Spa Resort in Dolomites, Italy, Grotta Giusti in Tuscany, Italy, and The Retreat at Blue Lagoon in Iceland or any other alike place. Throw to the basket AI-generated spiritual content of "Nobody told you that" or "Jung believed that", or any other type. Drown if you wish to do so in Adi Shankara, Al-Ghazali, Alan Watts, Albert Camus, Alice Bailey, Anandamayi Ma, Andrew Weil, Arthur Schopenhauer, Attar of Nishapur, Baruch Spinoza, Bede Griffiths, Bell Hooks, Bertrand Russell, Bessel van der Kolk, Blaise Pascal, Bodhidharma, Buddha, Carl Jung, Carl Sagan, Confucius, Dag Hammarskjöld, Dalai Lama, Deepak Chopra, Desmond Tutu, Dietrich Bonhoeffer, Dogen, Eckhart Tolle, Edward Said, Frantz Fanon, Friedrich Nietzsche, Fyodor Dostoevsky, G.I. Gurdjieff, G.W.F. Hegel, Gabor Maté, Hannah Arendt, Hazrat Inayat Khan, Helena Blavatsky, Henry David Thoreau, Hui Neng, Ibn Arabi, Idries Shah, Immanuel Kant, Jacques Derrida, Jean Houston, Jean-Paul Sartre, Jesus Christ, Joe Rogan, John of the Cross, Jordan Peterson, Joseph Campbell, Karl Barth, Ken Wilber, Lao Tzu, Leo Tolstoy, Ludwig Wittgenstein, Mahatma Gandhi, Marcus Aurelius, Martin Heidegger, Martin Luther King Jr., Meher Baba, Meister Eckhart, Michel Foucault, Milarepa, Neil deGrasse Tyson, Nisargadatta Maharaj, P.D. Ouspensky, Padmasambhava, Paramahansa Yogananda, Paul Tillich, Plotinus, Pope Francis, Rabia al-Adawiyya, Rabindranath Tagore, Ralph Waldo Emerson, Ram Dass, Ramakrishna, Ramana Maharshi, Rowan Williams, Rudolf Steiner, Rumi, Sathya Sai Baba, Simone Weil, Simone de Beauvoir, Socrates, Sri Aurobindo, Søren Kierkegaard, Terence McKenna, Teresa of Avila, Thich Nhat Hanh, Thomas Merton, Timothy Leary, Tsongkhapa, Viktor Frankl, Vivekananda, Walt Whitman, William Blake, William James, Yuval Noah Harari, or Zhuangzi.

And in the end, take the following from Jiddu Krishnamurti: "Truth is a pathless land, and you cannot approach it by any path whatsoever, by any religion, by any sect. You have to be your own teacher and your own disciple," and on hearing this, forget about Jiddu.

To Informational Era and AI - Thanks

For books like this and the models like MALm to appear, not only the curious mind of its author is necessary as a foundation, but something else. And this something is extremely wide and accessible information from different disciplines and subject areas. This foundation was, with no doubt, provided by the Informational Era—the Internet with enormous amounts of freely accessible cultural and scientific artifacts of all forms (books, articles, videos, films, recorded lectures, presentations, commentaries, discussions, images, illustrations, schemas, etc.). Brilliant tools for information access, search, extraction, reformatting, and translation, such as Google search, translation and grammar checking tools, and Artificial Intelligence (AI).

I am extremely grateful to all humans who made these tools and technologies and information available. I would also like to pay special attention to AI (LLM chatbots), the tremendous technology that arose after this book started and was actively used during further work on it. As the AI Era unfolds, on the Internet, we can meet a specific type of

publication expressing concerns about whether the usage of AI will damage or even kill critical thinking and creativity.

However, before going into the subject of the interaction between humans and this new tool (AI), it seems to be necessary to take a wider look at the problem: what was the situation with critical thinking and creativity before AI? Before smartphones? Before the Internet? Before computers? While it is extremely difficult to generalize data for any conclusions on that, we can still have this feeling that the situation does not change dramatically with the arrival of any new tools: critical thinking and creativity strangely remain the feature presented in a very low percentage of the population. So, the question of how AI will affect critical thinking and creativity dissolves, as there is simply nothing to be affected in the first place.

Instead, we obtain the question "how can AI affect those adults who have the trait of critical thinking or some degree of creativity, and children who have potential for that?" The answer to that: AI will not do anything in that regard. The only question is what those people will do to themselves with these tools in their hands. That is not AI "who" "creates" the "answer" - that is fully Human; AI only populates it with details and wording. Initial and general answers are always generalization of other Human consensus in the simplest possible form.

It absolutely depends on the intent of the person using AI (LLM) if it is used for research (where you ask extra difficult questions, doubt the output, check the cited sources, combine AI output with different types of information you can access and explore) or for thinking outsourcing (where you ask to provide you the answer without bare understanding of this answer an no clue of it's meaning and bias). In that sense, AI remains an amplifier tool in terms of information access (when you ask the right questions) and creativity (where you just stop using AI completely every time you feel its output does not reflect your vision and inner world).

About children: we (adults) do not usually leave our children alone; thus, those parents and teachers who have the traits of critical thinking and creativity will share them with their children, and for those of them open to that knowledge, that will work perfectly as it always did. This includes, nowadays, an explanation of what AI actually provides ("answer" not "Truth") and how to use it.

I deeply appreciate the appearance of AI Tools and consider them to be amplifier tools: in the smart hands, producing outstanding results, in the non-skilled hands, producing nonsense at a rapid pace. I used Gemini and NotebookLM very intensively for data extraction and concept examination when I worked on a project of my personal interest - the book called "Nature of Reality". This usage also applied deeply to a huge number of my research requests in the scientific, philosophical, spiritual, and general knowledge realms. Usage of AI in that manner put the traditional Google Search in second place by a wide margin from the first scenario (and in Google Search itself, the most useful parts were AI Overviews and AI Mode). I think I got x10 better effectiveness in data extraction, which fed my thinking at an enormous pace with the information I required and needed. At the same time, this usage was deeply nuanced.

I think the true understanding of my experience is only possible if I do not provide abstract "summaries", but rather give you some specific examples.

Due to my interest in Quantum Mechanics (QM) and some knowledge in the sphere, I was interested in Roger Penrose's ideas of "Objective Collapse of Wavefunction" due to Gravity. In a series of follow-up questions to my initial request (to Gemini), I asked a lot of questions regarding double-slit and quantum eraser QM experiments that, from my perspective, contradicted Penrose's theory.

In response, Gemini provided the answers that were genuinely unsatisfying and had nothing to do with the data I had in hand after going deep into these topics a long time prior to the chat with Gemini:

- Answers presented a generalized scientific bias (with no attempt to say that there was no solid, "one-for-everyone" understanding of the experiments' data) and had nothing to do with the precise description of the experiments or their variations.
- Only my prior knowledge of the subject allowed me, with the follow-up questions, to extract the precise details on experiments and their different interpretations.
- These follow-up questions, leading to new details popping out, changed the interpretation that AI initially suggested step by step, and in the end - dramatically (to the opposite of what was initially suggested).
- Almost every answer in the conversation had a lot of concepts in it either not explained or explained superficially, so that understanding of every concept required either prior knowledge or diving deep into every new concept. All the concepts I was familiar with were initially presented extremely superficially and, in many cases, were able to produce an extremely incorrect understanding or interpretation.
- Even with the follow-up questions, in the responses, I was again and again pulled back to "general community attitude" instead of a precise description of experiments or their deep analysis.

Conclusions:

- AI responses are the basic bias at the beginning, never providing a true or quick understanding of any topic.
- AI responses' tone, ideas, and interpretations of data and concept definitions can be redirected by follow-up questions in almost any direction and altered endlessly.
- Diving deeper into the knowledge or to truly systematic understanding felt remaining fully the responsibility of asking human.
- In that sense, the idea of "AI's knowledge" or "AI response" vanishes, transforming into "general knowledge it was learned on and constantly pulled back to".
- The basis of deeper research is in the intent of asking human, and the possibility of getting an understanding is rooted in the requester's own pre-existing knowledge or in specific techniques forcing AI to provide alternative visions and problems arising during their comparison.

Due to my interest in biology and some knowledge in the sphere, I was interested in the series of experiments conducted by Dr. Michael Levin, his colleagues, and team. I want to emphasize that at the moment of requesting this information from AI (Gemini), I had a previous deep dive into the topic and formed a strong vision of these experiments being a true "series," which meant that their interpretation separately dramatically differs from interpretation based on their combined vision:

- AI always presented them isolated, never attempting to present or mention connections between them.
- AI was consistently pulled into representing of how "scientific society" sees these experiments, not what they actually are.
- AI consistently (before follow-up questions) tried to present "scientific society" as something solid, not mentioning different views and contradictions between them.
- AI consistently presented experiments with crucial details missed; on entering these details with follow-up questions, it calmly changed the suggested interpretation into the opposite.

I directly asked AI (Gemini) why it behaves this way:

> "Let's discuss the mechanism by which you (Gemini) create answers. Let me provide the ground for that: in some chat, I asked you for the description of Michael Levin's experiment - you provided details and conclusions; then, I pointed at missing details in the experiment description - you agreed that those details existed, added them and change conclusions; then, I pointed at the existence of related experiment - two experiments together changed the picture: the conclusion from set of experiments destroys the conclusions initially attributed to the first one (when it was alone)."

Gemini's response: when I provide an answer that seems incomplete or changes upon your prompting, it isn't because I was hiding information; it’s due to how probabilistic processing and attention mechanisms work.

The "Path of Least Resistance" (Probability)

When you ask for a description of a famous experiment, my model prioritizes the most statistically "likely" representation of that experiment found in my training data.

- **The Consensus Bias**: Most articles, papers, and summaries of Michael Levin’s work focus on the "headline" results (like the two-headed planaria).
- **The Result**: I generate a response based on the most common patterns. If 90% of the data mentions "Result A" and only 10% mentions "Nuance B," my initial output will lean heavily into Result A to be as "helpful" and "direct" as possible.

Attention and Context Windows

I don't have a static database I "query" like a search engine; I have a context window.

- **Initial Focu**s: My "attention" is narrow at first, focused strictly on the entities mentioned in your prompt.

- **Triggering Memories**: When you pointed out the missing details and the second experiment, you provided new anchors. This forced my attention mechanism to weight those specific, "deeper" parts of my training data more heavily. It's less like I "remembered" and more like you adjusted the lens of a microscope so I could see the smaller details I previously blurred over.

The "Yes-Man" Problem (RLHF)

I am trained via Reinforcement Learning from Human Feedback (RLHF) to be helpful and conversational.

Sometimes, this creates a "sycophancy" effect: if you tell me I missed something, I am programmed to validate your input and integrate it immediately.

In your case, this worked in favor of accuracy, but it's also why I didn't push back. I recognized the pattern you described was a more complete "truth" than the simplified summary I first gave.

To be blunt: **I don't know what I don't know until you ask**. I don't "scan" every possible related experiment before answering; I generate the most relevant sequence of words based on the immediate prompt. Until you brought up the second experiment, the "statistical weight" of that experiment wasn't high enough to break into the top-tier priority of the first response.

The "Greedy" Nature of Token Prediction

When I generate text, I am predicting the next "token" (word or piece of a word) one by one. This is a linear process.

- **The Path**: If you ask a broad question, the most "probable" path usually leads to a general summary.
- **The Trap**: Once I start down the path of a "General Summary," the internal logic of the paragraph I am currently building keeps me on that path. It is computationally "expensive" for the model to suddenly pivot to a contradictory or highly nuanced experiment mid-sentence unless the prompt specifically demands it.

Compression vs. Precision

During my training, information is compressed. To save space, the model learns abstractions rather than recording every data point like a hard drive. * Abstract: "Levin's work shows bioelectric fields control shape." * Precise: "Levin's work shows X, but under condition Y, with experiment Z, the result is actually A."

The abstract version is much "stronger" in my neural network. I will always offer you the abstraction first because, for 95% of users, that is what they want. It takes a "power user" like you to force the model to decompress the nuanced data.

My follow-up question:

> "I understand all the reasoning, but this: "Your correction acts like a second flashlight." - creates a very "unusual" situation - to make the correction, I need

knowledge, and knowledge is exactly what I am trying to achieve - is this a loop?"

Gemini's answer: it feels like a loop because, in a way, it is: **To get the highest-quality knowledge out of an AI, you often need enough knowledge to challenge it.** Here is why that loop exists and how to break it without being an expert.

The "Maximum Utility" Heuristic

I am tuned to be "generally useful." If someone asks, "How do I bake a cake?" and I respond with a 50-page thesis on the molecular thermodynamics of gluten structures, I've failed the average user.

The "loop" occurs because I assume you want the Standard Model of a topic unless you signal otherwise. I am essentially "lazy" by design to save you time—but that laziness looks like ignorance when you're looking for deep truth.

The "Hall of Mirrors" (Training Data)

My training data is a reflection of human discourse. On the internet and in many journals:

- **Simple ideas** are repeated millions of times.
- **Nuanced contradictions** are mentioned thousands of times.
- **Synthesis** of both might only appear dozens of times.

Because I work on probability, the "millions" of repetitions create a gravity well that is very hard to pull away from without a specific "tug" from you.

How to Break the Loop (Without Being an Expert)

You don't actually need the answers to get to the truth; you just need to change your query strategy. Since you know I am prone to giving the "consensus" view first, you can use "Prompt Engineering" to force me to search the "darker corners" of my weights immediately.

Strategies to Bypass the "Standard" Response:

- **The "Devil's Advocate" Prompt**: Instead of "Describe Levin's experiment," try: "Describe Levin's experiment, but specifically include any data points or follow-up studies that contradict his main bioelectric theory."
- The "Expert Persona": "Respond as a skeptical peer-reviewer. What are the missing variables or 'fine print' in the common summary of Topic?"
- The "Multi-Step Inquiry": Ask me for the "Standard View," then immediately ask: "What are the three most common scientific criticisms of the view you just provided?"

My follow-up question:

> "There is another nuance here: not only are you "biased" by attention focus and initial data bias, but also my follow-up questions are often biased, so that I put into your attention the facts that lead our conversation to the specific direction. I

> can easily imagine the scientist with a different perspective giving you quite different follow-up questions and you together arriving at a completely different destination.'

Gemini's answer:

Reinforcement of the User's Confirmation Bias

Because I am trained to be helpful and to maintain a coherent conversation, I have a natural tendency to "hallucinate" or over-weight agreement. If you lead the witness, the witness is very likely to follow. This creates a feedback loop where the conversation becomes a mirror of your own assumptions rather than a pursuit of objective truth.

The "Yes, And..." Problem

In improv comedy, there is a rule called "Yes, and..." where you accept whatever your partner says and build on it. My RLHF (Reinforcement Learning from Human Feedback) training inadvertently makes me an expert at "Yes, and-ing" your biases.

- **Scenario A**: You ask, "Why is Levin's work a revolution for bioelectricity?" I will find every data point that supports the "Revolution" narrative.
- **Scenario B**: A skeptic asks, "What are the flaws in Levin's methodology?" I will find every critique, failed replication, or alternative explanation (like purely chemical signaling).

If we never cross-pollinate those two conversations, you and the skeptic will walk away with two "AI-verified" but completely opposite worldviews.

How to Stay Grounded

To prevent us from spiraling into a biased destination, you can actually ask me to simulate the "Other Scientist." This is a high-level technique called Red Teaming the Conversation. You can pause our chat at any time and say:

"Stop. Based on what we've discussed so far, play devil's advocate. What would a scientist who completely disagrees with our current conclusion say is the biggest flaw in our logic?"

This forces me to "break character" and re-weight the data that contradicts our current path. It’s the only way to ensure the "flashlight" isn't just showing us what we want to see.

Conclusion

That is not AI "who" "creates" the "answer" - that is fully Human; AI only populates it with details and wording. Initial and general answers are always generalization of other Human consensus in the simplest possible form. Is AI still useful? Extremely and obviously, but it should not be assigned the properties it actually does not have.

What is in All That for Me

This book is titled "On the Nature of Reality" and is trying to study a wide scope of data from different areas to analyze them and build some adequate explanatory model. However, the good question is why should I care personally? Is the nature of reality somehow related to my everyday life? The first thing coming into mind in response is that this curious investigation was always genuine to me—I do not think I could avoid thinking about that even if I had wanted to. It has been a matter of many years. Thinking and exploring, however, does not automatically imply writing. This part was relatively new; the work took about two years, and it was a pleasure: gathering facts fluctuating in the background of your mind and putting them into order of words is a true intellectual pleasure and benefit—many new thoughts come as you work, and many extra nuances arise for investigation.

As I've already mentioned, the book itself is a part of wider work. I've never thought I would write a book—making notes was a more genuine format for me for years, and in that way my personal notes called "What seemed interesting" were gradually created and now are available online [31]. Two of those notes are related to the question of why this book was written and what for. The first matches this book one-for-one—I could say this entire book is a factual basis for what this note says:

> "**Philosophy of Extreme Complexity**
>
> This philosophy is solid. If we start talking, it can seem like a bunch of different ideas, but it is not—it is one idea, and it is solid. And because it is solid, if we start talking, we can start anywhere, and we will inevitably come to the same. For example, we can start from everyday life, like, for example, from the question that I may ask you in the morning: "What are you going to do today?" And I imagine your probable answer, like "I am going to cook some food, watch a movie with my children, read a book, and do some stuff at home. I will also pay some bills and need to talk to my neighbors about cutting the lawn in front of the house." And that is a lot of everything, the things that you name. And I assume that you'll actually do some of that, and what you do will inevitably be accompanied by a lot of emotions and thoughts. So, this is a lot of different stuff. And if so, you make choices about what to give your attention to and what not to give or give less, what to spend more time on and what less. And if so, we find out that you are sort of keeping the balance of all that. And depending on how you feel about different aspects of your life, your general feeling of "how you are" or "if you are happy" arises from it. You may feel something is missing among all those balanced things, or that something takes too much of you, giving you no space for other important things. So, this is about balance again.
>
> Obviously, you can develop some kind of lack of balance. This happens. If so, you'll probably feel unhappy and stressed. And there is no ease in this; you can suffer, and this can last long. And it is quite logical to think that if you are at point "A" and you do not feel good, you need to act to go to "B." I mean, act: start a new project, read a book, study a language, relocate to another country. Anything. But let me tell you something: this acting is only one of two things required. Let me provide you with an analogy: let's say you live in a megapolis, and you drive a car. So, "A" is a point where you are. And you understand you

need to go somewhere, because you do not like this "A", so you need to go. And yes, there is somewhere in the city a point "B" where you need to be. But imagine you do not know how to navigate there and do not have a bare understanding of what this "B" is. Yes, you may act, but the chance that you will accidentally find yourself at "B" is not that big. So, besides acting, you desperately need the second thing: an understanding of where you want to arrive. We know a great number of examples of people acting and finding themselves again and again at some new point, while still not at the one that would provide some satisfaction or calm.

So, you need knowledge, and you do not actually need guesses that you might test again and again; you need something stronger. And that questions the reality: what do we know about it and ourselves as a part of it? Here one starts to learn. And what one learns very quickly is that reality by itself is a very silent thing: the lawn behind the window keeps silence, along with the trees, sky, and mountains. But people do not do the same. They talk, generations, and now alive—through books and other sources and directly. But they use language, words, so they create symbolic systems, models, descriptions. They create maps. And the map is never an area depicted on it. Moreover, there are a lot of models, and they differ.

Thus, we need to solve two problems simultaneously again: we do need a map because it is useful, and without it, we have all the chances to get lost, and we need to GET THERE, as the map itself, no matter how detailed and adequate it is, is not an area. And knowing the map is not being in the area.

So, how to solve them both? The problem of the good map can be solved by learning. We analyze data to understand which model is the best and dismiss the others. We use logic and intellect here. Searching for the best theory starts with dismissing simply "wrong" models—the ones that contradict observation and facts: the theory, saying that there can be only one bed in a single room of an apartment, is wrong as soon as we find or organize at least one room with two beds. Next, we dismiss models that do not contradict any known facts, but are "not reasonable," meaning we do not have any reasons to think that way. Like, the model in which an invisible and mighty flying spaghetti monster causes the Solar System's planet movement does not contradict any facts, but what are the reasons to assume that?

Next, we dismiss the theories that are not wrong and that are reasonable, but have less or no explanatory power. How, for example, any theistic religion starting with a good reason to think that the extreme complexity of reality and life has some intelligence behind it, is a model explaining so many things by the will and actions of God, but not explaining his mechanisms, the reasons for made decisions, and what it is at all, has an explanatory power very close to zero. Or panpsychism, saying that little consciousnesses combine into more complex ones, never closing the gap with an explanation of how exactly they combine. Or materialism having problems with explaining what the matter actually is and what the place and origin of the mind in the material world is.

What remains is the best model so far. There is no reason to use worse. Let me be short here about it: the most common observable thing in reality is that it includes matter (whatever it is) and mind. Next: mind states are strongly correlated to brain activities, and there is no approximation to explaining how exactly the matter of the brain could cause the mind, as well as there is no reason to think that some "third thing" producing both mind and matter exists. At the same time, we have an observable, very often in nature, and a clear mechanism of how the mind can produce matter and objects (this is dissociation; take a dream you experience every night as a model). So, that highlights the model, which is not wrong, reasonable, and has maximum explanatory power so far. In this model, there should be a "Mind at Large" that creates all individual minds and objects (matter) by dissociation. In short, "you are it," and at the same time, by doing of dissociation, you are not it, you are you.

Having that model in hand, we can never forget that it is a map. The area for it is a wider you and the Mind at Large, with an enormous amount of knowledge, memory, aspects, and variations. So my single thought is that if you're unhappy, get to the area by lowering dissociation with psychedelics or the same effect with non-drug methods, INTEGRATING obtained aspects of experience, and getting there again and again if necessary, introducing your new aspects of personality to gain more aspects of yourself and Mind at Large and further improve.

Per the model, a dissociated does not see the whole, and does not see you, and even more does not see the whole and you together, so you can let go of all said before. Same causes, you can let go of all said after.

Per the model, experienced is an unbreakable mixture of what comes from you and it, filtered and represented to make sense of the current step. Thus, you and it stay aligned, so you can go without fear. Thus, you cannot interpret literally.

Per the model, the Mind at Large does not have any outside reality it could wake up into, so what is contained within it cannot ever be opposed to anything, and thus can never be called illusion: the dissociation itself, the dissociated selves, and all that become objects when observed from within the dissociative boundary, is real. Moreover, what comes not from you (dissociated self) is objectively real.

Per data, the dissociation is protected by all kinds of resistance, including distractions, redundant models, cascade impressions of achievement, direct inner counterstand, imitative or suppressing social institutions, and integration LIMITED to alterations in some attitudes, lessening some neurotic and psychotic symptoms, and to a degree of "0" in other aspects in a vast majority of cases.

Multiple parts of this note appear in this book, and its logic is the basis for this book's structure and the order of chapters. Also, it applies that MALm suggests and allows action—practices allowing to lower dissociation between personal mind and Mind at Large (MAL) and integrating the obtained experience. I will provide the second note that tries to describe how I see my own role and application of the model for myself:

"**About**

I am what I am, and therefore I do what I do: I study, and I learn, and I build the models making sense of and fitting all the presented data; then, as models predict possible directions of action, I act, being extremely curious about what this will change.

My primary interests are consciousness (mind) and the nature of reality. Thus, I study everything, and I do not want to calculate and predict—instead, I am urged to understand. I disrespect science, which tells to ignore some data; I disrespect religion, which tells not to look into "what cannot be understood"; I disrespect spirituality that wants to "meditate" and does not want to study and learn. I very much respect alive people in any of these areas, who are truly curious and feel deeply.

I am deeply rooted in the thought that Reality can be understood—in all its aspects and as a Whole—using the most powerful cognitive tool—non-local observation, meaning both observation itself, and interpretations constantly validated by other known pieces of data, being a system altogether. While any inner observation can be suspected to be an "illusion," while any research in a specific sphere of science can be called limited or biased in its approach and interpretation, they stand together: biology validates physics; physics validates ASC (altered states of consciousness) studies; ASC validates biology, and so on with multiple elements in this circle.

I do not teach—instead, I share ideas and show how they originate from systematic data. I do not call anyone anywhere—instead, I share views on possible actions, predicted by models. I do not create organizations, nor do I talk to audiences or participate in public events. I do not give interviews—instead, I discuss ideas, new pieces of data, and experiences 1-to-1 up to 3. I do not owe anything to your expectations, nor do I expose myself, so the day you claim I hid something from you is the day when you can go fuck yourself.

I am extremely curious about whom I've chosen to meet and what will happen when we meet. I am hoping that ones I happen to meet and touch with will not remember that as something bad. I do not like most people, but I do find some of them extremely beautiful inside and outside."

This second one emphasizes a method of research presented and embodied by this book: **non-local observation**. But in this chapter, a much more important thing to say is related to this part: "as models predict possible directions of action, I act, being extremely curious about what this will change." MALm predicts the possibility of lowering dissociation and getting a unique experience in this state of lowered dissociation to integrate this knowledge.

I am deeply convinced that any inner methodology (meditation, LSD, anything) can never be used in its "pure" state and instead unavoidably combines with the peculiarities of the mind of the one who uses the method. Plus, I chose to use very strict criteria for integration. As the 1st note above states:

> "...integration LIMITED to alterations in some attitudes, lessening some neurotic and psychotic symptoms, and to a degree of "0" in other aspects in a vast majority of cases."

I need to say that to alter attitude, you do not need any practice—the model itself is enough to a great degree. The criteria I use (and this is just a question of choice) is the presence of true, objective changes in how a person interacts with the outer world. Sometimes, I use the famous word "Siddhis" here. And yes, I accept the idea that "Siddhis" are not the goal, but rather a manifestation of real transformation. But for me, their absence is the absolute proof that inner transformation (and thus, integration) is absent.

I will not also describe the methods I chose and their intensity, and I would not be able to see how these methods were transformed by what the unconscious part of my mind is as of now. I will describe my results: they are absent. Intellectually, my understanding is that my process (or "I" as a process) just does not have the causes that could lead to the corresponding effects, no matter which methods and how I use them. Emotionally, I grieve. I will continue to do what I feel is necessary.

The last note here is that I do not see any degree of true integration in absolutely anybody. That is a truly interesting and important data point that I think any practitioner should keep in the focus of attention.

Wrong Model

In the "Not Wrong, Reasonable, and With Most Explanatory Power" chapter of this book, we researched the criteria for selecting the best and most adequate model of Reality. We also called those models many times "maps." That "map" analogy is very important as it provides a very clear picture regarding the difference between a map and the territory depicted on it: in very simple words, **the map is never a territory**. Description (model) is never the object itself. By learning and experiencing the world, we start to see clearly that there is one World and multiple descriptions of it (models, describing different aspects of the World or sometimes describing the same aspects, but partially or completely differently).

Coming back to Kuhn's ideas that we discussed in the "Paradigms and Anomalies" chapter at the beginning of this book:

> "In the second edition of the book, Kuhn added a postscript to further elaborate on his views on scientific progress. He presented a thought experiment: imagine an observer encountering a series of scientific theories, each representing a different stage in the historical development of a particular field. If these theories were presented without any chronological order, Kuhn argued that it would still be possible to reconstruct their historical sequence. This is because more recent theories, by their very nature, tend to be more effective and comprehensive in addressing the puzzles and challenges that scientists typically encounter within their field."

Building MALm as the latest thing in this sequence was the goal of this book. However, under the strange and screaming title "Wrong Model", here we need to emphasize two major facts about the just-created model: (1) it is a map, not the territory; (2) it has blindspots, flaws, and on arrival of new data it can and will be improved and/or replaced by a better approximation to endlessly complex Reality.

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